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Fine-tuning of Neural Network Models for the Improved Identification and Quantification of Proteolytic Cleavage Events from Mass Spectrometry-Based Proteomics Data

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- **Softwareentwicklung:** Unterstützung bei der Erstellung, Anpassung und Fehlerbehebung von Programmcode, einschließlich agentischer Programmierunterstützung.
- **Textarbeit und Literaturrecherche:** Unterstützung bei der Formulierung, sprachlichen Überarbeitung, Verfeinerung und Übersetzung von Texten sowie bei der Literaturrecherche.
- **Grafische Gestaltung:** Erstellung einzelner Icons für Präsentationen und Workflow-Abbildungen.

Die Verantwortung für den verwendeten Code, die wissenschaftlichen Aussagen und die endgültige Fassung der Arbeit liegt bei mir.

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Abstract

Protein N-termini provide information about protein maturation and proteolytic processing, supporting studies of disease mechanisms and therapeutically relevant proteases. HUNTER enriches N-terminal peptides. In the analysed datasets, heavy dimethylation labelled free N-termini and lysine side chains. Identifying these peptides relies partly on models that predict retention time and fragment intensities. This thesis investigated whether adapting these models to HUNTER data could improve the identification of N-terminally heavy-dimethylated peptides in data-independent acquisition analyses. An automated DIA-NN benchmark pipeline, a HUNTER-compatible Carafe2 workflow and an external retention-time fine-tuning framework were developed. Comparisons across four datasets used separate training and held-out search files and evaluated peptide identifications alongside uniquely assigned N-terminal positions.

Integrated DIA-NN tuning reduced N-terminal site coverage for N-terminally heavy-dimethylated peptides in all four datasets. In contrast, the integrated Carafe2 workflow increased site coverage by approximately 7–52% relative to the unchanged-model DIA-NN baseline. The Carafe2 gain extended from N-terminally heavy-dimethylated peptides and precursor groups to uniquely assigned N-terminal sites, providing broader coverage of protein N-termini for biological interpretation. However, differences in models, spectral libraries and database configurations prevent attributing the gains to model adaptation alone. Both workflows also changed which N-terminally heavy-dimethylated peptides were identified, with additional identifications accompanied by losses from the baseline set. Even similar total counts after DIA-NN tuning concealed substantially different peptide sets. External retention-time fine-tuning provided greater control over training and evaluation. Heavy-focused sampling and loss weighting slightly improved predictions for selected modification groups, but the selected external workflows did not consistently increase coverage of N-terminally heavy-dimethylated peptides over the baseline or corresponding integrated workflow. Together, these findings show that adaptation must be assessed through identification outcomes as well as prediction accuracy. The developed workflows support this evaluation and provide a basis for broader N-terminal profiling in studies of proteolytic processing.

Zusammenfassung

Protein-N-Termini geben Aufschluss über die Proteinreifung und proteolytische Prozessierung und unterstützen damit die Untersuchung von Krankheitsmechanismen und therapeutisch relevanten Proteasen. HUNTER reichert N-terminale Peptide an. In den untersuchten Datensätzen wurden freie N-Termini und Lysin-Seitenketten durch isotopisch schwere Dimethylierung markiert. Die Identifizierung dieser Peptide stützt sich unter anderem auf Modelle zur Vorhersage der Retentionszeit und der Fragmentintensitäten. Diese Arbeit untersuchte, ob die Anpassung dieser Modelle an HUNTER-Daten die Identifizierung von N-terminal schwer dimethylierten Peptiden in datenunabhängigen massenspektrometrischen Analysen verbessern kann. Dafür wurden eine automatisierte DIA-NN-Pipeline für vergleichende Analysen, ein HUNTER-kompatibler Carafe2-Workflow und ein Framework für das externe Fine-Tuning von Retentionszeitmodellen entwickelt. Die Vergleiche umfassten vier Datensätze mit getrennten Messdateien für das Training und die Bewertung durch Datenbanksuchen. Neben der Anzahl identifizierter Peptide wurden eindeutig zugeordnete N-terminale Positionen ausgewertet.

Das integrierte Fine-Tuning in DIA-NN verringerte in allen vier Datensätzen die Anzahl identifizierter N-terminaler Positionen von N-terminal schwer dimethylierten Peptiden. Der integrierte Carafe2-Workflow identifizierte dagegen etwa 7–52% mehr N-terminale Positionen als die DIA-NN-Baseline mit unveränderten Modellen. Der Zugewinn umfasste somit neben zusätzlichen Peptidformen auch eine breitere Erfassung eindeutig zugeordneter N-terminaler Positionen. Aufgrund von Unterschieden in Modellen, Spektralbibliotheken und den Einstellungen der Datenbanksuche lässt sich dieser Zugewinn jedoch nicht allein auf die Modellanpassung zurückführen. Beide Workflows veränderten zudem, welche Peptide identifiziert wurden: Zusätzlich identifizierten Peptiden standen andere gegenüber, die nur mit der DIA-NN-Baseline identifiziert wurden. Auch nach dem DIA-NN-Fine-Tuning unterschieden sich die identifizierten Peptide trotz ähnlicher Gesamtzahlen deutlich. Das externe Framework erlaubte es, die Trainingseinstellungen gezielt anzupassen und die Vorhersagegenauigkeit anhand separater Validierungsdaten zu prüfen. Die stärkere Berücksichtigung bestimmter schwer dimethylierter Peptidformen im Training und eine stärkere Gewichtung ihrer Vorhersagefehler verbesserten die Vorhersagen für ausgewählte Modifikationsgruppen geringfügig. Die ausgewählten externen Workflows erhöhten die Zahl der identifizierten N-terminal schwer dimethylierten Peptide jedoch nicht durchgängig gegenüber der DIA-NN-Baseline oder dem jeweiligen integrierten Workflow. Die Ergebnisse zeigen, dass der Nutzen einer Modellanpassung sowohl an der Vorhersagegenauigkeit als auch an den tatsächlich identifizierten Peptiden beurteilt werden muss. Die entwickelten Workflows ermöglichen diese Vergleiche und bieten eine Grundlage, um N-Termini bei der Untersuchung proteolytischer Prozessierung umfassender zu erfassen.

List of Abbreviations

bwUC3	bwUniCluster 3.0
CPU	Central processing unit
DDA	Data-dependent acquisition
DIA	Data-independent acquisition
ESI	Electrospray ionisation
FDR	False discovery rate
HCD	Higher-energy collisional dissociation
Heavy-N	N-terminal heavy dimethylation
HI	Hydrophobicity index
GPU	Graphics processing unit
GRAVY	Grand average of hydropathy
HUNTER	High-efficiency undecanal-based N-termini enrichment
LC	Liquid chromatography
LC-MS/MS	Liquid chromatography tandem mass spectrometry
LR	Learning rate
iRT	Indexed retention time
KDE	Kernel density estimation
MAE	Mean absolute error
MBR	Match-between-runs
M^{pro}	SARS-CoV-2 main protease
MS	Mass spectrometry
MSE	Mean squared error
PTM	Post-translational modification
RT	Retention time
RAM	Random-access memory
R^2	Coefficient of determination
TAILS	Terminal amine isotopic labeling of substrates
VRAM	Video random-access memory

1. Introduction

Proteins are the primary functional molecules of biological systems and perform structural, catalytic, and regulatory roles. However, their functional state is not determined by amino-acid sequence or total abundance alone. Post-translational modifications (PTMs) and processing events can lead to distinct proteoforms (Smith et al., 2013) that differ in their activity, subcellular localisation, interaction partners, or stability. Characterising these molecular states is therefore important for relating proteomic measurements to biological function.

Proteomics is the systematic study of the proteins present in biological systems. Mass spectrometry (MS) is a central analytical technology in proteomics and is commonly coupled with liquid chromatography (LC). MS-based proteomics enables the large-scale identification and quantification of proteins and can also provide information on their chemical state, cellular context and localisation (Aebersold & Mann, 2016). Specialised MS-based approaches, such as N-terminomics, enable the identification of protein termini generated by proteolytic cleavage. This allows proteolytic processing to be investigated systematically at the proteome level. Such analyses can provide insights into protease activity and its biological consequences (Koudelka et al., 2022). The following sections of the Introduction outline the biological and pharmaceutical relevance of proteolysis, the analysis of protein N-termini by N-terminomics, and the computational challenges associated with their identification in data-independent acquisition (DIA) proteomics.

1.1. Proteolytic processing: biological and pharmaceutical relevance

Proteolytic processing is not limited to complete protein degradation. It is an irreversible post-translational modification in which proteases cleave peptide bonds at specific positions within a protein sequence (Klein et al., 2018; Koudelka et al., 2022). Limited proteolysis can generate stable protein fragments that have new N- and C-termini and are functionally distinct from the uncleaved protein (Klein et al., 2018; Koudelka et al., 2022). It can thereby regulate protein maturation, activity, stability and localisation, as well as cellular signalling (Klein et al., 2018). Protein abundance alone does not necessarily reflect the biological impact of proteolytic processing (Klein et al., 2018). Detecting protein termini is therefore important for determining whether, and at which position, a cleavage event has happened (Koudelka et al., 2022).

Proteases can themselves be therapeutically relevant targets. The SARS-CoV-2 main protease (M^{pro}) cleaves the viral polyproteins pp1a and pp1ab at 11 sites, which leads to the release of non-structural proteins required for viral replication (Owen et al., 2021). Nirmatrelvir (PF-07321332)

inhibits M^{pro} and thereby interferes with this essential step in viral replication (Owen et al., 2021). In combination with ritonavir, nirmatrelvir is marketed as Paxlovid and was authorised for use in the European Union in January 2022 (Lamb, 2022). Beyond its role in viral polyprotein processing, M^{pro} can also target host proteins. Using TAILS, an N-terminomics approach for the enrichment and identification of protein N-termini, Pablos et al. (2021) identified 102 high-confidence cleavage sites across 101 human protein substrates. This study illustrates how cleavage-site-resolved analyses can support the characterisation of a therapeutically relevant protease by identifying potential host substrates and their cleavage sites.

Cleavage-site measurements can also show how pharmacological treatment affects proteolytic processing. The complement system is a proteolytic cascade that can be targeted with eculizumab, a therapeutic antibody that binds complement component 5 (Demir et al., 2025). Demir et al. (2025) used N-terminomics to analyse plasma samples from patients before and after eculizumab treatment. They also measured total protein abundance in the same samples. For complement proteins, changes in the abundance of N-termini correlated only weakly with changes in total protein abundance (Demir et al., 2025). Thus, measuring N-termini can reveal treatment-induced changes in proteolytic processing that may be missed when only total protein abundance is considered (Demir et al., 2025).

Together, the M^{pro} and eculizumab examples represent two complementary pharmaceutical uses of cleavage-site information: characterising a therapeutically targeted protease and measuring changes in proteolytic processing following a pharmacological intervention. More broadly, proteases represent an important class of therapeutic targets across diverse disease areas. However, validating new targets requires an understanding of their physiological substrates and roles in disease, while similarities between protease active sites can make selective inhibition difficult (Drag & Salvesen, 2010). Their systematic identification can reveal protease substrates and cleavage sites and thereby help to relate protease activity to biological function (Koudelka et al., 2022). This principle motivates the specialised analytical approaches introduced in the following section.

1.2. From bottom-up proteomics to N-terminomics

MS-based bottom-up proteomics enables a systematic, large-scale analysis of proteomes by measuring the peptides generated from proteins (Aebersold & Mann, 2016). The following subsections introduce the general bottom-up proteomics workflow, describe its adaptation for the specialised analysis of protein N-termini by N-terminomics, and then explain the principles of LC-MS/MS analysis used to measure the resulting peptide mixtures.

1.2.1. Bottom-up proteomics

Two principal strategies for MS-based protein analysis are top-down and bottom-up proteomics. Top-down proteomics analyses intact proteins and can therefore preserve information about molecular features that occur together within individual proteoforms, such as PTMs or proteolytic

processing. In bottom-up proteomics, proteins are enzymatically digested and analysed through the resulting peptides. When this strategy is applied to a mixture of proteins, it is commonly referred to as shotgun proteomics (Zhang et al., 2013). This strategy is experimentally and computationally more feasible and is the most widely used approach in MS-based proteomics (Aebersold & Mann, 2016; Zhang et al., 2013).

A typical bottom-up workflow begins with the extraction of proteins from a biological sample. Depending on the sample preparation strategy, proteins may first be solubilised and denatured. Before enzymatic digestion, reduction and alkylation are commonly performed to disrupt disulfide bonds and prevent their reformation (Varnavides et al., 2022). Digestion is most commonly performed with trypsin, which cleaves peptide bonds primarily on the C-terminal side of lysine and arginine residues and thereby generates a largely predictable population of peptides (Zhang et al., 2013). In a conventional workflow, the resulting complex peptide mixture is injected into the LC system after digestion and peptide cleanup for separation and subsequent MS/MS analysis. Identified peptides are used to infer the proteins present in the sample, while peptide-level quantitative measurements can be combined to estimate protein abundance (Aebersold & Mann, 2016; Zhang et al., 2013).

Bottom-up proteomics thus translates a complex protein sample into an analytically accessible peptide population. Comprehensive analysis of protein N-termini, however, requires adaptations to this general workflow. The following subsection introduces N-terminomics and strategies for the selective enrichment of N-terminal peptides.

1.2.2. N-terminomics and N-terminal peptide enrichment

Protein N-termini provide information about distinct proteoforms and their molecular origin (Demir et al., 2022). They may represent canonical protein starts or arise from alternative translation initiation, initiator methionine removal, or proteolytic processing. Protein N-termini may additionally carry modifications such as acetylation (Demir et al., 2022). Proteolytically generated N-termini are referred to as neo-N-termini (Demir et al., 2022). Global profiling of protein N-termini can therefore provide insights into protein maturation and proteolytic regulation (Demir et al., 2022). MS-based N-terminomics enables the proteome-wide identification of these termini (Demir et al., 2022; Koudelka et al., 2022).

In conventional shotgun proteomics, protein N-terminal peptides are difficult to analyse comprehensively (Perrar et al., 2019). They constitute only a small fraction of the complex peptide mixture generated by enzymatic digestion, and neo-N-terminal peptides may additionally be present at low abundance (Perrar et al., 2019; Demir et al., 2022). Peptide identification by database search involves comparing measured MS/MS data with candidate peptides generated from protein sequences. Neo-N-terminal peptides generally match the digestion protease's specificity at only one terminus. A semi-specific search requires the protease's cleavage rule to be met at one end of a candidate peptide, while the other end may vary (Perrar et al., 2019; Weng et al., 2019). Selective enrichment increases the proportion of N-terminal peptides in the analysed mixture and facilitates

their identification (Perrar et al., 2019). Dedicated data analysis accounts for their variable termini, so enrichment and analysis together support comprehensive N-terminome profiling (Perrar et al., 2019).

Several strategies have been developed to enrich protein N-terminal peptides, including charge-based separation, positive selection, and negative selection (Perrar et al., 2019). Terminal Amine Isotopic Labeling of Substrates (TAILS), which was used in the M^{pro} study described in Section 1.1, is one established negative-selection method (Kleifeld et al., 2011; Pablos et al., 2021).

High-efficiency Undecanal-based Enrichment of N-termini (HUNTER), which is of particular relevance to this thesis, likewise enriches N-terminal peptides by negative selection (Figure 1) (Weng et al., 2019; Demir et al., 2022). Following protein extraction, reduction and alkylation, the proteins are purified using paramagnetic SP3 beads. Before digestion, free α-amines at protein N-termini and ε-amines of lysine side chains are blocked by reductive dimethylation. The HUNTER datasets investigated in this thesis used heavy stable-isotope dimethylation for this purpose; the modification is defined for peptide or protein N-termini and lysine side chains (UniMod, 2026). In this thesis, Heavy-N denotes heavy dimethylation at the N-terminus. After removal of excess labelling reagents, the proteins are enzymatically digested. Digestion generates internal and C-terminal peptides with free α-amines at their new N-termini, whereas the original protein N-termini remain dimethylated or naturally blocked (Weng et al., 2019; Demir et al., 2022). Before enrichment, an aliquot of the complete digest may be analysed as a pre-HUNTER sample to assess dimethyl-labelling efficiency (Weng et al., 2019). The newly generated free peptide amines are then modified with undecanal, increasing the hydrophobicity of the internal and C-terminal peptides. These peptides are retained on reversed-phase material, while the protein N-terminal peptides pass through and are collected (Weng et al., 2019; Demir et al., 2022). The enriched N-terminal peptide fraction is subsequently desalted and analysed by LC-MS/MS, followed by a semi-specific database search and N-terminus-specific annotation (Weng et al., 2019; Demir et al., 2022).

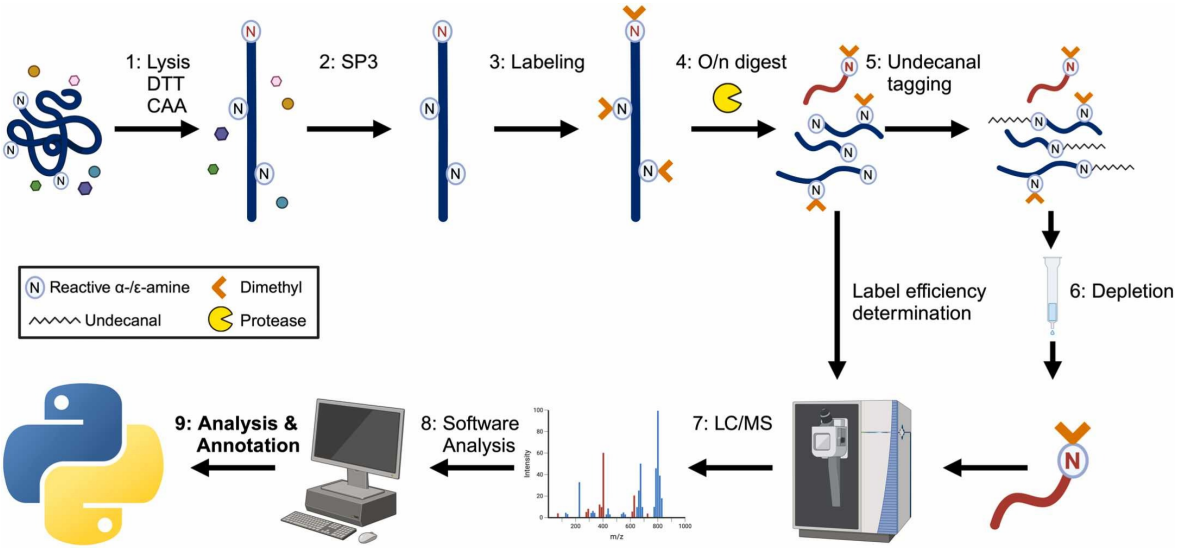


Figure 1.: Schematic overview of the HUNTER workflow for enrichment and identification of protein N-terminal peptides. Protein lysates are reduced, alkylated, purified with SP3 beads, and their primary amines are reductively dimethylated before proteolytic digestion. An aliquot of the digest can be used to assess labelling efficiency. Newly generated internal and C-terminal peptide amines are modified with undecanal and depleted by C18 chromatography, whereas labelled protein N-terminal peptides remain in the flow-through for nanoLC-MS/MS analysis and computational annotation. Reproduced from Stockert et al. (2024).

1.2.3. LC-MS/MS analysis and data acquisition

Whether generated by a conventional bottom-up workflow or enriched by HUNTER, peptide mixtures are analysed by LC-MS/MS. To improve peptide separation prior to mass spectrometric analysis, peptide mixtures are commonly subjected to online reversed-phase liquid chromatography. In nanoflow LC, peptides interact with a hydrophobic stationary phase, typically C18, and elute at different times as the proportion of organic solvent in the mobile phase increases (Zhang et al., 2013; Ludwig et al., 2018). The time at which a peptide elutes under defined chromatographic conditions is referred to as its retention time (RT). RTs provide additional peptide-specific information that can support computational identification across LC-MS/MS measurements (Ludwig et al., 2018). At the column outlet, electrospray ionisation (ESI) transfers the eluting peptides from the liquid phase into gas-phase ions that can enter the mass spectrometer (Aebersold & Mann, 2016).

The mass spectrometer measures ions according to their mass-to-charge ratio (m/z). An MS1 survey scan records the precursor ions that co-elute at a given point in the chromatographic gradient. For tandem mass spectrometry, precursor ions are isolated and fragmented. The resulting fragment ions are recorded in MS2 spectra, which provide sequence-specific evidence for peptide identification (Aebersold & Mann, 2016). In addition to identification, precursor- or fragment-ion signals recorded across chromatographic elution can be used for peptide quantification (Aebersold & Mann, 2016). The principal HUNTER DIA datasets analysed in this thesis were acquired using hybrid quadrupole-Orbitrap mass spectrometers of the Orbitrap Exploris series. These instruments use a quadrupole to select precursor ions, an ion-routing multipole for higher-energy collisional

dissociation (HCD), and an Orbitrap for high-resolution mass analysis (Bekker-Jensen et al., 2020). The exact LC and MS acquisition parameters are reported in Materials and Methods Chapter 3.

The acquisition strategy determines which precursor ions are selected for fragmentation. In data-dependent acquisition (DDA), precursor ions are selected from an MS1 survey scan, commonly based on their signal intensity, and sequentially isolated and fragmented (Aebersold & Mann, 2016; Zhang et al., 2013). In data-independent acquisition (DIA), precursor ions are not individually selected. Instead, the instrument cycles through consecutive isolation windows across a defined m/z range and fragments all precursor ions within each window (Aebersold & Mann, 2016; Ludwig et al., 2018). Because several co-eluting precursors can be isolated together, DIA produces multiplexed MS2 spectra containing fragment ions from multiple peptides.

Thus, LC-MS/MS combines chromatographic separation with precursor- and fragment-ion measurements to identify and quantify peptides in complex mixtures. Interpreting the multiplexed signals acquired by DIA requires dedicated computational analysis.

1.3. Computational analysis of HUNTER DIA data

DIA can provide more consistent peptide detection and quantification across samples than DDA because it systematically samples a defined precursor mass range (Ludwig et al., 2018). Identifying N-terminal peptides in DIA data requires plausible precursor candidates and information about their expected signals. The following sections explain how these candidates are defined for HUNTER samples, represented in spectral libraries and evaluated by DIA-NN.

1.3.1. Database searches and the HUNTER candidate space

In a database search, peptide identification is restricted to a predefined search space: the candidate peptides and precursors considered when interpreting the measured data. Starting from protein sequences, digestion specificity and permitted missed cleavages define the candidate peptide sequences. A peptidoform is a peptide sequence with specified modifications at defined positions (LeDuc et al., 2022). Different modification states of the same sequence therefore give rise to distinct peptidoforms. Allowed charge states turn these peptidoforms into candidate precursors. Peptide-length and mass-to-charge filters can further restrict this space (Wen et al., 2025b). The resulting search space determines which peptidoforms can be identified in principle.

HUNTER data require more flexible peptide boundaries than a conventional fully specific bottom-up search. The N-terminus of an enriched peptide reflects a protein N-terminus that existed before experimental digestion. This can be a canonical protein start or a terminus created by protein maturation or other proteolytic processing. Its C-terminus is generally generated by the protease used for experimental digestion. An N-terminally semi-specific search therefore permits a variable N-terminal sequence boundary while retaining the protease specificity at the C-terminus (Weng et al., 2019; Demir et al., 2022).

HUNTER labelling also affects which candidates must be considered. Free protein N-termini and lysine side chains are dimethylated before digestion, as described above. Trypsin cleaves dimethylated lysines inefficiently, resulting in an ArgC-like C-terminal cleavage pattern (Demir et al., 2022). The search must also distinguish dimethylated protein N-termini from naturally blocked termini, such as acetylated species. Dimethylated lysines must also be accounted for (Demir et al., 2022).

Selective enrichment and semi-specific searching address complementary problems. Enrichment increases the proportion of N-terminal peptides in the measured sample, while semi-specific settings allow peptides with variable N-terminal boundaries to be considered (Weng et al., 2019). Allowing a variable N-terminal boundary also increases the number of candidate sequences compared with a fully specific search. Including many implausible candidates can make correct assignments harder to distinguish from incorrect ones and reduce the number of confident identifications (Noble, 2015). A useful search space therefore includes plausible HUNTER peptidoforms without unnecessary candidates.

1.3.2. Spectral libraries for DIA analysis

For subsequent library-based DIA analysis, the candidate precursors defined by the search space require information about their expected chromatographic and mass-spectrometric properties. This information can be represented in a spectral library. Library entries commonly include the modified peptide sequence, precursor charge and m/z, expected RT, and selected fragment-ion m/z values and relative intensities (Ludwig et al., 2018). Together, these properties describe the signals expected for a candidate precursor and provide the information needed to search for it in DIA data.

The properties represented in a spectral library can be obtained empirically or predicted computationally. Empirical spectral libraries contain spectra and RT information from measured and confidently identified peptides. They therefore provide experimentally observed properties, but require suitable empirical data and are limited to the precursors observed during library construction (Ludwig et al., 2018). Predicted spectral libraries, also known as in silico spectral libraries, are generated using computational models. These models predict properties such as RTs and fragment-ion intensities from peptide sequence and modification state. Deep-learning models can thereby generate spectral libraries directly from protein-sequence databases without a separate DDA library experiment (Gessulat et al., 2019; Yang et al., 2020).

1.3.3. Peptide identification with DIA-NN

Several software tools are available for DIA data analysis, including Spectronaut, Skyline and DIA-NN (Fröhlich et al., 2024). DIA-NN combines spectral-library generation, peptide identification and quantification in one environment (Demichev et al., 2020). The Academia edition of DIA-NN is publicly available for non-profit academic research (Demichev, 2026b). This accessibility may have

contributed to DIA-NN's widespread use in DIA proteomics (Fröhlich et al., 2024) and, together with its integrated workflow, motivated its use in this thesis.

Spectral-library entries represent candidate hypotheses. The predicted spectral-library workflow in DIA-NN evaluates these hypotheses in two consecutive stages: generation of a predicted spectral library from a sequence database, followed by analysis of the DIA raw data using that library (Demichev et al., 2020) (Figure 2). In the second stage, multiple co-eluting precursors may be fragmented within the same DIA isolation window. The resulting MS2 signals therefore contain fragments from several peptide species. Peptide-centric analysis addresses this complexity by using the expected chromatographic and mass-spectrometric properties stored for each candidate in the spectral library to extract and evaluate corresponding signals in the DIA data. Precursor and fragment m/z values define where signals are extracted. Expected fragment-intensity patterns and RTs then help assess whether the extracted signals are consistent with the candidate (Ludwig et al., 2018; Demichev et al., 2020).

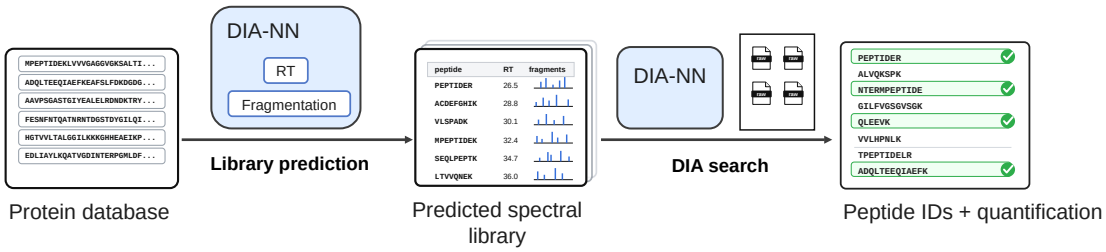


Figure 2.: Simplified two-stage predicted spectral-library workflow for DIA-NN analysis. A protein-sequence database is converted into candidate precursors whose RTs and fragment-intensity patterns are predicted to generate a spectral library. Then DIA-NN evaluates the library candidates against the DIA raw data through signal extraction, scoring and statistical error control (Demichev et al., 2020).

DIA-NN uses neural networks to combine the extracted evidence into a discriminant score for each candidate (Demichev et al., 2020). Neither an accurate RT prediction nor a single matching fragment is therefore sufficient for peptide identification. Candidate precursors derived from the protein-sequence database are treated as targets. DIA-NN evaluates them alongside artificial decoy candidates, which provide a reference for incorrect assignments (Demichev et al., 2020). At a given score threshold, the retained target and decoy assignments are used to estimate the false discovery rate (FDR) (Elias & Gygi, 2007; Demichev et al., 2020). The q-value assigned to an identification corresponds to the lowest estimated FDR at which that identification would be retained (Demichev et al., 2020). It therefore describes error control for a set of accepted identifications and should not be interpreted as the probability that an individual peptide assignment is correct (Elias & Gygi, 2007). After filtering at a chosen q-value threshold, accepted precursors can be quantified and linked to proteins or protein groups (Demichev et al., 2020). DIA-NN reports q-values for different identification levels, including precursor- and protein-level q-values (Demichev et al., 2020). The identification level and threshold used for filtering must therefore be reported. The DIA-NN version and q-value filters used in this thesis are specified in Chapter 3.

In this predicted spectral-library workflow, peptide-property predictions enter the analysis during spectral-library generation, before identification takes place. For HUNTER data, the library must represent the relevant N-terminal peptidoforms and their expected signals. DIA-NN also provides

documented options for adapting its RT and fragmentation predictors (Demichev, 2026b). This makes DIA-NN a practical system for examining the broader question of whether peptide-property model adaptation can improve the identification of HUNTER peptides.

1.4. Peptide-property model adaptation

The accuracy of the predicted RTs and fragment-ion intensities in a predicted spectral library affects the number of peptides identified at a controlled FDR (Searle et al., 2018; Wen et al., 2025b). Models pretrained on broad reference datasets may provide less accurate predictions when peptide modifications or chromatographic conditions differ from those represented during training (Bouwmeester et al., 2026). Peptide-property models can therefore be adapted to specific modifications and experimental conditions of the intended application.

1.4.1. Transfer learning and fine-tuning

Peptide-property models learn relationships between peptide sequence, chemical modifications and experimentally observed properties such as RT or fragment-ion intensities. Training a model from scratch starts with randomly initialised parameters. At this stage, the model has not yet learned these relationships and does not provide reliable predictions. During training, its parameters are then iteratively adjusted to improve prediction accuracy. Transfer learning instead starts from parameters learned in a related source domain. Fine-tuning is the continued optimisation of such a pretrained model using target-domain data and can reduce the amount of new training data required (Bouwmeester et al., 2026).

Transfer learning was used to extend Prosit to fragment-intensity prediction for cross-linked peptides in Prosit-XL (Kalhor et al., 2025). pDeep3 used few-shot fine-tuning, a data-efficient approach requiring only a small number of training examples, to adapt fragment-intensity prediction to instrument settings and fragmentation types that were not represented during pretraining (Tarn & Zeng, 2021). These examples illustrate that the target domain may be defined by the peptide population, the experimental conditions or both, without changing the underlying prediction task. In HUNTER, this peptide population includes N-terminally semi-specific peptides carrying isotopic dimethyl labels. Different isotopic dimethyl forms of the same peptide can elute at slightly different times (Boutilier et al., 2012). These RT differences, together with the possibility that pretrained models have not encountered these labelled peptidoforms during training, motivate model adaptation to HUNTER data.

Many current peptide-property models use neural networks, including those available through the AlphaPeptDeep framework (Zeng et al., 2022). As illustrated in Figure 3, the peptide sequence and its modifications are first encoded as numerical features. A modification can therefore change the numerical representation even when the amino-acid sequence remains the same. During training, the model learns how these features relate to the measured property. Neural-network layers process the representation, and an output layer predicts the property of interest, as illustrated for

RT in Figure 3. The input encoding differs between models: DIA-NN represents modifications as sequence tokens, whereas Carafe2, which uses AlphaPeptDeep models, supplies sequence and modification features separately (Demichev, 2026b; Wen et al., 2026; Zeng et al., 2022).

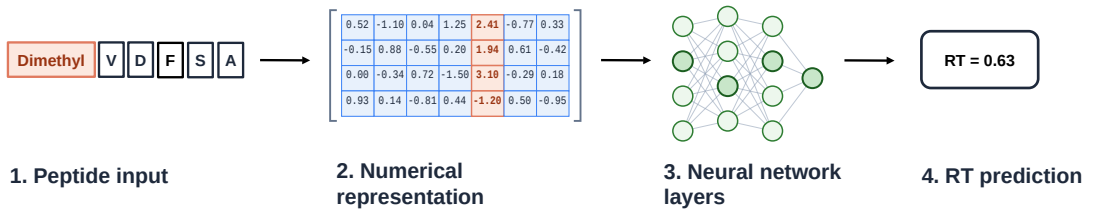


Figure 3.: Conceptual illustration of RT prediction by a peptide-property model. A peptide sequence and modification are converted into numerical features, processed by neural-network layers, and used to predict RT. These layers contain trainable parameters, which can either be updated or kept fixed during fine-tuning. The diagram shows the general principle; the input representation varies between models.

During fine-tuning, either all model parameters or only those in selected components may be updated. Keeping some components fixed preserves their previously learned parameters while allowing others to adapt to the target data. Both approaches were compared in this thesis. Updating more parameters gives the model greater flexibility. However, especially when training data are limited, the model may adapt too closely to the training examples and predict new peptides less accurately than those used for training. This is known as overfitting.

1.4.2. Model adaptation workflows

The approaches considered here all use target-domain observations to update the parameters of peptide-property models, but differ in how closely this process is integrated with the subsequent DIA analysis. Adaptation may be performed through the DIA analysis software, in a separate spectral-library generation workflow or through direct access to the underlying model.

DIA-NN provides options for fine-tuning its RT and fragmentation predictors. Its GitHub documentation explicitly lists dimethylation at peptide N-termini and lysine residues (UniMod:255) among the modifications for which fine-tuning can further improve detection (Demichev, 2026b). It also cautions that fine-tuning can be sensitive to the quality of the training library. Therefore, it recommends verifying that the adapted model performs better than the pretrained model (Demichev, 2026b). Assessing whether fine-tuning improves peptide identification requires comparing DIA results obtained using fine-tuned and pretrained models. An automated workflow can coordinate these analyses and apply the same evaluation procedure across tuning settings and datasets.

Carafe is a separate framework for generating experiment-specific in silico spectral libraries from DIA-derived training data. It uses confidently identified precursors from an initial DIA search to fine-tune pretrained AlphaPeptDeep models for RT and fragment-intensity prediction. The adapted models then predict these properties for candidates generated from a target protein database, and the predictions are assembled into an experiment-specific in silico spectral library. Carafe does not perform the downstream DIA analysis itself. Instead, it exports the resulting library in formats supported by DIA analysis software such as DIA-NN (Wen et al., 2025b). Carafe2 is a

later implementation of this approach and served as the starting point for the HUNTER-specific workflow in this thesis (Wen et al., 2026).

Beyond its use within Carafe and Carafe2, AlphaPeptDeep provides an open-source framework with pretrained models for RT and fragment-intensity prediction that can be adapted through transfer learning (Zeng et al., 2022). More generally, when a pretrained model can be accessed and modified, fine-tuning can also be implemented directly in custom code without relying on a predefined training workflow. This greater control also requires the training objective and the procedure for selecting the adapted model to be defined explicitly.

1.4.3. Model training and evaluation

Adapting peptide-property models to HUNTER peptides and experimental conditions is central to this thesis. This subsection introduces the principles of model training and evaluation, using RT prediction as an example. The training dataset contains peptide sequences, their modifications and experimentally measured RTs. During training, these examples are divided into groups called batches. For each peptide in a batch, the model receives the sequence and modification information and predicts its RT. The measured RT is not provided as an input to the model but serves as the reference for evaluating the prediction. A batch loss is calculated from the resulting prediction errors and guides updates to the model parameters.

For a peptide with measured RT y and predicted RT $\hat{y}$, this prediction error is expressed as the residual $r = y - \hat{y}$. The loss function ℓ assigns a contribution to each residual. Averaging these contributions across the B training examples in a batch gives the mean batch loss:

$$\mathcal{L}_{\text{batch}} = \frac{1}{B} \sum_{i=1}^{B} \ell(r_i) \tag{1.1}$$

Different loss functions ℓ place different emphasis on small and large residuals. For mean absolute error (MAE), the loss increases linearly with $|r|$, whereas for mean squared error (MSE), it increases quadratically. MSE therefore gives more weight to large residuals, while MAE limits their relative influence. Huber loss combines both approaches by using a quadratic contribution for residuals within a threshold δ and a linear contribution for larger residuals (Huber, 1964). It therefore provides a balance between sensitivity to moderate errors and reduced influence of unusually large deviations (Figure 4).

Training losses and evaluation metrics can be calculated in the same way. However, they serve different purposes. During training, the model parameters are updated to minimise the batch loss. In contrast, evaluation metrics quantify model performance without directly contributing to the parameter updates. The behaviour of the model also depends on hyperparameters such as the learning rate (LR), batch size, training duration and which model components are adapted (Zeng et al., 2022; Bouwmeester et al., 2026). Validation data can be used to select hyperparameters and model checkpoints. In contrast, a held-out test set remains unseen during parameter fitting

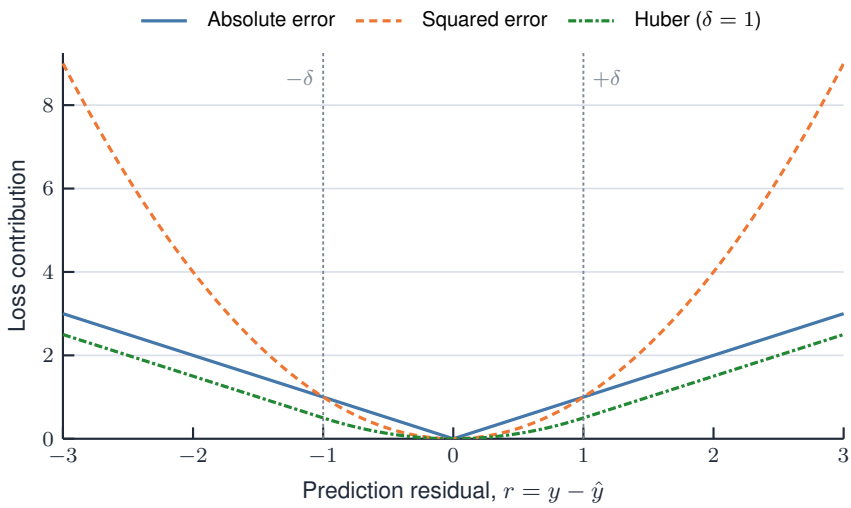


Figure 4.: Comparison of loss functions for retention-time prediction. The curves show how the residual $r = y - \hat{y}$ contributes to absolute error, squared error and Huber loss. Squared error increases quadratically, while absolute error increases linearly. Squared error therefore gives large residuals greater relative influence. Huber loss is quadratic for $|r| \leq \delta$ and linear beyond this threshold (Huber, 1964). The curves use $\delta = 1$ for illustration.

and model selection and is used only for the final evaluation of the selected model (Bouwmeester et al., 2026).

The quality and composition of the training data can affect model adaptation. Incorrect peptide identifications can associate an observed RT with the wrong peptidoform, while measurement noise reduces the precision of the training targets. If peptide or modification groups are unevenly represented, abundant groups contribute more strongly to the mean batch loss and can also dominate an overall evaluation metric. A low overall validation error can therefore hide larger prediction errors for groups with fewer examples. Model selection should consider overall validation error together with performance for groups relevant to the intended application. For the adaptation of peptide-property models to HUNTER data, this includes assessing RT-prediction errors for heavy-dimethylated peptidoforms.

1.4.4. RT alignment and evaluation

RT is one of the properties predicted for spectral libraries used in DIA searches and is a central focus of the model-adaptation experiments in this thesis. Absolute RT in minutes is specific to an LC experiment rather than an intrinsic and directly transferable peptide property. It depends on factors including the chromatographic system, column and gradient and can additionally shift between runs (Escher et al., 2012; Bouwmeester et al., 2026). Consequently, the same peptidoform can have different RT values across experiments. When observations from different LC experiments are combined into a training dataset, their RT values need to be mapped to a common coordinate.

Alignment and normalisation serve related but distinct purposes in constructing such a coordinate. Alignment maps experiment-specific RT values to a common coordinate so that observations from different experiments become comparable. This coordinate can be established using standard peptides, peptides observed in multiple experiments or retention values predicted from peptide sequences by an external model. Normalisation subsequently rescales the aligned values to a defined numerical range, for example from 0 to 1 (Zeng et al., 2022; Wen et al., 2025b). Applying the same numerical range to unaligned RT values does not by itself make them comparable across experiments.

Escher et al. introduced indexed retention time (iRT) as a dimensionless peptide-specific value obtained by expressing an observed RT relative to a set of reference peptides. The measured RTs of these reference peptides provide the calibration needed to place other observed peptide RTs on the iRT scale (Escher et al., 2012). When dedicated reference peptides are unavailable, peptides observed in multiple experiments can serve as empirical anchors. Their observed RTs link the experimental RT scales to an internal reference coordinate (Wilburn et al., 2023). This coordinate depends on the available data and the selected reference and is therefore not standardised iRT.

A prediction model can establish a common coordinate by generating reference values for suitable peptides in each experiment. Because these predictions are expressed on the same scale, measured RTs can be aligned even when the experiments do not contain the same peptides. Chronologer is an open-source model that provides such a reference by predicting the hydrophobicity index (HI) of a peptide (Wilburn et al., 2023). HI expresses peptide retention in terms of the acetonitrile percentage under defined reversed-phase conditions (Krokhin & Spicer, 2009).

In addition to alignment for model training, RT alignment is also performed during DIA analysis. DIA-NN uses peptides detected in the sample to align the reference RT values in the spectral library with the RTs observed in each run (Demichev et al., 2020).

RT-prediction error and FDR-controlled peptide identifications describe different levels of performance. More accurate RT predictions can improve peptide identification, but the extent of this effect depends on the dataset and on the other peptide properties used for scoring (Wen et al., 2025b). Whether a reduction in RT error results in more FDR-controlled identifications therefore remains an empirical question. For HUNTER-based N-terminomics, model adaptation is useful if it improves the identification of Heavy-N peptides. These peptides provide information about protein N-termini and proteolytic processing (Weng et al., 2019; Koudelka et al., 2022). Evaluating model adaptation in this context can help establish its usefulness for HUNTER-based studies of proteolysis.

2. Aims

N-terminomics enables the study of proteolytic processing through the identification of protein N-terminal peptides. The accuracy of peptide-property predictions used to identify these peptides can depend on peptide modifications and experimental conditions. Improved identification of these peptides could support proteolysis research, including studies of disease mechanisms and potential therapeutic targets.

The aim of this thesis was to investigate how peptide-property models can be adapted to data-independent acquisition (DIA) data from enriched N-terminal peptides. This included testing whether these adaptations improve the identification of N-terminal peptides carrying a heavy dimethyl label.

To address the aim, four objectives were defined:

1. To build an automated DIA-NN pipeline for reproducible evaluation of different tuning settings across four datasets of enriched N-terminal peptides. This pipeline should then be used to establish a baseline using pretrained models without fine-tuning.
2. To compare results obtained using integrated model tuning in DIA-NN and Carafe2 with the baseline on DIA runs not used for tuning.
3. To develop and evaluate an external workflow for fine-tuning the retention-time (RT) models used by DIA-NN and Carafe2.
4. To assess whether changes in RT-prediction accuracy translate into improved identification of Heavy-N peptides.

In this thesis, integrated tuning refers to the model-adaptation procedures provided by DIA-NN and Carafe2. External RT fine-tuning refers to a separately developed training workflow for the RT models used by these tools.

3. Materials and Methods

This thesis reanalysed previously acquired DIA data. The final benchmark consisted of four HUNTER datasets, while additional datasets were used only during pipeline development. Sample preparation and LC-MS/MS acquisition were not performed as part of this thesis. This chapter documents the datasets, their partitioning, search settings, model-adaptation procedures, evaluation methods and computational environments used for the reported analyses.

3.1. DIA datasets

3.1.1. Dataset provenance

The datasets had two analysis roles (Table 1). Raw files from the three *C. elegans* experiments MD046–MD048 were combined into one benchmark dataset. The other three benchmark datasets were T_EX0097XX, T_Hamilton and MG054. MM68 and QE00774X were only used for pipeline development and were not included in the final benchmark.

Table 1.: Datasets, organisms, mass spectrometers and analysis roles.

Dataset	Organism(s)	Mass spectrometer	Analysis role
MM68	*A. thaliana*	Orbitrap Exploris 480	Pipeline development
QE00774X	*H. sapiens*	Q Exactive Orbitrap	Pipeline development
MD046–MD048	*C. elegans*	Orbitrap Exploris 480	Final benchmark
T_EX0097XX	*H. sapiens*, *A. thaliana* and *E. coli*	Orbitrap Exploris 480	Final benchmark
T_Hamilton	*H. sapiens*, *A. thaliana* and *E. coli*	Orbitrap Exploris 480	Final benchmark
MG054	*H. sapiens*	Orbitrap Exploris 480	Final benchmark

All final benchmark samples had previously been enriched for N-terminal peptides using the HUNTER approach (Stockert et al., 2024). Corresponding pre-HUNTER acquisitions were not included in the final benchmark. Within each experimental condition, MD046–MD048 contained biological replicates of the same *C. elegans* strain cultured on separate plates. New biological samples were prepared for each of the three experiments. T_EX0097XX, T_Hamilton and MG054 contained technical replicates within each experimental condition. Here, the same starting material was divided into separate samples before HUNTER processing. Each sample was then processed separately.

The prefix `T_` denotes TripleMix datasets containing material from *H. sapiens*, *A. thaliana*, and *E. coli*. The mixture compositions are shown in Table 2. Combining three proteomes was expected

to increase the number and diversity of potentially identifiable N-terminal peptides and therefore provide a broader dataset for evaluating the HUNTER search.

Table 2.: TripleMix compositions (%).

Dataset	Mixture	*A. thaliana*	*E. coli*	*H. sapiens*
T_EX0097XX	A	5	30	65
T_EX0097XX	B	20	15	65
T_Hamilton	A	65	20	15
T_Hamilton	B	65	5	30

3.1.2. LC-MS/MS acquisition settings

The final benchmark datasets were all acquired on the same Orbitrap Exploris 480, but differed in their LC systems and DIA acquisition settings (Table 3).

Table 3.: LC-MS/MS acquisition settings of the final benchmark datasets.

Parameter	MD046–MD048	T_EX0097XX	T_Hamilton	MG054
LC system	Evosep One	Vanquish Neo	NCS	NCS
MS acquisition time (min)	44	53	60	60
MS1				
Scan range (m/z)	350–1400	380–980	350–1400	350–1400
Resolution	120,000	60,000	120,000	120,000
AGC target (%)	300	300	300	300
Maximum injection time (ms)	45	Automatic	45	45
DIA-MS2				
Isolation windows	79	38	79	79
Window widths (m/z)	14/7/14	16	14/7/14	14/7/14
Window overlap (m/z)	1	0	1	1
Resolution	30,000	30,000	30,000	30,000
AGC target (%)	1000	1000	1000	1000
Maximum injection time (ms)	54	Automatic	54	54
NCE	28	25	28	28

Note. AGC = automatic gain control; NCE = normalised collision energy; NCS = Nano/Cap System. Window widths of 14/7/14 m/z refer to successive groups of isolation windows.

All final acquisitions used positive-mode nanospray ionisation. T_Hamilton and MG054 also used the same analytical gradient and DIA-MS settings, although their sample-loading programs differed.

Thermo RAW files were converted to mzML using MSConvertGUI version 3.0.21229-9668f52 (developer build) on Windows. Conversion used 32-bit binary encoding, index writing, TransProteomic Pipeline compatibility and the vendor peak-picking filter `vendor msLevel=1-`. zlib, gzip and Numpress compression were disabled. All subsequent searches used the converted mzML files.

3.1.3. Training-test split by raw file

The raw files of each final benchmark dataset were divided into a training set and a held-out test set. Training files provided empirical data for model adaptation and construction of the libraries used in the benchmark searches. The held-out files did not supply training observations for these steps. The same test files were used to compare baseline and adapted searches. Where a dataset contained multiple experimental conditions, each condition was represented in the test set. The complete raw-file assignments are listed in Tables 17 and 18.

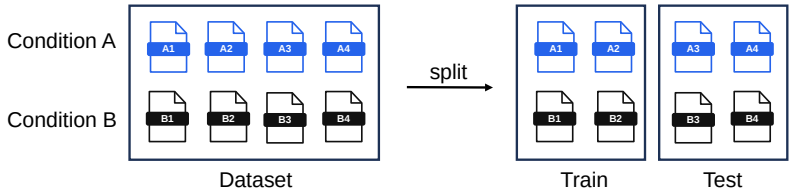


Figure 5.: Schematic illustration of a raw-file split for two experimental conditions. The actual training and test allocations for the final benchmark datasets are reported in Table 4.

Table 4.: Training and held-out test raw-file allocations for the final benchmark datasets.

Dataset	Training raw files	Test raw files
MD046–MD048	30	6*
T_EX0097XX	4	2
T_Hamilton	18	6
MG054	6	6

Note. * All six test raw files in the combined MD046–MD048 dataset originated from MD048.

3.2. DIA-NN search configuration

3.2.1. Protein databases

The final benchmark searches used protein sequence databases in FASTA format for the organisms in each dataset. The *C. elegans* FASTA for MD046–MD048 and the human FASTA for MG054 were downloaded from UniProt. The two TripleMix datasets used the combined FASTA, that is routinely used for TripleMix analyses in the lab. It contained reviewed UniProtKB/Swiss-Prot sequences from *H. sapiens*, *A. thaliana* and *E. coli* K12 (Table 5).

Table 5.: Protein sequence databases used for the final benchmark searches. Counts refer to native FASTA records before adding entrapment sequences.

Dataset	Organism	Entries	Contains isoforms	UniProt release
MD046–MD048	*C. elegans*	26,651	Yes	2026_01
MG054	*H. sapiens*	83,526	Yes	2026_01
TripleMix	*H. sapiens*	20,420	No	2025_04
	A. thaliana	16,682	No	2025_04
	E. coli K12	4,531	No	2025_04

Some searches during initial local pipeline development used smaller FASTAs containing only reviewed UniProtKB/Swiss-Prot entries. The FASTA files and corresponding dataset configurations are indexed through the overview repository described in Section 3.6.2.

3.2.2. Entrapment analysis

To examine false-hit behaviour alongside identification counts, we used an entrapment database (Wen et al., 2025a). For each native protein, we added one shuffled version of its amino-acid sequence to the FASTA. During shuffling, the first amino acid and all arginine positions were kept fixed. The remaining amino acids were shuffled within the same protein. This kept the arginine cleavage sites used by the `R*` search unchanged. The shuffled entries received the prefix `ENT_`. The baseline and DIA-NN tuning searches for MD046–MD048, T_EX0097XX and T_Hamilton used these entrapment FASTAs.

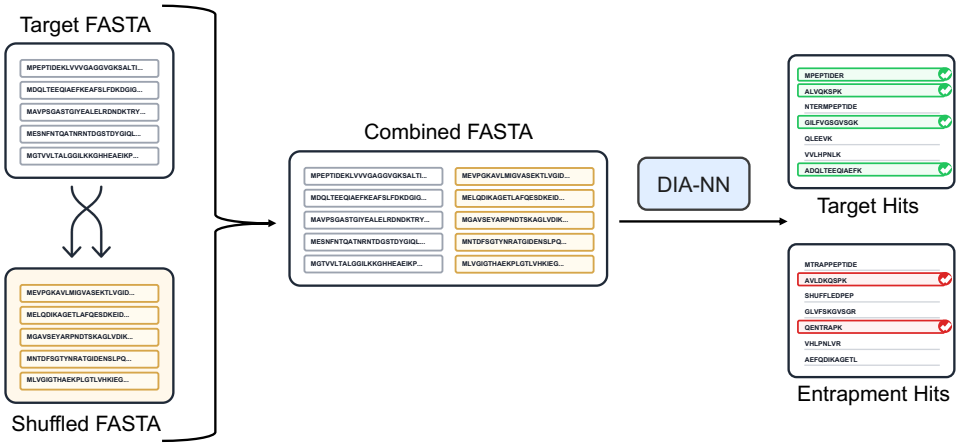


Figure 6.: Entrapment workflow for the three entrapment-enabled DIA-NN benchmark datasets. One shuffled version of each target protein sequence was added to the FASTA. Target and entrapment hits from the DIA-NN search against the combined FASTA were counted separately. The entrapment proportion provided an external error diagnostic alongside DIA-NN's internal FDR estimates.

Entrapment was evaluated using precursor groups defined by peptide sequence, N-terminal modification status and charge (Table 11). Unlike complete DIA-NN precursor identities, these groups combined additional modification states, including methionine oxidation. Each group was

counted once per raw file, and counts were summed across files for dataset-level comparisons. DIA-NN decoys were excluded and the precursor q-value cutoff was applied before aggregation. This analysis included all N-terminal modification groups. Groups were classified from the combined protein assignments of their accepted detections. Target groups mapped only to unshuffled proteins, whereas entrapment groups mapped only to shuffled `ENT_` proteins. Groups with both target and entrapment assignments were classified as mixed and excluded from the proportion.

Here, Q is the DIA-NN precursor q-value and q is the chosen cutoff. The entrapment proportion was calculated as

$$\text{Entrapment proportion} = \frac{N_{\text{entrapment hits}}(Q \leq q)}{N_{\text{target hits}}(Q \leq q) + N_{\text{entrapment hits}}(Q \leq q)}. \tag{3.1}$$

The proportion was reported as a percentage and used to compare false-hit behaviour across searches. It provides an external error diagnostic, not a direct FDR estimate.

3.2.3. DIA-NN search settings

The baseline and all other final DIA-NN searches of the held-out test files used the shared settings summarised in Table 6. Deviations from these settings are stated explicitly. The main benchmark searches used DIA-NN version 2.5.0 (Demichev et al., 2020; Demichev, 2026a).

Table 6.: Shared DIA-NN search settings for the final benchmark.

Parameter	Setting
Digestion	`--semi`; semi-specific at either terminus, `R*`, one missed cleavage
Peptide / precursor limits	Length: 7–30 residues; charge: 1–4 Precursor m/z: 300–1800; fragment m/z: 200–1800
Mass tolerances	MS1: 10 ppm; MS2: 20 ppm
Fixed modifications	C: carbamidomethyl (UniMod:4, $+57.021464$ Da) K: heavy dimethyl (UniMod:510, $+34.063117$ Da)
Variable modifications	M: oxidation (UniMod:35, $+15.994915$ Da) Peptide N-terminus: heavy dimethyl (UniMod:510, $+34.063117$ Da) Peptide N-terminus: acetyl (UniMod:1, $+42.010565$ Da)
Variable-modification limit	Maximum two per precursor
Match-between-runs (MBR)	`--reanalyse`; enabled
DIA-NN report output	`--qvalue 1` Subsequent acceptance filter: precursor `Q.Value` ≤ 0.01

With MBR enabled, DIA-NN searched the files in two passes. The first pass used the input library to identify precursors and generate an empirical library. The second pass searched the same files using this empirical library (Demichev, 2026b).

3.3. Integrated model tuning

Integrated tuning used the model-adaptation procedures provided by DIA-NN and Carafe2. Only the training files in Table 4 supplied experimental observations for adaptation. The developed workflows are presented in Section 4.2.

3.3.1. DIA-NN

DIA-NN's tuning options and parameters are described in its GitHub README (Demichev, 2026b). DIA-NN provides integrated tuning of its RT and fragmentation predictors through `--tune-rt` and `--tune-fr`, respectively. The main comparison adapted both predictors in all four benchmarks. RT-only tuning (`--tune-rt`) was additionally tested for MD046–MD048 and MG054. Restricted RT-plus-fragment tuning for T_EX0097XX, T_Hamilton and MG054 used `--tune-restrict-layers`. According to the documentation, this option keeps recurrent-layer weights and amino-acid embedding weights except those for cysteine fixed (Demichev, 2026b).

Training libraries were generated with DIA-NN 2.5.0 using the default profiling mode. They contained empirically aligned RTs and retained the fragment-ion intensities from the input library (Demichev, 2026b). The library-export q-value threshold (`--out-lib-qvalue`) remained at its default of 0.05.

Tuning used ten learning rates (`--tune-lr`): 10^{-5} and 1.1×10^{-4} to 9.1×10^{-4} in increments of 10^{-4}. Models were generated at three distillation levels (model sizes), d0–d2. Subsequent searches used libraries generated with the 1.1×10^{-4}/d0 models. Predicted libraries were generated with `--predictor` using the adapted models.

Three profiles were used for the DIA-NN searches: balanced, fast and slow. They differed only in their RT-window settings (Table 7). Balanced retained the defaults, fast narrowed the windows and slow disabled RT-windowed searching. The main benchmark used the balanced profile. For MG054, all three profiles were compared with both RT-only and RT-plus-fragment libraries, reusing the corresponding balanced-generated libraries.

Table 7.: Project profiles for DIA-NN RT-window settings.

Profile	RT-window settings
balanced	Default settings, without additional profile options
fast	`--rt-window-mul 1.7 --rt-window-factor 100`
slow	RT-windowed search disabled with `--no-rt-window`

For the MBR searches described in Section 3.2.3, the library-export q-value threshold (`--out-lib-qvalue`) controlled inclusion in the empirical library used in the second pass based on global precursor q-values. Higher thresholds allowed more precursors into this library. This threshold was separate from the precursor q-value filter applied to the final identification reports.

A development sweep on MD048 test files EX012141 and EX012142 combined three library-export q-value thresholds (`--out-lib-qvalue`: 0.05, 0.20, 0.50) with four report-output thresholds (`--qvalue`: 0.01, 0.05, 0.20, 0.50). The twelve searches used a separate 1:1 entrapment FASTA and predicted-library context.

A training-data comparison used six MD048 or 30 MD046–MD048 training files, native-only FASTAs and separately generated libraries searched on the same six held-out MD048 files.

A baseline-only control replaced UniMod:510 with UniMod:255 for fixed lysine and variable peptide-N-terminal dimethylation, retaining the explicit mass shift of $+34.063117$ Da.

The MG054 version comparison used baseline searches with DIA-NN 2.3.0, 2.5.0, 2.6.0 and 2.7.0 on the same six held-out files. Matched RT-plus-fragment searches were completed for 2.5.0–2.7.0; the 2.3.0 tuning attempt did not yield a final search.

3.3.2. Carafe2

The Carafe2 comparison used a modified implementation based on Carafe2 v2.2.0 (Wen et al., 2025b; Wen et al., 2026; Wen, 2026). The HUNTER-specific extensions are presented in Section 4.2.2. The runs used PyTorch 2.5.1+cu121 and PeptDeep 1.1.0.

For each benchmark dataset, Carafe2 fine-tuned its RT and fragmentation models using precursors identified by DIA-NN in the training files listed in Table 4. Decoy and entrapment identifications were not used for training. Carafe2 used an FDR threshold of 0.01, a fragment mass tolerance of 20 ppm and a peak-correlation threshold of 0.75. At least four matched fragments and four high-quality fragments were required. Extracted ion chromatograms were smoothed over three points, with a 1 min RT-window offset. Fragment predictions used the Exploris instrument setting and the NCE recorded in the spectra: 25 for T_EX0097XX and 28 for the other datasets. Pretrained RT and fragmentation models were loaded and adapted with the settings in Table 8. MS2 masking was enabled.

Table 8.: Training settings for integrated Carafe2 adaptation.

Parameter	RT model	Fragmentation model
Initial learning rate	10^{-4}	10^{-4}
Epochs	40	20
Warm-up epochs	10	10
Batch size (except MG054)	512	512
Batch size (MG054)	128	128
Random seed	2024	2024

The RT and fragmentation model weights from the last training epoch were used for library prediction. Library generation used the digestion, modification and precursor limits in Table 6. The export retained up to 20 fragments between m/z 200 and 1800 and required at least two fragments per precursor. The libraries were exported in DIA-NN-compatible Parquet format and

searched with DIA-NN 2.5.0 on the held-out mzML files of each dataset using the settings in Table 6.

3.4. Controlled external fine-tuning of RT models

External RT fine-tuning used custom training code for the DIA-NN and Carafe2 RT models. The code is linked through the overview repository described in Section 3.6.2.

3.4.1. RT dataset construction

RT observations were obtained only from DIA-NN analyses of the training raw files at `Q.Value` ≤ 0.01. DIA-NN decoy identifications and entrapment hits were not included in the RT dataset. Within each source, a peptidoform was defined by its sequence, modification identities and modification sites. The four sources contributed 91,908 observations corresponding to 79,000 distinct peptidoforms across sources (Table 9).

Table 9.: Source observations in the four-source RT dataset.

Source	RT observations
MD046–MD048	38,193
T_EX0097XX	22,973
T_Hamilton	24,330
MG054	6,412
Total	91,908

A deterministic split assigned 15% of peptidoforms to the test partition and 10% of the remaining peptidoforms to validation. The split kept the proportions of source groups, heavy-dimethyl groups and `R*` cleavage classes similar across the three partitions. The split was defined at the peptidoform level, so different modification forms of the same sequence could occur in different partitions. The resulting partitions contained 60,435 training, 6,715 validation and 11,850 test peptidoforms. The peptidoform-level split is illustrated in Figure 7.

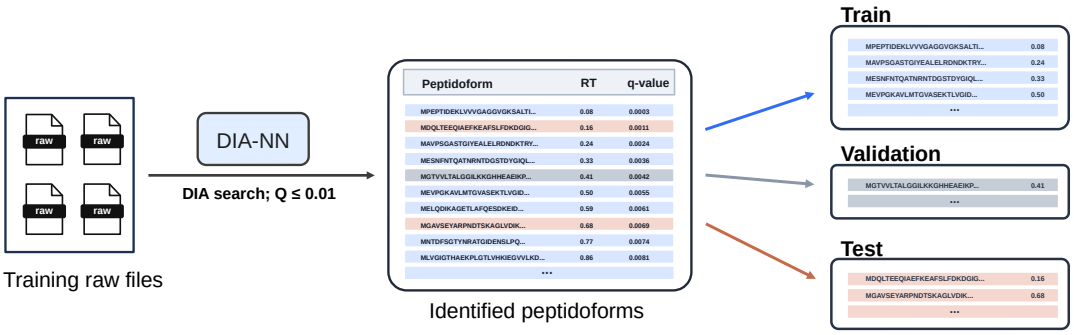


Figure 7.: Peptidoform-disjoint data construction for external RT-model evaluation. RT observations were retained from the training raw files in Figure 5 at $\texttt{Q.Value} \leq 0.01$. Held-out raw files did not contribute. Each peptidoform was assigned to one train, validation or test partition. Colours indicate partition assignment. Entries and RT values are schematic.

3.4.2. RT target construction

The main external RT experiments used the four-source RT dataset described above. Earlier comparisons used MD046–MD048, T_EX0097XX and MG054 data. For the four-source RT dataset, RT mappings were fitted only on training observations and applied unchanged to validation and test observations. The following RT target definitions were used in the main external fine-tuning experiments. They were based on alignment to human data, mapping to model predictions, or scaling without alignment.

Joint-anchor RT jointly aligned the sources to the human reference using shared peptidoforms. Robust linear alignment fitted each source separately to this reference. In both methods, the aligned RTs were divided by the MG054 reference time of 60.10 min.

For the Chronologer HI target, unmodified peptides served as anchors. Their measured RTs were paired with HI predictions from the Chronologer model (Wilburn et al., 2023). A mapping was fitted for each source and applied to all observed RTs, including those of modified peptidoforms. The anchor HI range was then used to scale the targets to a common 0–1 coordinate.

DIA-NN space used the output scale of the pretrained DIA-NN RT predictor as the target. Source-specific mappings aligned observed RTs to this scale using unmodified training peptides as anchors. No further scaling was applied.

Two control targets used observed RTs without alignment. The global 60-min control divided RTs from all sources by the same reference time of 60.10 min. The source-native control divided each RT by the reference time of its own source.

After the split (Section 3.4.1), repeated observations of each peptidoform were transformed to the selected RT target and combined by their median. Implementation details are available in the code linked under “RT alignment and target construction” in the overview repository (Section 3.6.2).

3.4.3. RT models and training

The external DIA-NN experiments started from the `rt.d0.pt` TorchScript model and `dict.txt` distributed with DIA-NN 2.5.0. Two unused embedding rows were assigned to peptide-N-terminal and lysine UniMod:510 states and initialised from the corresponding light-dimethyl representations. The code is linked under “External DIA-NN RT fine-tuning” in the overview repository (Section 3.6.2).

The Carafe2 branch used Carafe2’s RT training code, starting from the generic pretrained AlphaPeptDeep RT model with its native modification features (Zeng et al., 2022). For matched four-source comparisons, both model families used the same peptidoform partitions and RT targets while retaining their own modification encodings.

Comparisons varied the RT target, training sources, trainable model components and learning rate. Models were trained on either the combined training data from all four benchmark datasets or the training data from a single dataset. We fine-tuned either the entire model or selected parts, including embeddings, output layers and layers that process modification information. We also tested different learning rates for different model parts. The common training settings are listed in Table 10.

Table 10.: Common settings for external RT fine-tuning.

Parameter	Setting
Optimiser	Adam
Reference training loss	MAE
Requested epochs	40
Batch size	512
Warm-up epochs	10
Learning-rate schedule	Cosine decay after warm-up
Random seeds	2024, 2025, 2026
Early stopping	Disabled

Additional comparisons tested Huber loss for both model families and MSE for DIA-NN. Loss weighting and oversampling were varied for peptidoforms carrying heavy-dimethyl labels only at the peptide N-terminus, only at lysine residues, or at both. The initial four-source comparisons used no weighting or oversampling of heavy-dimethyl groups. In these comparisons and the subsequent learning-rate refinements, each epoch processed the full training dataset. Later comparisons used fixed budgets of 120 or 294 optimiser steps per epoch, drawing training examples with replacement. The code and experiment configurations, including learning rates, loss parameters and weighting and sampling settings, are linked through the overview repository (Section 3.6.2).

3.5. Evaluation

Evaluation covers three levels: RTs in spectral libraries, model performance during and after training, and peptide identification in DIA-NN searches. These comparisons distinguish changes

in predicted RTs, prediction accuracy against measured RTs and identification performance on held-out raw files.

3.5.1. RT comparisons in spectral libraries

The Python tool `diann-speclib-explorer` was developed to extract precursor entries and their RTs from binary DIA-NN spectral libraries. It supports RT-distribution analyses across libraries and modification groups. Paired comparisons examine shared peptidoforms between libraries. For RT comparisons, entries with the same modified peptide sequence are grouped across charge states. The code is linked under “DIA-NN spectral-library explorer” in the overview repository (Section 3.6.2).

3.5.2. Model performance evaluation

Training and validation losses were examined where available. The integrated DIA-NN learning-rate comparison had no independent validation-loss curves. According to the developer, the logged value was a batch-wise training loss (Demichev, 2026c).

For model evaluation, predicted RTs were mapped from their target scale to observed RT minutes using a separate mapping for each source. For the four-source RT dataset, these mappings were determined from training data and applied unchanged to validation and test predictions. First, the MAE introduced in Section 1.4.3 was calculated separately for each source, in observed RT minutes. For the source-balanced MAE, these source-specific MAEs were averaged with equal weight. Each source therefore contributed equally, regardless of its number of observations. The Heavy at both MAE uses the same source-balanced calculation, restricted to peptidoforms carrying heavy dimethylation at both the N-terminus and lysines.

For the heavy-macro MAE, the source-balanced MAE was calculated separately for three groups: peptidoforms with heavy dimethylation only at the N-terminus, only at lysine residues, or at both. The three group-specific MAEs were then averaged with equal weight:

$$\mathrm{MAE}_{\text{heavy-macro}} = \frac{1}{3}\sum_{g=1}^{3}\left(\frac{1}{S}\sum_{s=1}^{S}\mathrm{MAE}_{s,g}\right). \tag{3.2}$$

Here, $\mathrm{MAE}_{s,g}$ is the MAE for group g in source s, and S is the number of data sources. Both sources and heavy-dimethyl groups contributed equally, regardless of their numbers of observations.

The RT-target comparison used three data sources; the subsequent four-source comparisons used all four benchmark datasets. This changed the number of sources included in the average, but not the calculation of the scores. For the RT-target comparison, source-balanced MAEs were also averaged equally across the fourteen training-data combinations within each seed. Additional

metrics were calculated, including the highest source-specific MAE and separate MAEs by source, cleavage specificity and heavy-dimethyl group.

During external DIA-NN and Carafe2 RT fine-tuning, validation was performed after each epoch. Within each training run, the saved model with the lowest validation MAE on the respective RT target scale was generally selected. Some DIA-NN heavy-focused comparisons used predefined combinations of overall validation errors and validation errors for heavy-dimethyl groups to select the saved model. After selecting a model from each training run, different training configurations were compared using the source-balanced validation MAE in minutes. Heavy-focused comparisons used the heavy-macro validation MAE. Exact selection settings are provided with the code and experiment configurations through the overview repository (Section 3.6.2).

Externally fine-tuned DIA-NN and Carafe2 RT models were evaluated on the held-out model-level test partition of the four-source RT dataset (Section 3.4.1). Model-test prediction exports for the RT models adapted through integrated Carafe2 tuning were unavailable. Model-level test results were also used to select external DIA-NN RT models for subsequent searches on held-out raw files.

3.5.3. Identification performance and N-terminal coverage

For each dataset, baseline and adapted searches used the same test files defined by the raw-file split in Section 3.1.3. DIA-NN searches used the shared settings in Table 6. Identification counts were derived from the final precursor report (`final_report.parquet`) and the corresponding search FASTA. Identification counts included precursor detections with `Q.Value` ≤ 0.01. DIA-NN decoys were excluded, and detections had to be assigned only to unshuffled proteins in the search FASTA. Accepted identifications were summarised at three levels: precursor groups, N-terminal peptides and N-terminal sites (Table 11). This distinction accounts for the fact that several peptide forms can support the same N-terminal position in a protein.

Table 11.: Identification levels and aggregation rules.

Level	Counted once for each	Aggregated differences
Precursor group	Peptide sequence, N-terminal modification status and charge	Other modifications, including methionine oxidation
N-terminal peptide	Peptide sequence and N-terminal modification status	Charge states and other modifications
N-terminal site	Protein and peptide starting position	Peptide lengths, C-terminal boundaries, charge states and modification states

The main comparisons focused on Heavy-N identifications, recognised by a leading UniMod:510 annotation. These identifications were counted at all three levels.

Cleavage specificity was assigned using the search FASTA and the `R*` rule. Fully specific peptides matched this rule at both ends, semi-N peptides only at the C-terminus and semi-C peptides only at the N-terminus. Protein starts and ends were treated as specific boundaries. All three classes were included in the Heavy-N comparison. Heavy-N peptides with conflicting or unclassifiable

cleavage-specificity assignments were excluded before aggregation at all three levels. Additional peptide-level analyses compared N-terminal modification groups (Heavy-N, acetylated and free) and cleavage-specificity classes within Heavy-N.

Filters were applied before aggregation. Within each dataset and search condition, objects were counted either once across all held-out raw files or once per raw file, with counts summed across files. Tables and figures state which counting approach was used. All peptide matches in the reported proteins were checked, and only unique protein and starting-position assignments contributed to site counts. Ambiguous or unmapped assignments were reported separately.

Entrapment proportions used the precursor-group level, as described in Section 3.2.2. Run-level statistics were taken from `final_report.stats.tsv`.

To assess changes in RT agreement, absolute RT errors, $|\mathrm{RT} - \mathrm{Predicted.RT}|$, were compared for precursor detections accepted in both searches within the same test file. For each shared precursor detection, the change in absolute error was calculated as adapted minus baseline. The median of these changes was reported in seconds for each test file. Negative values indicate smaller RT errors after adaptation.

Python adaptations of TermineR's report-processing and N-terminal annotation functions were used for these analyses and for the final plots and tables (Cosenza-Contreras et al., 2025). MAPLOT, an interactive visualisation tool developed in this thesis, used the same functions for quick comparisons and quality control. It provided interactive views of identification overlap, N-terminal modification groups and run-level quality metrics across searches. The analysis scripts, Python adaptations of TermineR and MAPLOT are linked through the overview repository (Section 3.6.2).

3.6. Computational infrastructure and reproducibility

3.6.1. Computing systems

Computational analyses were performed across three computing systems with different roles (Table 12). Initial pipeline development and analyses were run on the locally available systems Braavos, a CPU server, and Wynaut, a GPU server. bwUniCluster 3.0 (bwUC3) was incorporated into the workflow later in the project and provided access to substantially greater CPU and memory resources for large DIA-NN searches. These searches were run with the Slurm job scheduler. Braavos was used for pipeline development, smaller searches and MAPLOT, while Wynaut was used for all GPU-intensive tasks, such as external RT-model training and Carafe2 inference. DIA-NN ran in an Apptainer container on bwUC3, whereas analyses on Braavos and Wynaut were performed in native Conda environments.

Hardware and queue specifications were taken from the bwHPC documentation, available at `https://wiki.bwhpc.de/e/BwUniCluster3.0`. The table reports the node type rather than an

Table 12.: Computing systems used in this thesis.

System	Processing hardware	Operating system	Role in this thesis
Braavos	AMD EPYC 7713; 100 available logical CPUs; initially approx. 340 GB RAM, later 590 GiB RAM	Ubuntu 24.04 LTS	development, DIA-NN searches and MAPLOT
Wynaut	Intel Xeon Silver 4210; 36 available logical CPUs; 125 GiB RAM; three NVIDIA GeForce RTX 2080 Ti GPUs with 11 GiB VRAM each	Ubuntu 24.04 LTS	RT-model training and Carafe2 inference
bwUniCluster 3.0 (bwUC3) high-memory node	two AMD EPYC 9454 processors; 96 physical cores and 192 logical CPUs; 2,304 GiB RAM	RHEL 9.4	large DIA-NN library-generation and search jobs

individual job allocation. The cluster searches used Apptainer 1.5.3-1.el9 and a Debian-12 image containing DIA-NN 2.5.0.

Analysis outputs were consolidated on shared storage of the Huesgen laboratory (Figure 8). Braavos and Wynaut wrote directly to the mounted storage, while bwUC3 outputs were transferred using `rsync`. MAPLOT accessed either the consolidated storage or the Braavos output directories.

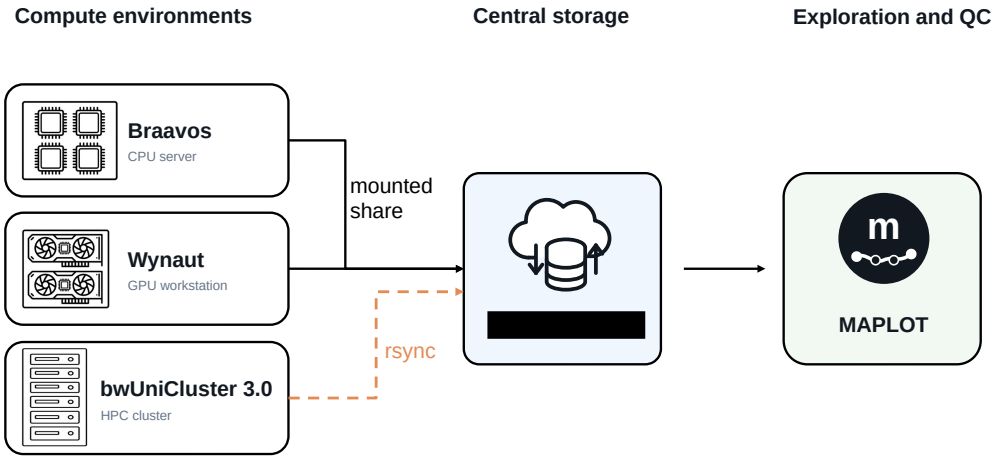


Figure 8.: Computational data flow and central result storage. Braavos and Wynaut wrote to mounted shared storage, while bwUC3 outputs were transferred using `rsync`. MAPLOT accessed the consolidated outputs for comparison and quality control.

3.6.2. Code and supporting materials

The code and supporting materials used in this thesis can be accessed through the HUNTER DIA model-adaptation overview repository. It links to the individual software repositories and their documentation. Software dependencies and environment details are documented in the respective repositories. `https://github.com/mmmddd98/hunter-dia-model-adaptation`.

4. Results

4.1. Retention shifts and precursor composition in HUNTER data

4.1.1. N-terminal dimethylation was associated with later retention

First, we examined whether N-terminal heavy dimethylation was associated with a measurable RT shift in the analysed data. For this comparison, we used identifications from the baseline search of six held-out MD048 files. The search used a library generated with DIA-NN 2.5.0's unchanged predictors and the final `R*` semi-specific digestion and modification settings (Table 6). Within each run, we matched identified precursors with free N-termini to their N-terminally heavy-dimethylated counterparts (UniMod:510). The two precursors had the same stripped sequence and charge. N-terminal heavy dimethylation was their only encoded modification difference, allowing us to compare their RTs. The same precursor pair could occur in several runs. For each pair, we averaged the RT of each form across the runs in which both were identified. This yielded 277 matched precursor pairs for comparison (Figure 9).

In 262 of the 277 pairs, the dimethylated form eluted later. Across all pairs, the median RT shift was 0.41 min towards later elution (interquartile range: 0.21–0.69 min). This comparison shows a modification-dependent RT shift among the paired identifications.

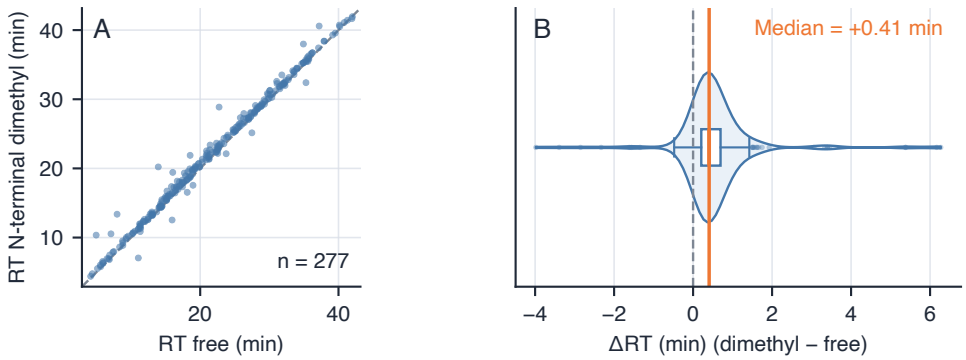


Figure 9.: Retention times of free and N-terminally heavy-dimethylated forms in MD048. Identifications came from the untuned baseline DIA-NN search of the six held-out MD048 files. Pairs shared stripped sequence and charge; neither form had other encoded modifications. Both forms were non-decoy and assigned only to unshuffled *C. elegans* proteins at precursor $Q \leq 0.01$. The 1,214 within-file pairs were averaged over the available files for each sequence-charge pair ($n = 277$). Pairs need not occur in all six files. (A) Free versus dimethylated RT, with a dashed identity line. (B) Distribution of $\Delta\mathrm{RT} = \mathrm{RT}_{\mathrm{dimethylated}} - \mathrm{RT}_{\mathrm{free}}$, with dashed zero and solid median lines.

4.1.2. Semi-specific precursors dominated the theoretical search space

We next compared the composition of theoretical and accepted precursors under the final `R*` semi-specific settings (Figure 10 and Table 6). The theoretical set was reconstructed from the unshuffled *C. elegans* search FASTA. The identified set came from the unchanged DIA-NN baseline search of six held-out MD048 files. In both sets, each combination of modified peptide sequence and charge was counted once, even if it matched several proteins or was identified in several files.

The reconstruction yielded 198.4 million theoretical precursors, of which 97.1% were semi-specific and 2.8% fully R*-specific. Semi-specific precursors outnumbered fully R*-specific ones by about 34 to 1 in the theoretical set. Across the six files, the baseline searches yielded 261,658 accepted precursor identifications. Counting each precursor only once across files left 49,947 distinct precursors. Among these identified precursors, 45.2% were semi-specific and 54.5% fully R*-specific. Semi-specific precursors thus dominated the theoretical search space but not the accepted identifications. Of the 22,588 accepted semi-specific precursors, 17,336 had a semi-specific N-terminus. This class represents the variable N-terminal peptides enriched by HUNTER. The large candidate set highlights why HUNTER enrichment matters: it increases the proportion of N-terminal peptides in the measured sample. At the same time, identifying their variable starting positions still requires a semi-specific search capable of handling the large candidate space.

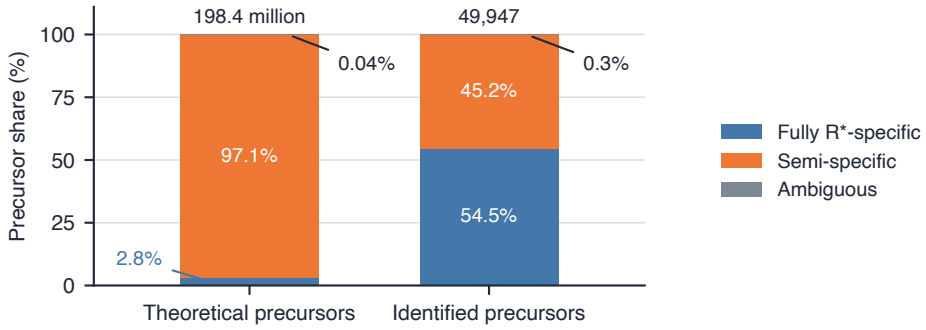


Figure 10.: Theoretical and identified precursor composition in the MD046–MD048 *C. elegans* data. Both bars show the composition of distinct precursors, defined by modified peptide sequence and charge. The left bar shows theoretical precursors reconstructed from the unshuffled *C. elegans* sequences in the search FASTA under the `R*` settings (Table 6). A precursor matching several proteins counts once. The theoretical count is a reconstruction. The right bar shows precursors accepted by the unchanged DIA-NN search in at least one of six held-out MD048 files. Only IDs with `Decoy = 0` and precursor `Q.Value` ≤ 0.01 assigned solely to unshuffled *C. elegans* proteins are included. A precursor identified in several files counts once. Both bars combine N- and C-terminal semi-specific classes. Ambiguous precursors have conflicting cleavage classes among different protein matches in the FASTA (left) or DIA-NN report (right).

4.2. Automated library workflows enabled held-out comparisons

To establish an unchanged-model baseline and evaluate integrated model tuning in DIA-NN and Carafe2, we built an automated DIA-NN benchmark pipeline and a HUNTER-compatible Carafe2 workflow. For each comparison, only training files supplied data for model adaptation, and the final searches used the same held-out files. Building the DIA-NN pipeline and establishing its baseline addressed the first objective. We then used both workflows to compare integrated tuning with that baseline, as specified in the second objective (Chapter 2).

4.2.1. DIA-NN integrated tuning and the automated benchmark pipeline

DIA-NN's standard predicted-library workflow generates a library from a protein FASTA and then searches DIA files with it, as shown earlier in Figure 2. To create the tuned DIA-NN workflow, we added its integrated tuning stage before predicting the library for the final search (Figure 11). After adaptation, library prediction and the DIA search followed the same sequence as in the standard workflow.

The tuning stage itself consisted of two steps. First, DIA-NN searched the designated training files with a library predicted by its unchanged models. This search produced an identification report and an empirical tuning library. Second, DIA-NN used the tuning library to update the parameters of either its RT predictor alone or both its RT and fragmentation predictors. With the adapted models, it predicted a new library from the same FASTA and candidate settings for the search of the held-out files. Details of the tuning-library profiling and predictor settings are given in Section 3.3.1.

We combined the standard and tuned DIA-NN workflows in one automated benchmark pipeline. It included two final-search conditions:

1. **Unchanged-model baseline:** The initial predicted library was used to search the held-out files without model tuning.
2. **Tuned runs:** One or more libraries predicted with adapted models were used to search those same held-out files.

To run these conditions reproducibly, we mapped the inputs and outputs of DIA-NN's separate commands. A modular script selected each dataset's FASTA and its training and held-out files, set the chosen parameters, and started the library-generation, training-search, tuning and final-search calls in order. The pipeline could run several tuning variants, such as RT-only or RT-plus-fragment adaptation at selected learning rates. It could also repeat the final searches with balanced, fast or slow RT-window settings, without preparing each DIA-NN call by hand (Table 7).

The pipeline collected commands, libraries, reports and logs by stage and variant. It tracked progress, elapsed time and RAM use, so we could follow long searches and reuse completed stages after an interruption. The resulting baseline and tuned reports supported comparisons

of RT metrics, identification yields and overlap on the same held-out files (Section 4.4). The benchmark pipeline code is linked through the overview repository described in Section 3.6.2.

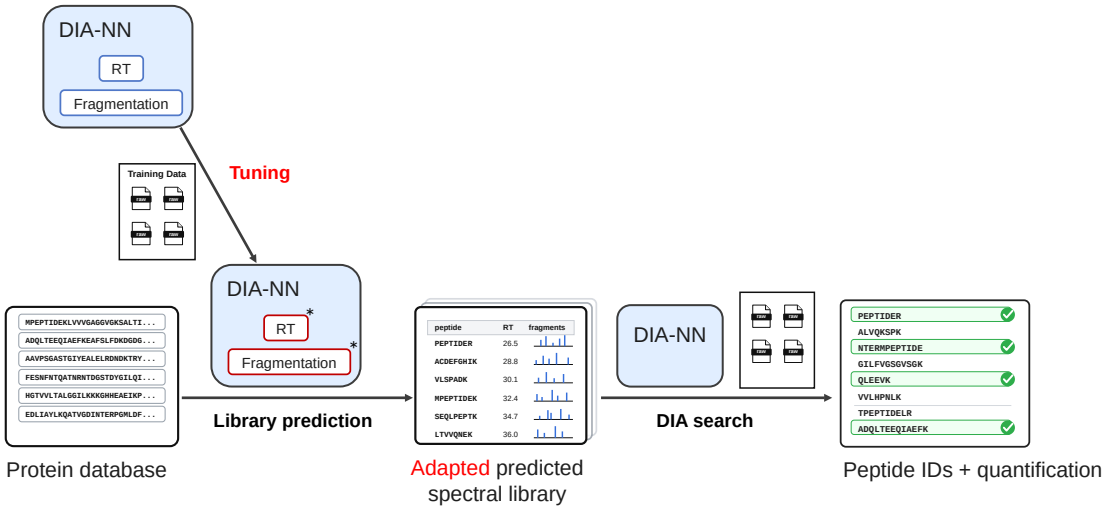


Figure 11.: DIA-NN workflow with integrated tuning. The schematic shows adaptation of the RT and fragmentation predictors, library prediction with the adapted models, and the final search of held-out DIA files. Before adaptation, DIA-NN searched the training files to generate an empirical tuning library. This preparatory search is not drawn. Asterisks mark the adapted predictors.

4.2.2. HUNTER-compatible Carafe2 workflow

We built a second workflow around Carafe2's library-generation approach, introduced in Section 1.4.2. As in the tuned DIA-NN workflow (Figure 11), DIA-NN performed the training-file search and the final search of held-out files. Between these two DIA-NN searches, Carafe2 adapted its own models and generated the spectral library (Figure 12).

The upper route in Figure 12 begins with the DIA-NN training-file search described in Section 4.2.1. The same search also produced DIA-NN's empirical tuning library. Carafe2 filtered the identifications from that search and combined them with matching spectra from the training mzML files. Then, Carafe2 used these observations as training data to adapt the pretrained AlphaPeptDeep RT and fragmentation models.

In the lower route, Carafe2 used the adapted models to predict a spectral library for HUNTER-compatible candidates from the protein database. DIA-NN then used this library to search the held-out files and report identifications and quantification. The training and search settings are given in Section 3.3.2.

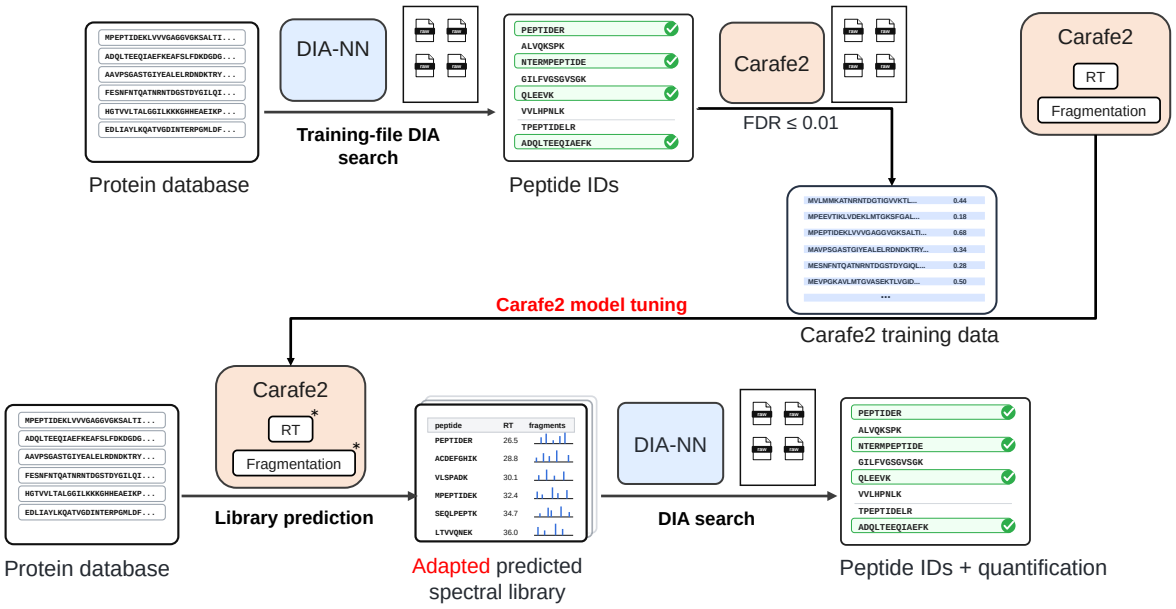


Figure 12.: Carafe2 model adaptation and spectral-library generation for HUNTER data. The upper route begins with the DIA-NN training-file search also used for DIA-NN tuning. Carafe2 filters the resulting identifications at FDR ≤ 0.01 and combines them with matching spectra from the training files. These data are used to adapt the pretrained RT and fragmentation models shown at the upper right. Asterisks mark the adapted models. In the lower route, Carafe2 uses these models to predict a spectral library from the protein database. DIA-NN uses the library to search the held-out files and report identifications and quantification.

Carafe2 v2.2.0 could use DIA-NN identifications and matching spectra for model adaptation, but it did not directly support the combination of semi-specific candidates and N-terminal modification states required here (Wen, 2026). To make this route work with HUNTER data, we extended its training-data import and library-generation code (Table 13).

Table 13.: HUNTER-specific extensions to Carafe2 v2.2.0. The fork added the capabilities needed to generate a DIA-NN-searchable library containing semi-specific and N-terminally modified peptides.

Required function	Extension in this work
Generate HUNTER candidates	Added semi-specific digestion of the protein database while retaining fully specific peptides.
Read training identifications	Extended DIA-NN report import to recognise N-terminal modification annotations and combinations of UniMod modifications.
Predict and export modified peptides	Added N-terminal heavy dimethylation (UniMod:510) and acetylation (UniMod:1) to the model inputs and exported library.
Produce large libraries	Added chunked prediction, saved intermediate results for resuming, and reduced memory use during library assembly.

We linked the Carafe2 and DIA-NN steps with scripts that automated preparation of the matching training inputs, Carafe2 execution and setup of the final DIA-NN search with the predicted library. Carafe2 model adaptation and library prediction ran on the GPU server Wynaut, whereas the final DIA-NN searches ran on the CPU systems Braavos or bwUC3 (Table 12). Before each final search, we manually copied the required files from Wynaut to the respective CPU system. Training and held-out files remained separate throughout. The workflow scripts and our HUNTER-compatible Carafe2 fork are linked through the overview repository described in Section 3.6.2.

4.3. Search-space growth constrained spectral-library handling

4.3.1. Semi-specific settings expanded the candidate space

The semi-specific candidates described in Section 4.1 could enlarge predicted libraries and increase the disk space and RAM needed to generate, store and load them. We therefore used *A. thaliana* as an example to estimate how digestion specificity, modification settings and FASTA size changed the number of possible precursor instances (Figure 13).

Using the full native *A. thaliana* FASTA, we counted 4.2 million theoretical precursor instances with fully specific R∗ digestion. Allowing one non-enzymatic peptide terminus raised the count to 142.5 million. Allowing heavy-dimethylated or acetylated peptide N-termini alongside the free form raised it to 409.3 million. Allowing variable lysine dimethylation and increasing the limit from two to three variable modifications raised it further to 1.19 billion instances, more than 280 times the fully specific starting value.

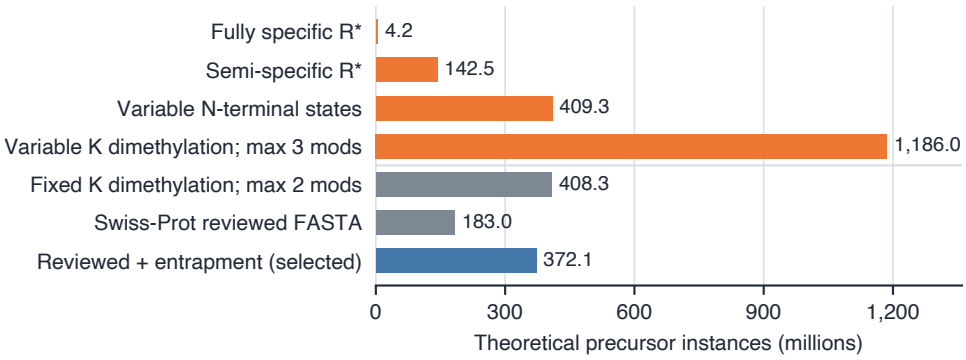


Figure 13.: Theoretical *A. thaliana* precursor instances under successive search settings. Bars count modified peptide-sequence and charge combinations per protein context, rather than entries in a generated DIA-NN library. All scenarios use R∗ cleavage and at most one missed cleavage; peptide and precursor limits follow Table 6. From top to bottom, each bar retains the preceding settings except for the changes named at left. The first bar is fully specific, and the others are semi-specific. The first five bars use the full native FASTA, the sixth uses the Swiss-Prot reviewed FASTA, and the blue bar adds 1:1 entrapment. The lower three bars use the final digestion and modification settings in Table 6; the blue bar is the selected Arabidopsis development setup.

To reduce the candidate space while retaining the N-terminal states needed for HUNTER, we fixed lysine dimethylation and limited variable modifications to two. This lowered the theoretical count to 408.3 million. At this point, the digestion and modification settings matched the final preset (Table 6). Using the Swiss-Prot reviewed FASTA lowered the count to 183.0 million, while adding 1:1 entrapment raised the selected Arabidopsis development setup to 372.1 million. The four final benchmarks used their own dataset-specific FASTAs.

4.3.2. Memory use limited the local workflow

An earlier *A. thaliana* library-generation run on Braavos exposed the memory constraint (Figure 14). It used the full FASTA and also permitted acetylation on lysine, unlike the selected development preset. While DIA-NN generated the predicted library, its process used up to 415.2 GB of RAM. Total system memory reached the 585 GB limit. Because almost no RAM remained available, the DIA-NN process was killed with exit code 137. The DIA-NN log reported approximately 550 million generated precursors for this run.

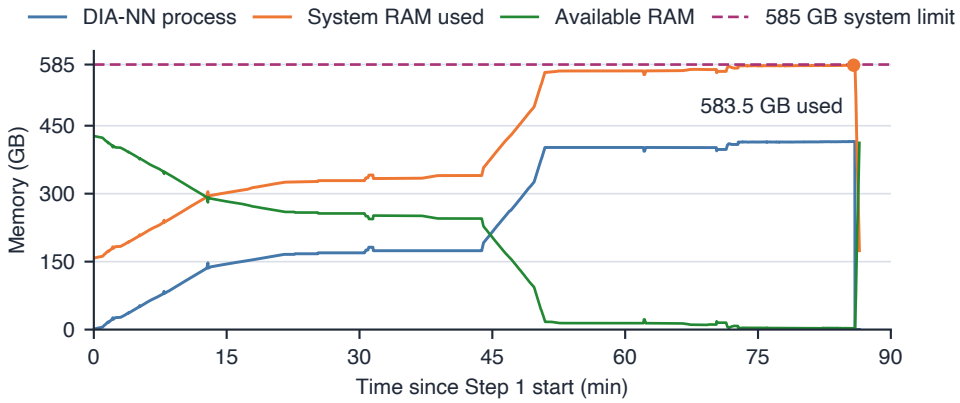


Figure 14.: RAM trace of an Arabidopsis DIA-NN library-generation attempt on Braavos. One-second measurements of DIA-NN process memory, total used system memory and available RAM. System memory reached the 585 GB limit, causing the DIA-NN process to be killed with exit code 137. The DIA-NN log reported 550,345,097 generated precursors for this run.

This run took place after Braavos's RAM allocation had already been increased from approximately 340 GB to 590 GiB (Table 12). The added RAM did not suffice for this broad search, so we used bwUC3 high-memory nodes for the final baseline and integrated-tuning runs of the automated DIA-NN benchmark pipeline. The following section compares their results on the held-out files.

4.4. Integrated DIA-NN tuning was evaluated against four held-out baselines

We used the automated DIA-NN benchmark pipeline to test whether integrated RT-plus-fragment tuning increased Heavy-N coverage. The unchanged and adapted DIA-NN 2.5.0 models were compared on the same held-out files in MD046–MD048, T_EX0097XX, T_Hamilton and MG054. All the searches used the final digestion and modification settings in Table 6; the workflow and training procedure are described in Sections 3.3.1 and 4.2.1.

4.4.1. N-terminal site coverage decreased after integrated tuning

We first compared Heavy-N identifications at the three levels defined in Table 11: precursor groups, N-terminal peptides and N-terminal sites. Each object was counted once across all held-out files

for each dataset and search condition. The peptide and precursor-group counts showed whether changes in site coverage were accompanied by changes at these two levels (Figure 15).

Heavy-N site coverage decreased after tuning in all four datasets. The reductions were 6.22% in MD046–MD048, 20.75% in T_EX0097XX, 21.05% in T_Hamilton and 0.74% in MG054. At the N-terminal peptide level, Heavy-N counts also fell in every dataset, by 6.97%, 19.72%, 21.62% and 0.16%. Precursor-group counts decreased in the first three datasets but increased by 38 in MG054 (+1.30%). Despite this small increase in precursor groups, MG054 had four fewer Heavy-N peptides and five fewer N-terminal sites after tuning. More precursor groups therefore did not mean greater N-terminal site coverage.

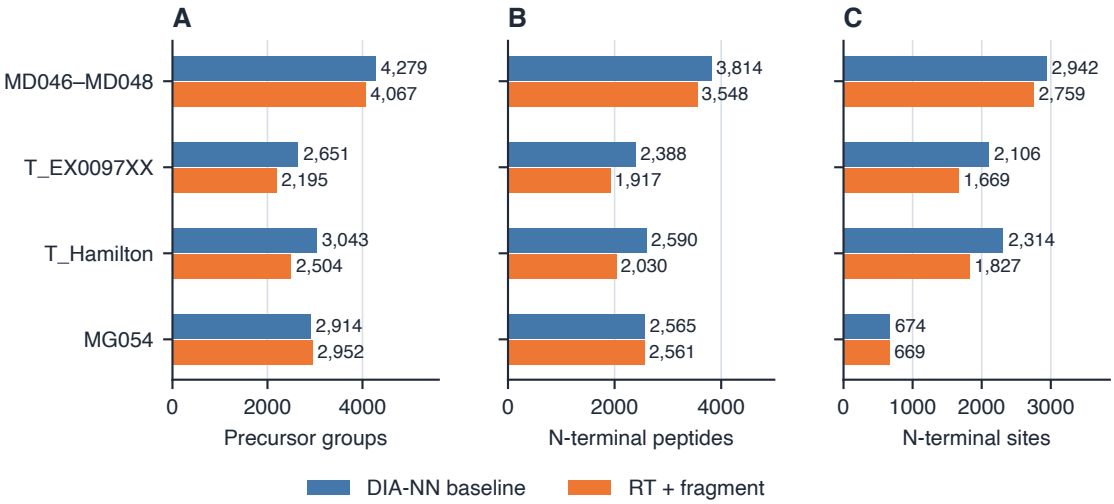


Figure 15.: Heavy-N coverage before and after integrated RT-plus-fragment tuning. Panels compare precursor groups (A), N-terminal peptides (B) and N-terminal sites (C). Non-decoy Heavy-N identifications at precursor `Q.Value` ≤ 0.01 were counted once across all held-out files per dataset and search condition, following the definitions in Table 11. Only identifications with an unambiguous R*-specific, semi-N or semi-C classification were included. Site counts additionally required a unique protein and starting-position assignment.

In MG054, 1,887 of 2,565 baseline Heavy-N peptides (73.57%) and 1,889 of 2,561 tuned peptides (73.76%) matched multiple candidate sites. Most matched several protein entries annotated with the same gene name. These peptides contributed to peptide and precursor-group counts but could not be assigned to a unique N-terminal site. Overlapping protein entries in the human FASTA therefore limited the number of peptides that could be assigned to a unique site. The complete mapping audit is given in Table 19.

4.4.2. Tuning changed the composition of identified N-terminal peptides

After comparing overall Heavy-N coverage, we examined whether tuning affected all peptide groups similarly. We compared cleavage specificity within Heavy-N peptides and N-terminal modification groups among all accepted peptides (Figure 16). Panel A divides Heavy-N peptides into R*-specific, semi-N and semi-C classes. Panel B shows the proportions of Heavy-N, N-terminally acetylated and free-N-terminal peptides. The Heavy-N subset in panel B corresponds

to all peptides shown in panel A. Both panels use the N-terminal peptide level. Each combination of peptide sequence and N-terminal modification status was counted once across all held-out files per dataset and search condition.

Semi-N peptides formed by far the largest Heavy-N class in every search (Figure 16A). Their counts and proportions decreased in MD046–MD048, T_EX0097XX and T_Hamilton. In contrast, the number of semi-N peptides in MG054 increased from 2,168 to 2,242, and their proportion rose from 84.52% to 87.54%. Despite this increase, the total number of Heavy-N peptides in MG054 decreased because the other classes declined. For example, the number of R*-specific peptides decreased from 356 to 298. The full class counts are given in Table 20.

The proportions of N-terminal modification groups changed only slightly after tuning (Figure 16B). Heavy-N proportions showed no consistent increase or decrease across datasets. MG054 differed clearly from the other datasets in both searches: peptides with a free N-terminus accounted for 42.13–44.36%, compared with approximately 86–88% in the other datasets. This difference suggests, that the HUNTER enrichment was more effective in MG054. The number of peptides in each modification group nevertheless decreased in every dataset (Table 21). Higher proportions therefore did not indicate more identified peptides.

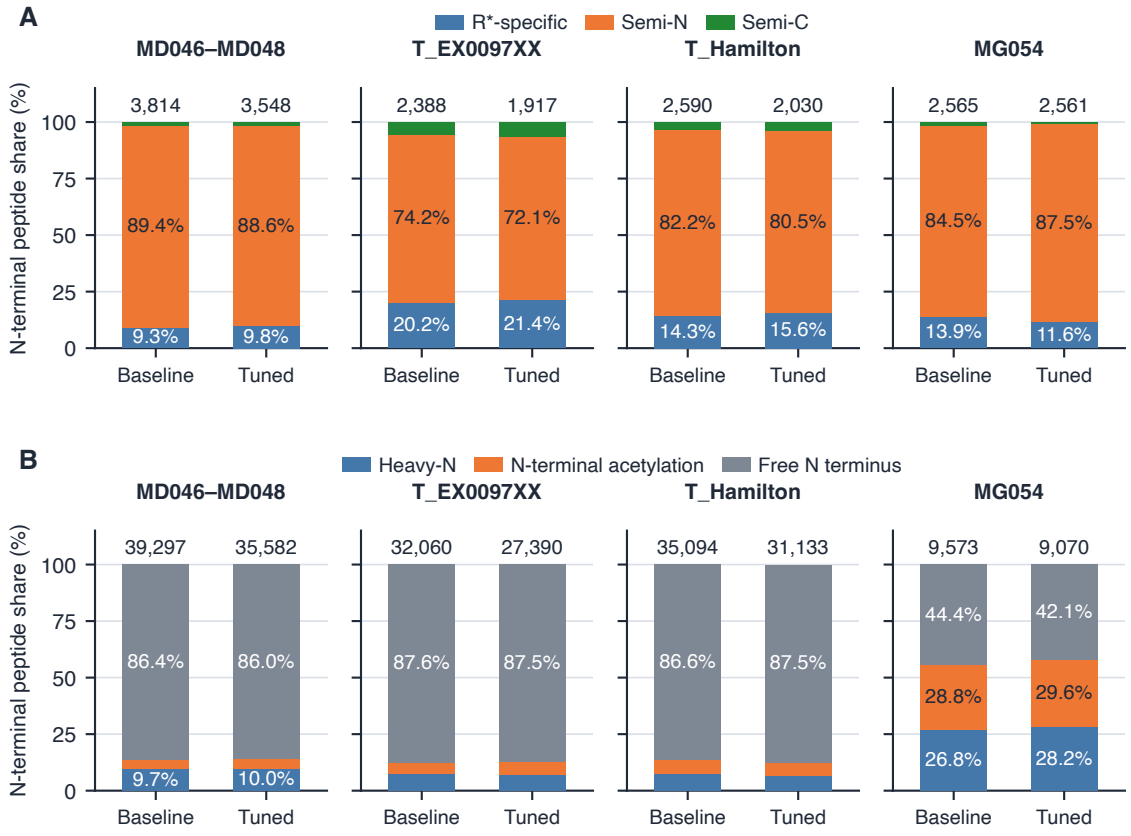


Figure 16.: N-terminal peptide composition before and after integrated RT-plus-fragment tuning. Panel A shows R* cleavage specificity within Heavy-N peptides, which form the Heavy-N subset in panel B. Panel B compares N-terminal modification groups among accepted peptides. Heavy-N peptides required an unambiguous R*-specific, semi-N or semi-C classification. Counts follow Table 11, with each object counted once across held-out files per dataset and search condition. Free denotes an unmodified peptide N-terminus. Numbers above bars give total peptide counts.

4.4.3. Tuning changed which N-terminal peptides were identified

The overall counts did not show whether tuning retained the same Heavy-N peptides or identified different ones. We therefore performed an intersection analysis to divide the N-terminal peptides into shared, baseline-only and tuned-only groups (Figure 17). Each Heavy-N N-terminal peptide was counted once across all held-out files within each dataset and search condition. Shared peptides were identified in both searches, but not necessarily in the same file.

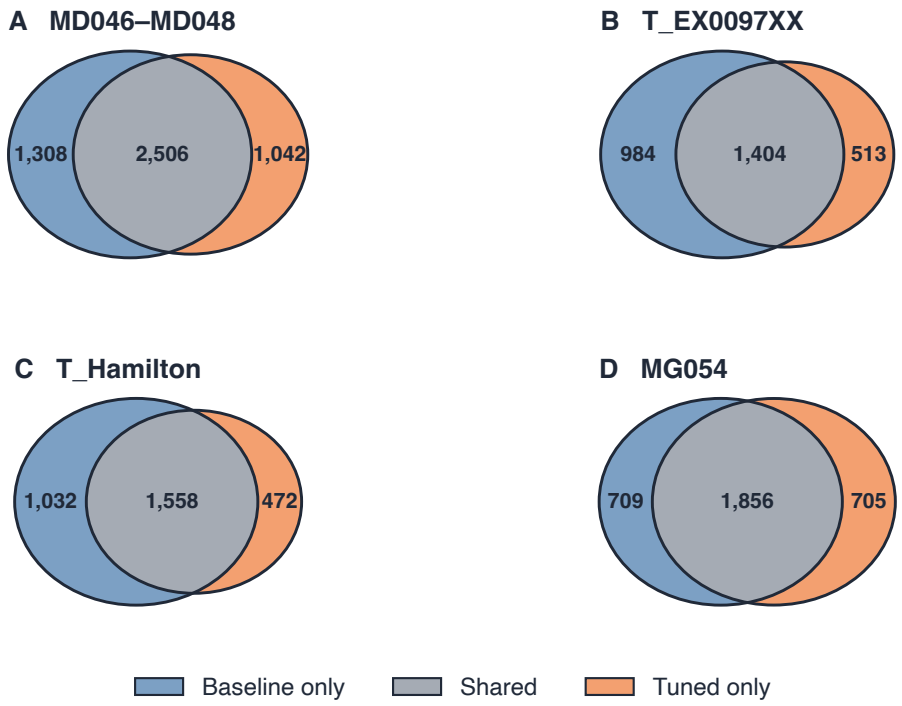


Figure 17.: Shared and exclusive Heavy-N identifications at the N-terminal peptide level. Each peptide was counted once across all held-out files per dataset and search condition, following Table 11. Blue and orange regions show peptides identified only in the baseline or tuned search, respectively; grey regions show peptides identified in both searches. Peptide selection follows Figure 15. Numbers give peptide counts. Areas are proportional to counts within each panel, but diagram sizes cannot be compared between datasets.

The two searches identified substantially different Heavy-N peptide sets in every dataset. In MD046–MD048, T_EX0097XX and T_Hamilton, 984–1,308 peptides were identified only in the baseline, compared with 472–1,042 only after tuning. Baseline-only counts were higher in all four datasets, consistent with the higher total Heavy-N peptide counts in the baseline searches (Figure 15). Even in MG054, where the total peptide count changed little, 709 peptides were baseline-only and 705 were tuned-only. Thus, similar total counts did not mean that the searches identified the same peptides.

These differences raised the question of whether baseline-only and tuned-only peptides differed in their properties. Such differences could indicate that tuning favoured peptides with certain characteristics. We therefore compared sequence length and hydropathy across the baseline-only, shared and tuned-only groups (Figure 18). We focused on MD046–MD048, where Heavy-N peptide counts decreased, and MG054, where the total count remained nearly unchanged although many

peptides were identified in only one search. Sequence length and hydropathy were compared at the N-terminal peptide level, using the groups shown in Figure 17. Hydropathy was expressed as the grand average of hydropathy (GRAVY), calculated from the Kyte-Doolittle residue scores (Kyte & Doolittle, 1982). Charge distributions were compared at the precursor-group level because this level distinguishes charge states.

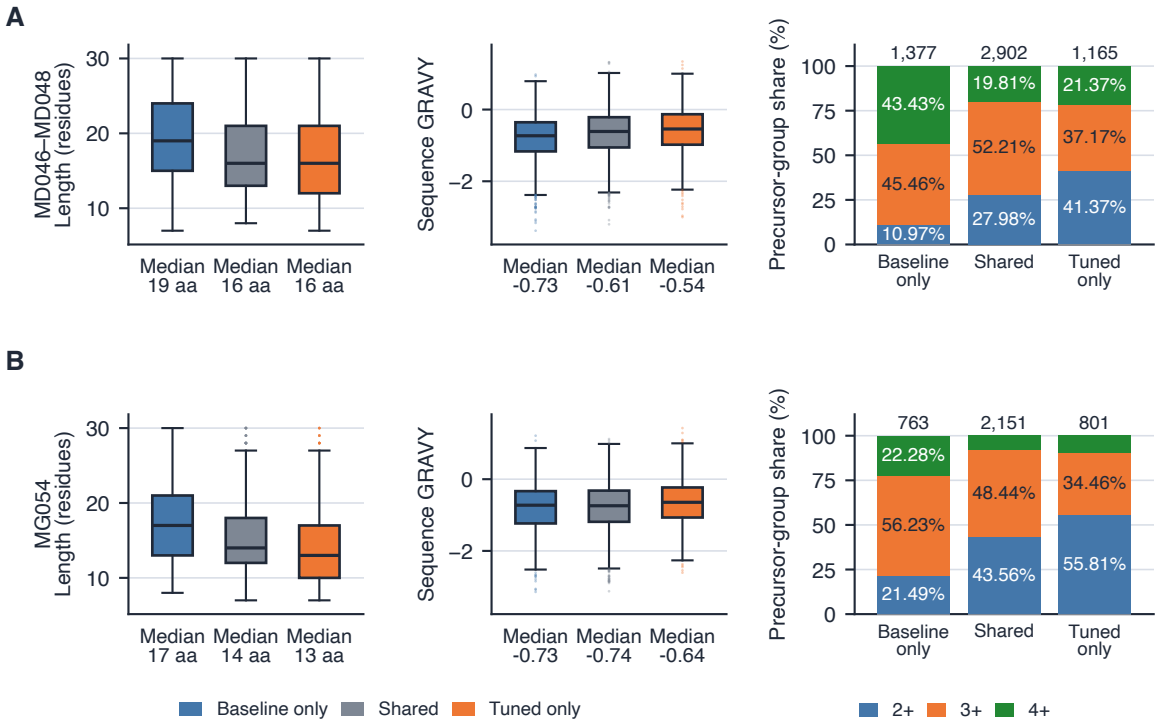


Figure 18.: Sequence properties and charge distributions of Heavy-N identifications. Rows show MD046–MD048 (A) and MG054 (B). Left and middle columns compare length and GRAVY at the N-terminal peptide level. The right column compares charge distributions at the precursor-group level. Shared and exclusive groups were defined separately at each level across all held-out files, following Table 11 and the selection in Figure 15. Sequence properties were calculated from unmodified sequences. One peptide containing X was excluded from GRAVY. Boxes show medians and interquartile ranges; whiskers follow the 1.5-IQR rule, with outliers shown as points. Medians are given below the boxes. Numbers above bars give precursor-group counts. The 1+ fractions were below 0.15% and are omitted from the legend.

Both datasets showed similar patterns in sequence length and charge distribution (Figure 18). Baseline-only peptides were longer than shared peptides, with median lengths of 19 versus 16 residues in MD046–MD048 and 17 versus 14 residues in MG054. Tuned-only peptides had the same or a shorter median length than shared peptides. The hydropathy distributions largely overlapped, although tuned-only peptides had slightly higher median GRAVY values in both datasets. At the precursor-group level, 4+ charges were more common in baseline-only groups than in shared groups, whereas 2+ charges were more common in tuned-only groups. For example, in MD046–MD048, 4+ charges accounted for 43.43% of baseline-only groups and 19.81% of shared groups. Overall, shared and exclusive identifications differed more clearly in length and charge distribution than in hydropathy.

4.4.4. RT prediction changes varied across datasets

The sequence and charge differences in Section 4.4.3 raised the question of whether baseline-only and tuned-only identifications also differed in RT agreement. For MD046–MD048 and MG054, we compared the predicted and observed RTs reported for baseline-only and tuned-only Heavy-N identifications (Figure 19). Groups were defined at the precursor-group level across all held-out files, as in the charge comparison in Figure 18. Each accepted complete precursor form contributed one RT observation per file. The comparison included 5,205 baseline-only and 4,429 tuned-only observations in MD046–MD048, and 2,573 and 2,394 in MG054.

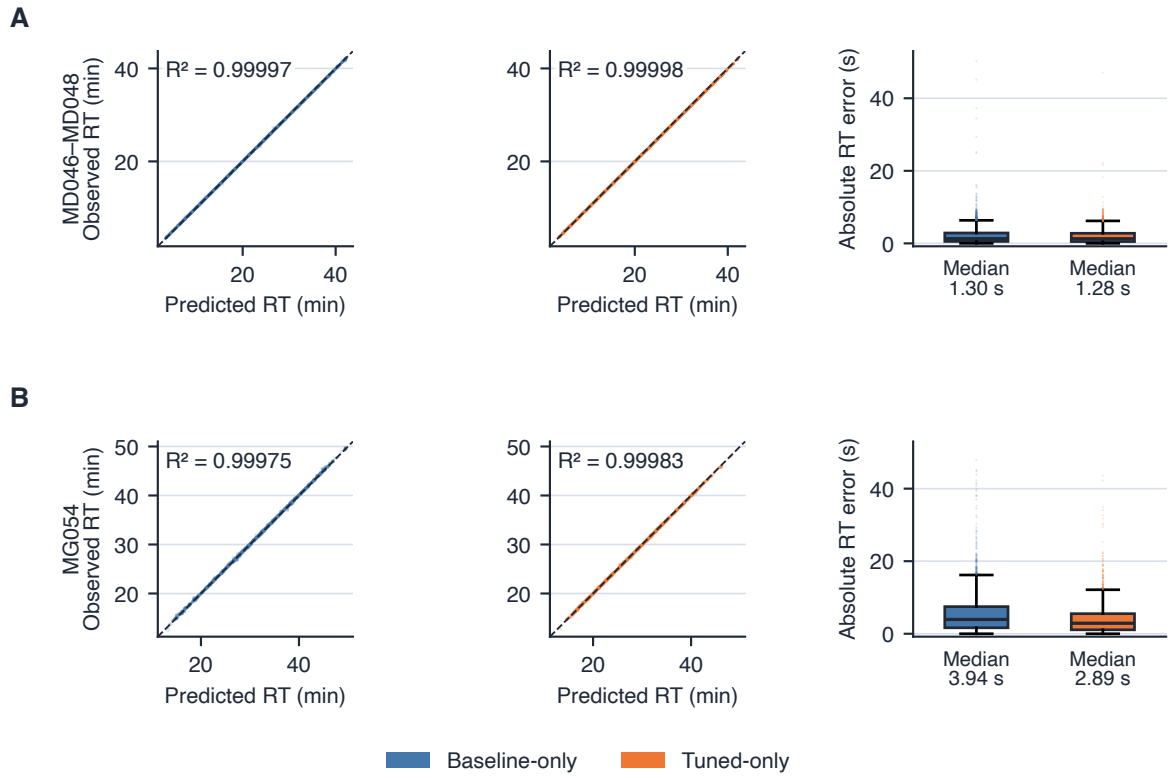


Figure 19.: RT agreement for baseline-only and tuned-only Heavy-N identifications. Rows show MD046–MD048 (A) and MG054 (B). Left and middle columns show predicted versus observed RTs for baseline-only and tuned-only groups. Dashed lines indicate equal predicted and observed RTs. The right column shows absolute RT errors in seconds. Groups were defined at the precursor-group level across all held-out files, following Table 11 and the selection in Figure 15. Each complete precursor form contributes one observation per file. R^2 uses observed RT as the reference. Boxes show medians and interquartile ranges; whiskers follow the 1.5-IQR rule, with outliers shown as points. Medians are given below the boxes.

Predicted and observed RTs agreed closely in both exclusive groups at the minute scale. In MD046–MD048, the median absolute RT errors were nearly identical: 1.30 s for baseline-only and 1.28 s for tuned-only observations. In MG054, tuned-only observations had a smaller median error than baseline-only observations, at 2.89 versus 3.94 s. Thus, RT agreement was similar between baseline-only and tuned-only identifications in MD046–MD048, whereas tuned-only identifications showed slightly closer RT agreement in MG054.

We also examined how tuning changed the RT predictions stored in the spectral libraries, using the extraction and charge-state grouping described in Section 3.5.1. The same complete modified peptide forms were paired between the baseline and tuned libraries, with all library records retained. For each pair, we calculated tuned minus baseline RT (Figure 20). The horizontal axis shows this difference in DIA-NN reference-RT units, not in minutes or seconds. Positive values indicate higher stored RT predictions after tuning, and negative values indicate lower predictions. Forms were grouped by the location of UniMod:510: at the N-terminus only, at lysines only, at both, or at neither.

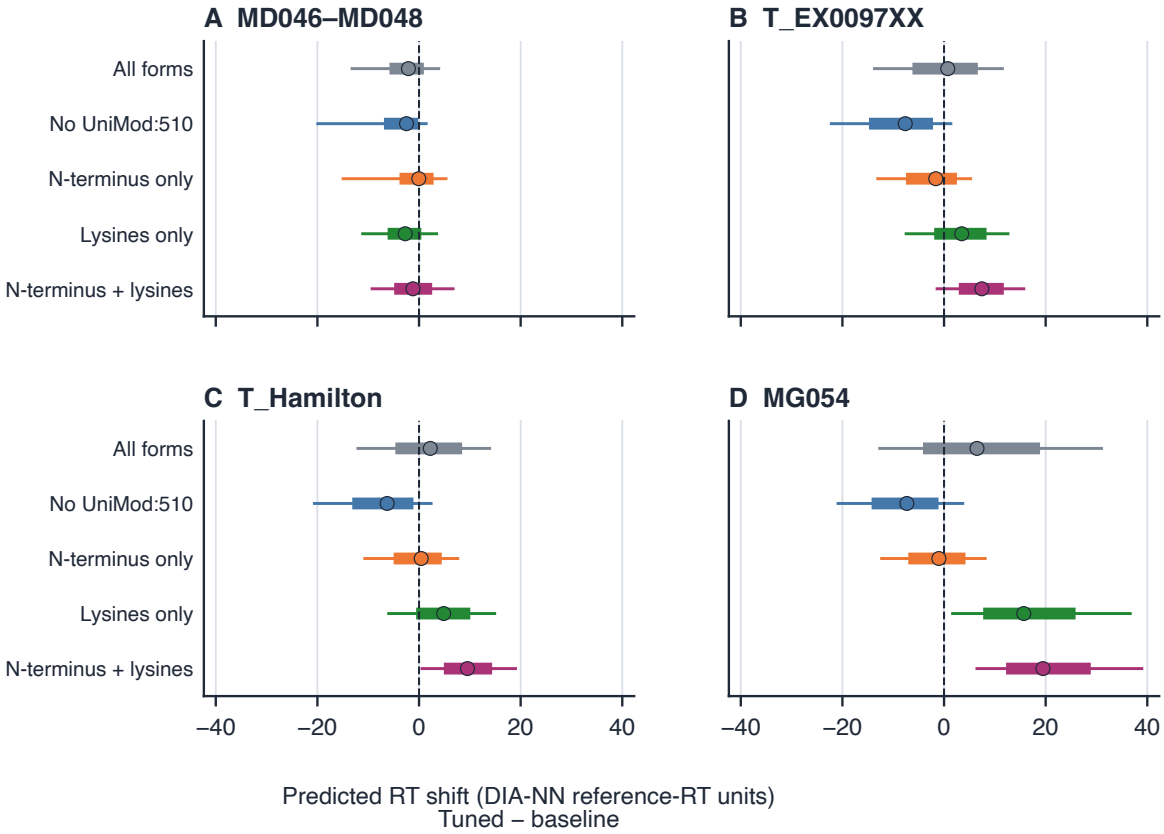


Figure 20.: RT prediction shifts in baseline and tuned spectral libraries. Panels show MD046–MD048 (A), T_EX0097XX (B), T_Hamilton (C) and MG054 (D). The horizontal axis gives tuned minus baseline RT in DIA-NN reference-RT units; the dashed line marks no change. Points show medians, thick intervals the interquartile range and thin intervals the 10th–90th percentiles. Groups distinguish UniMod:510 at neither site, the N-terminus only, lysines only, or both; All forms pools these groups. Exact modified peptide forms present in both libraries were paired after grouping charge states, following Section 3.5.1.

Tuning changed the stored RT predictions in all four datasets, with the largest shifts in MG054. Across all paired forms, the median absolute shift was 11.76 reference-RT units in MG054, compared with 3.55 in MD046–MD048. The direction of change also differed between modification groups. In T_EX0097XX, T_Hamilton and MG054, forms with lysine UniMod:510 shifted towards higher values in the median, whereas forms without UniMod:510 shifted towards lower values. In these three datasets, forms with UniMod:510 at both the N-terminus and lysines showed the largest positive median shifts. Median shifts for forms with UniMod:510 only at the N-terminus remained closer to zero in all four datasets. MD046–MD048 showed smaller median shifts, which

were negative in all modification groups. Together, these results show that tuning changed the library RT predictions differently across datasets and modification groups.

4.4.5. Entrapment provided an external check of the searches

To assess false-hit behaviour among Heavy-N identifications, we compared entrapment and target precursor groups before and after tuning (Figure 21). The comparison included the three datasets with a 1:1 entrapment FASTA and was restricted to peptides carrying UniMod:510 at the N-terminus. Only non-decoy identifications passing the DIA-NN precursor Q-value filter of $Q \leq 0.01$ were included. Target groups were assigned only to proteins in the original database, whereas entrapment groups were assigned only to the shuffled proteins added as controls (Section 3.2.2). Each group was counted once per held-out file, and counts were summed across files. MG054 was excluded because its FASTA contained no entrapment sequences.

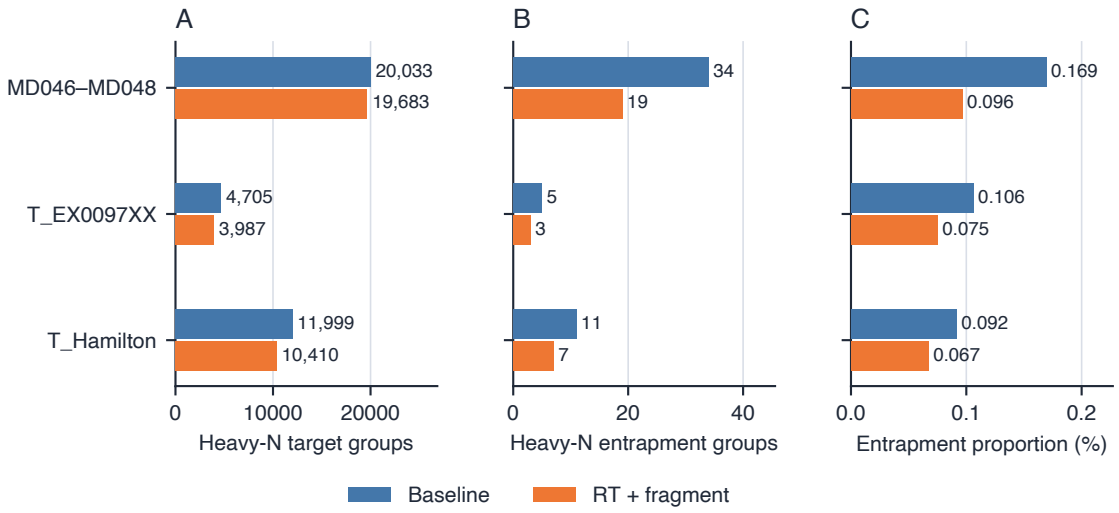


Figure 21.: Heavy-N entrapment before and after integrated RT-plus-fragment tuning. Panels show Heavy-N target precursor-group counts (A), entrapment counts (B) and the entrapment proportion in percent (C). Counts include non-decoy identifications at precursor Q ≤ 0.01 carrying UniMod:510 at the peptide N-terminus. Groups follow Table 11 and are counted once per held-out file, then summed across files. Mixed assignments to target and entrapment proteins are excluded as described in Section 3.2.2. Numbers beside the bars give counts or percentages.

Baseline searches yielded more Heavy-N entrapment detections and higher entrapment proportions than tuned searches in all three datasets. The absolute numbers of Heavy-N entrapment detections were small in both search conditions, ranging from three to 34 across the datasets. At the DIA-NN precursor Q-value threshold of 1%, all entrapment proportions remained below 0.20% (Figure 21). These proportions provide an external error diagnostic, not a direct FDR estimate.

4.4.6. Alternative configurations did not exceed baseline site coverage

Integrated RT-plus-fragment tuning did not increase overall Heavy-N coverage in the main comparison, as shown in Figure 15. We therefore tested alternative configurations within the integrated

DIA-NN tuning workflow and the final search to see whether they could improve coverage. These included adapting only the RT predictor, using more training files, updating fewer model layers and changing the RT window in the final search.

First, we compared RT-only and RT-plus-fragment tuning using the same six held-out MD048 files for the final searches (Figure 22). With the unshuffled FASTA, we used either six MD048 training files or 30 files from MD046–MD048. We also compared both tuning approaches in the final entrapment benchmark, which used 30 training files.

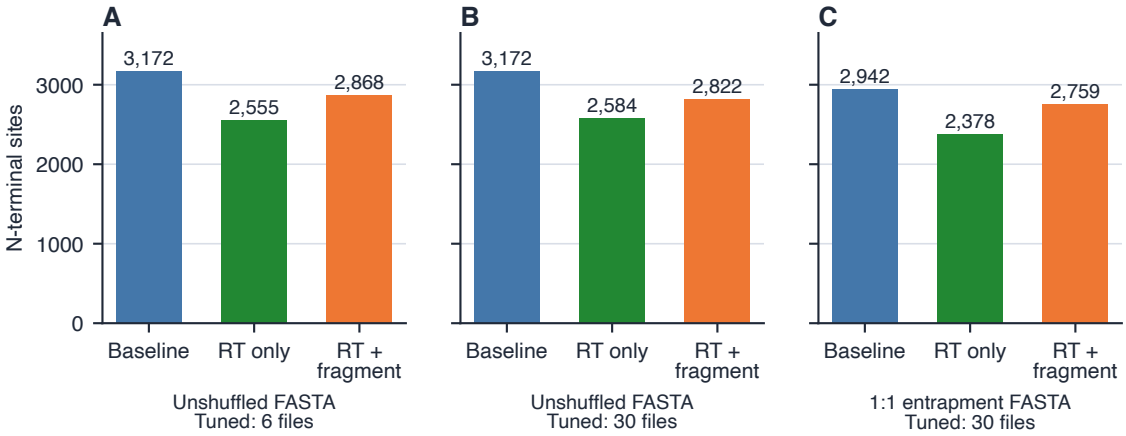


Figure 22.: Heavy-N N-terminal site coverage across MD048 tuning configurations. Sites were counted once across the same six held-out MD048 files, using the selection in Figure 15. Panels A and B use the same unshuffled FASTA with six and 30 training files, respectively. Panel C shows the final 1:1 entrapment benchmark with 30 training files. Training-file counts apply only to tuned searches; baseline searches used unchanged models. Counts at all three identification levels are given in Table 22.

RT-plus-fragment tuning identified more Heavy-N N-terminal sites than RT-only tuning in all three contexts, but both identified fewer sites than their matched baseline (Figure 22). This raises the question of whether adapting only the fragmentation predictor could increase coverage beyond the baseline, although this configuration was not tested. In the final entrapment benchmark, RT-plus-fragment tuning identified 2,759 sites and RT-only tuning identified 2,378, compared with 2,942 for the baseline.

Increasing the training set from six MD048 files to 30 MD046–MD048 files did not consistently improve site coverage with the unshuffled FASTA. RT-only tuning identified 29 more sites, whereas RT-plus-fragment tuning identified 46 fewer. Both remained below the unchanged baseline of 3,172 sites.

Searches with the larger 1:1 entrapment FASTA identified fewer Heavy-N N-terminal sites than searches with the unshuffled FASTA in all three variants, using 30 training files. Baseline coverage decreased from 3,172 to 2,942 sites, and site counts also decreased with RT-only and RT-plus-fragment tuning.

We also tested whether updating fewer model layers could improve Heavy-N coverage. For T_EX0097XX, T_Hamilton and MG054, we used DIA-NN's `--tune-restrict-layers` option during RT-plus-fragment tuning, as described in Section 3.3.1.

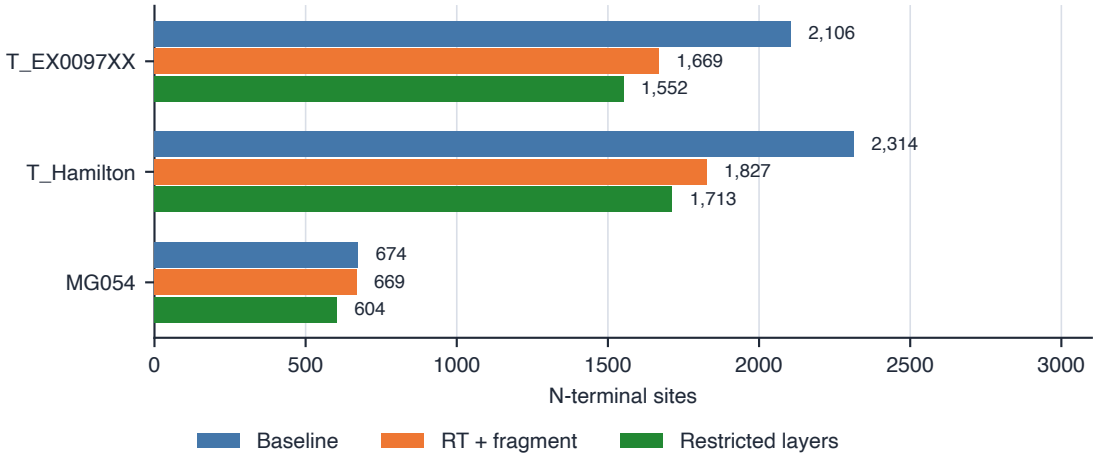


Figure 23.: Heavy-N N-terminal site coverage with restricted and unrestricted model tuning. Bars compare the unchanged-model baseline with unrestricted and restricted RT-plus-fragment tuning in each dataset. Sites were counted once across the same held-out files per dataset, using the selection in Figure 15.

Updating fewer model layers did not improve Heavy-N coverage. Restricted RT-plus-fragment tuning identified fewer N-terminal sites than both unrestricted tuning and the baseline in all three datasets (Figure 23). In MG054, for example, site counts decreased from 669 with unrestricted tuning to 604 with restricted tuning, compared with 674 for the baseline. Precursor-group and N-terminal peptide counts also decreased with restricted tuning.

We also tested whether the final RT search window affected MG054 coverage (Figure 24). We compared the `balanced`, `fast` and `slow` profiles defined in Table 7. For each tuning approach, we varied the RT-window profile while keeping the library and held-out files the same.

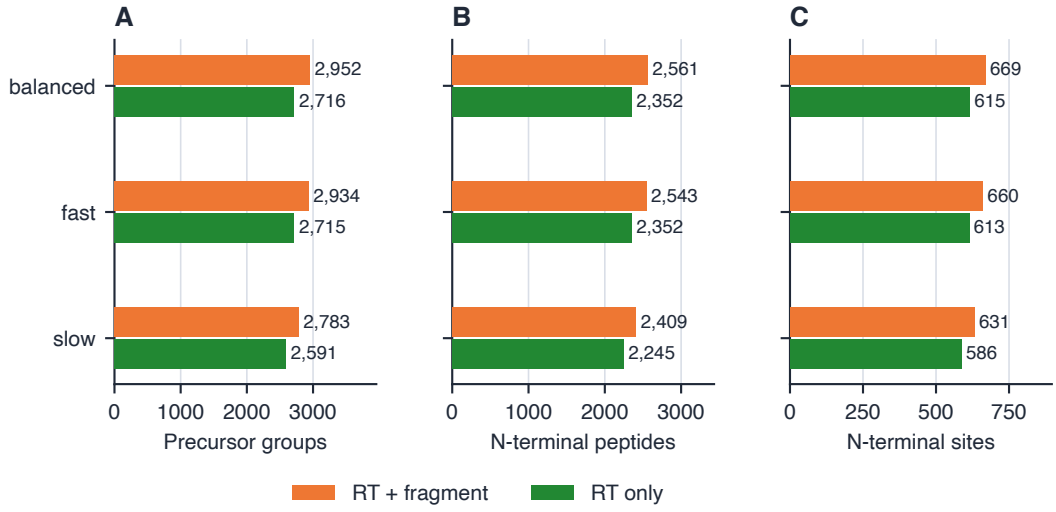


Figure 24.: Heavy-N coverage under MG054 final RT-window configurations. Panels show precursor groups (A), N-terminal peptides (B) and N-terminal sites (C), counted once across the six held-out files using the selection in Figure 15. The `balanced`, `fast` and `slow` profiles follow the RT-window settings in Table 7. Libraries were fixed within each predictor family; panels use separate count ranges.

Neither the `fast` nor the `slow` profile improved Heavy-N coverage over `balanced` (Figure 24). The `fast` profile produced similar counts, whereas `slow` yielded fewer identifications at all three levels with both tuning approaches. With RT-plus-fragment tuning, for example, N-terminal site counts decreased from 669 with `balanced` to 660 with `fast` and 631 with `slow`. RT-plus-fragment tuning identified more sites than RT-only tuning with every profile, but all six configurations remained below the baseline of 674 sites.

Overall, none of the tested configurations changes increased Heavy-N N-terminal site coverage beyond the matched unchanged-model baseline.

4.4.7. Larger empirical libraries increased Heavy-N coverage

We tested whether relaxing precursor selection for the empirical MBR library increased Heavy-N coverage. A separate two-file MD048 development experiment used library-export q-value thresholds (`--out-lib-qvalue`) of 0.05, 0.20 and 0.50, while the final report filter remained at precursor $Q \leq 0.01$ (Figure 25). The predicted input library, held-out files and search profile were the same in all three searches. The resulting empirical libraries were used in the second MBR pass, as described in Section 3.2.3.

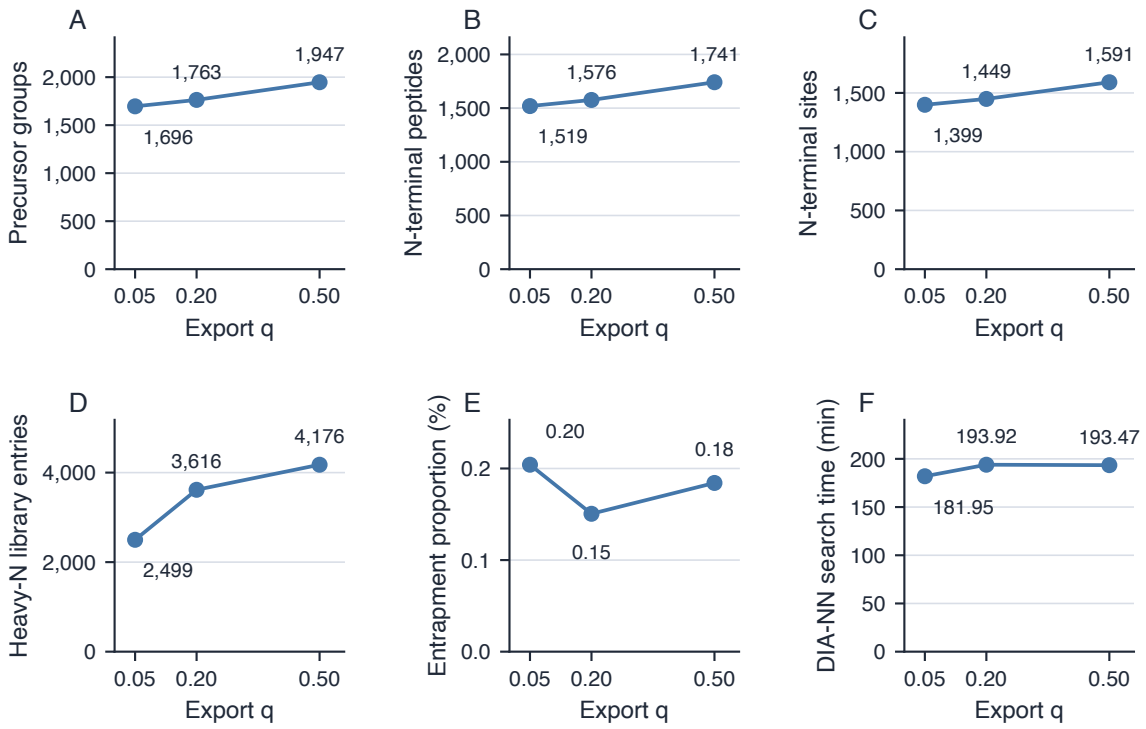


Figure 25.: Heavy-N coverage across library-export q-value thresholds. Export q denotes the library-export q-value threshold (`--out-lib-qvalue`). Panels show Heavy-N precursor groups (A), N-terminal peptides (B), N-terminal sites (C), empirical-library entries (D), the entrapment proportion (E) and total DIA-NN search time (F). Panels A–C count each object once across the two held-out MD048 files, using the selection in Figure 15. Panel D counts complete Heavy-N precursor entries, including decoys, in the empirical library used for the second MBR pass. Panel E shows the entrapment proportion for non-decoy precursor groups across all N-terminal modification groups. Panel F shows total DIA-NN search time, including both MBR passes.

Higher library-export q-value thresholds produced larger empirical libraries and more Heavy-N identifications at all three levels (Figure 25). For example, increasing the threshold from 0.05 to 0.50 increased N-terminal peptide counts from 1,519 to 1,741 and N-terminal site counts from 1,399 to 1,591. The increase in Heavy-N coverage was not accompanied by a consistent increase in entrapment proportions.

Most identifications were retained at higher thresholds, although some were no longer identified (Figure 42). Increasing the threshold from 0.05 to 0.50 added 276 Heavy-N N-terminal peptides, while 54 were not retained. Additional identifications outnumbered those not retained at all three levels in every comparison.

The searches at 0.20 and 0.50 took about twelve minutes longer than the search at 0.05 (Figure 25F).

4.4.8. DIA-NN versions differed in Heavy-N coverage

Additional DIA-NN versions were released during and after pipeline development. To keep the benchmark comparisons consistent, we retained version 2.5.0 for the main analyses. Nevertheless, we examined whether Heavy-N coverage and the response to integrated tuning differed between various DIA-NN versions. For MG054, we compared baselines across four versions and RT-plus-fragment tuning where available (Figure 26). Each version used the same human FASTA and six held-out files, with newly predicted spectral libraries. We also compared the Heavy-N N-terminal peptide sets between versions to determine which peptides were retained, added or no longer identified (Figure 43).

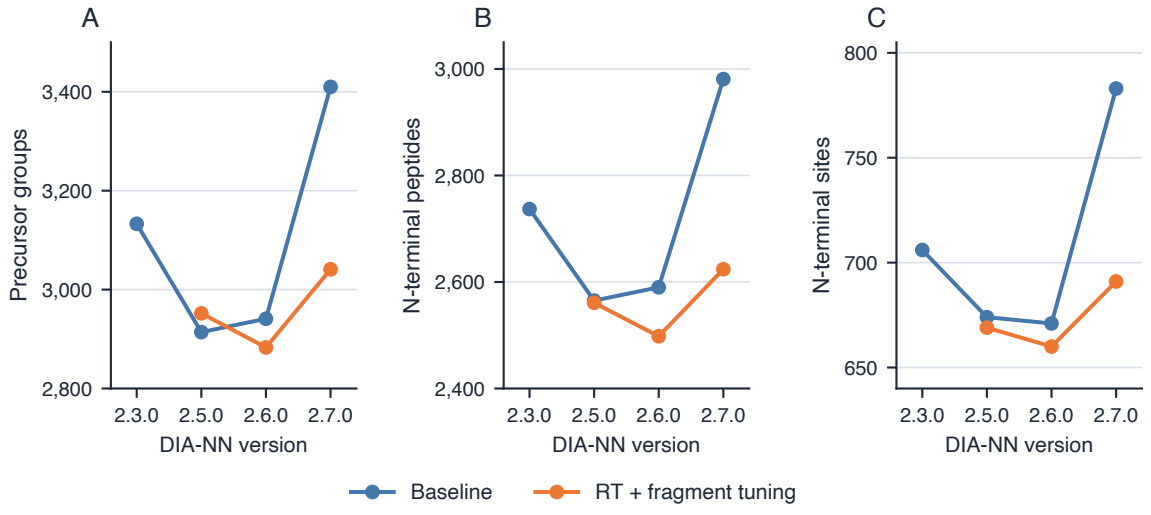


Figure 26.: Heavy-N coverage across DIA-NN versions in MG054. Panels show precursor groups (A), N-terminal peptides (B) and N-terminal sites (C), counted once across the six held-out files with the selection in Figure 15. Lines compare the unchanged-model baseline with RT-plus-fragment tuning. Y axes do not start at zero. No completed tuned search was available for version 2.3.0. Exact counts are given in Table 23.

Version 2.7.0 gave the highest baseline Heavy-N counts at all three levels (Figure 26). Compared with version 2.5.0, used in the main benchmark, N-terminal peptide counts increased from 2,565 to 2,981 and site counts from 674 to 783. Newer versions did not always identify more Heavy-N peptides and sites. RT-plus-fragment tuning identified fewer Heavy-N peptides and sites than the corresponding baseline in every tested version. In version 2.7.0, for example, site counts fell from 783 to 691, a reduction of 11.75%. The small precursor-group increase in version 2.5.0 was not reproduced in versions 2.6.0 or 2.7.0.

The versions also differed in which peptides they identified (Figure 43). For example, the version 2.7.0 baseline identified 534 peptides not found in version 2.6.0, while 143 peptides identified in version 2.6.0 were not retained. Therefore, the higher total counts did not simply represent the same peptides plus additional identifications.

The archived model comparison found byte-identical default RT-model files across all four versions. The observed baseline coverage differences therefore cannot be attributed to different default RT-model weights. The baseline predicted spectral libraries also contained the same complete precursor forms and bit-identical RT and fragment predictions (Table 24). The baseline coverage differences therefore cannot be explained by changes in these precursor forms or stored predictions.

4.4.9. Dimethyl encoding affected baseline coverage

We also tested whether using a dimethyl modification recognised by DIA-NN's predictor could increase Heavy-N peptide and site coverage. Following the developer and maintainer's recommendation, we declared heavy dimethylation as UniMod:255 instead of UniMod:510. We set its mass shift to $+34.063117$ Da, the same heavy-label mass used in the main benchmark. This kept the mass unchanged in the search while giving the predictor a modification it recognised (Demichev, 2026d). We compared this configuration with the UniMod:510 baselines used in the main benchmark (Figure 27).

The UniMod:255 configuration yielded substantially fewer Heavy-N identifications at all three levels in every dataset (Figure 27). Counts were approximately one-third to one-half of those obtained with UniMod:510. For example, N-terminal peptide counts fell from 3,814 to 1,618 in MD046–MD048, and site counts fell from 674 to 238 in MG054. This configuration did not improve coverage, and we retained UniMod:510 for the main benchmark.

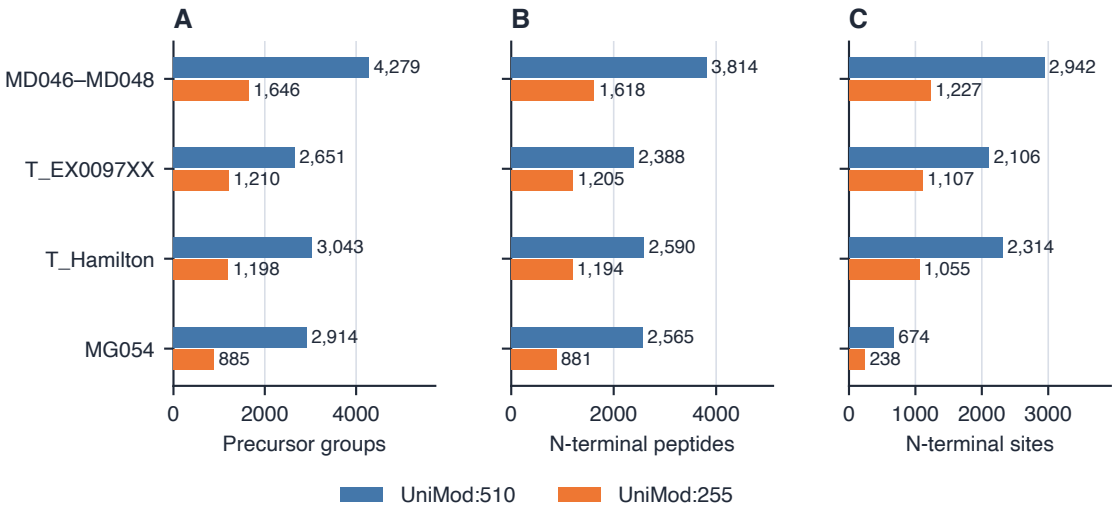


Figure 27.: Baseline Heavy-N coverage with UniMod:510 and UniMod:255. Panels show precursor groups (A), N-terminal peptides (B) and N-terminal sites (C), counted once across each dataset's held-out files with the selection in Figure 15. Both configurations used the same heavy-label mass of $+34.063117$ Da. N-terminal UniMod:255 was counted as Heavy-N in this control.

4.4.10. Training losses motivated an external RT workflow

The preceding comparisons in this section showed that integrated DIA-NN tuning did not increase Heavy-N site coverage. To learn more about predictor training, we examined the loss values recorded in the logs. Because their meaning was unclear, we asked the DIA-NN developer and maintainer. He described them as probably training loss per batch (Demichev, 2026c). Figure 28 shows these loss curves for the RT and fragmentation predictors in MD046–MD048 across ten learning rates.

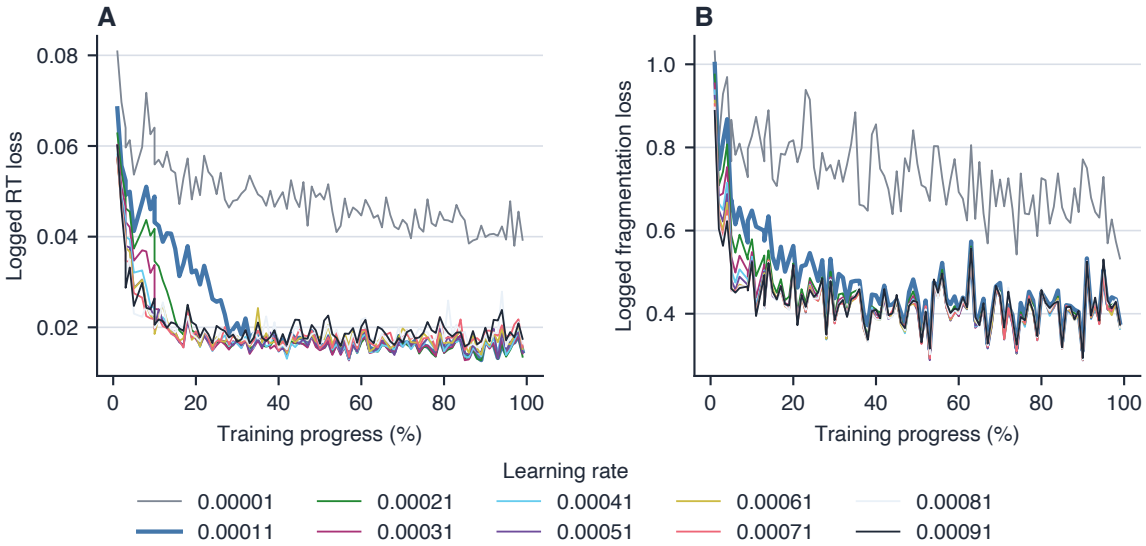


Figure 28.: Logged predictor loss across learning rates for MD046–MD048. Panels show the RT predictor (A) and fragmentation predictor (B) at distillation level d0. Blue highlights the learning rate of 0.00011 used for the final tuned libraries. Values were taken from the training logs; no validation-loss curves were available.

Logged losses decreased during training for both predictors (Figure 28). Except for the smallest learning rate, 0.00001, the curves reached similar final loss values. We selected 0.00011 for the final searches because it was among the learning rates with low final losses. However, no validation-loss curves were available to assess performance on unseen data. This motivated an external RT training workflow with separate training and validation outputs and control over the learning rate and training duration (Section 4.6).

4.5. Carafe2 expanded Heavy-N coverage

Integrated DIA-NN tuning did not increase Heavy-N site coverage in the preceding comparisons. We therefore evaluated whether the HUNTER-compatible Carafe2 workflow, introduced in Section 4.2.2, could increase coverage compared with the unchanged-model DIA-NN baseline. Carafe2 adapted its own RT and fragmentation models and generated a predicted spectral library for the final DIA-NN search. Both workflows were evaluated on the same held-out files within each dataset.

Carafe2 generated its predicted spectral libraries from unshuffled FASTAs without entrapment proteins. The predicted spectral libraries of the unchanged-model DIA-NN baseline contained entrapment candidates in MD046–MD048 and both TripleMix datasets. The final Carafe2 searches used FASTAs without entrapment in both TripleMix datasets and MG054. Only the final Carafe2 search for MD046–MD048 used the entrapment FASTA. Entrapment-only assignments and mixed assignments to target and entrapment proteins were excluded before evaluation. The comparison therefore includes differences in the predicted spectral libraries as well as model adaptation.

4.5.1. Heavy-N coverage increased at all three identification levels

We compared Heavy-N identifications at the three levels defined in Table 11: precursor groups, N-terminal peptides and N-terminal sites. Each object was counted once across all held-out files for each dataset and search condition (Figure 29).

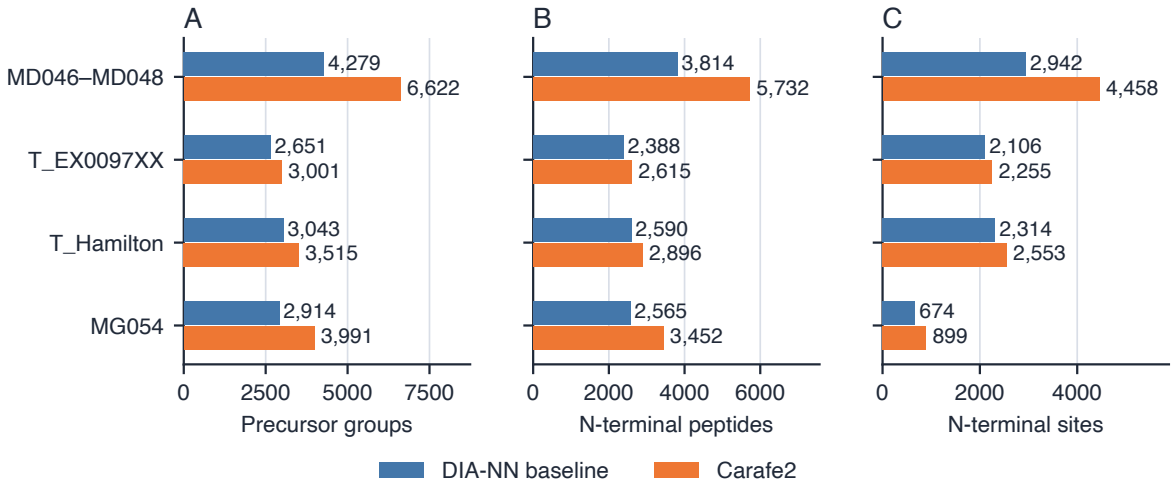


Figure 29.: Heavy-N coverage with the baseline and Carafe2 workflows. Panels compare precursor groups (A), N-terminal peptides (B) and N-terminal sites (C). Non-decoy Heavy-N identifications at precursor `Q.Value` ≤ 0.01 were counted once across all held-out files per dataset and search condition, following the definitions in Table 11. Only identifications with an unambiguous R*-specific, semi-N or semi-C classification were included. Site counts additionally required a unique protein and starting-position assignment.

Carafe2 identified more Heavy-N precursor groups, N-terminal peptides and N-terminal sites than the unchanged-model DIA-NN baseline in all four datasets (Figure 29). The largest relative increases in site coverage occurred in MD046–MD048, from 2,942 to 4,458 sites (+51.53%), and MG054, from 674 to 899 (+33.38%). Site counts also increased in T_EX0097XX and T_Hamilton, by 7.08% and 10.33%, respectively. These results show that the Carafe2 workflow consistently increased Heavy-N coverage across all three identification levels and all four datasets.

4.5.2. Specificity and N-terminal modification composition changed

After observing higher Heavy-N coverage with Carafe2, we examined whether the increase was similar across peptide groups. We compared cleavage specificity within Heavy-N peptides and N-terminal modification groups among all accepted peptides (Figure 30). Panel A divides Heavy-N peptides into R*-specific, semi-N and semi-C classes. Panel B shows the proportions of Heavy-N, N-terminally acetylated and free-N-terminal peptides. The Heavy-N subset in panel B corresponds to all peptides shown in panel A. Both panels use the N-terminal peptide level. Each combination of peptide sequence and N-terminal modification status was counted once across all held-out files per dataset and search condition.

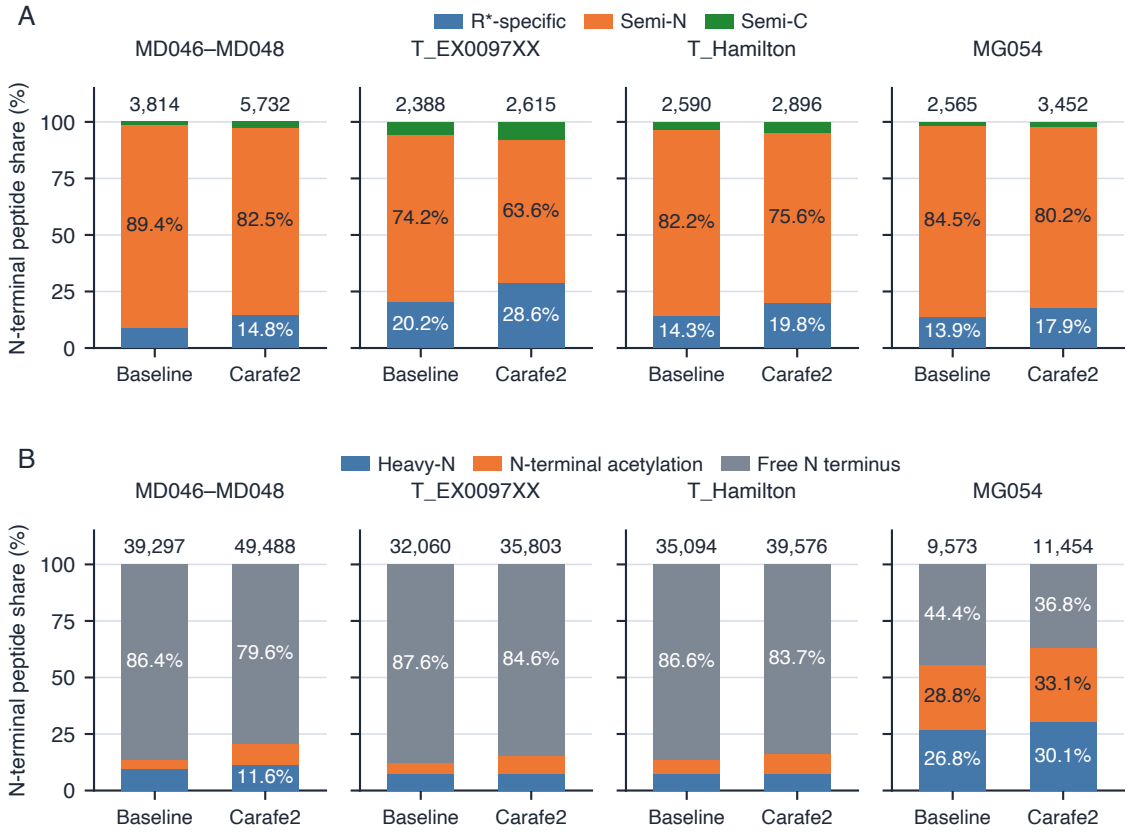


Figure 30.: N-terminal peptide composition in the baseline and Carafe2 searches. Panel A shows R* cleavage specificity within Heavy-N peptides, which form the Heavy-N subset in panel B. Panel B compares N-terminal modification groups among accepted peptides. Heavy-N peptides required an unambiguous R*-specific, semi-N or semi-C classification. Counts follow Table 11, with each object counted once across held-out files per dataset and search condition. Free denotes an unmodified peptide N-terminus. Numbers above bars give total peptide counts.

Semi-N peptides formed by far the largest Heavy-N class in every search (Figure 30A). Their proportions decreased with Carafe2 in all four datasets, although their counts increased in MD046–MD048 and MG054. For example, the semi-N count in MD046–MD048 increased from 3,408 to 4,731, while its proportion fell from 89.36% to 82.54%. In contrast, semi-N counts decreased in T_EX0097XX and increased only slightly in T_Hamilton. R*-specific Heavy-N peptides increased in both count and proportion in all four datasets. The Heavy-N gain therefore differed between cleavage-specificity classes.

Carafe2 also changed the proportions of N-terminal modification groups (Figure 30B). Both the number and proportion of acetylated peptides increased in every dataset. For example, their proportion in MD046–MD048 more than doubled, from 3.88% to 8.85%. Heavy-N peptide counts increased in all four datasets, but their proportions increased only in MD046–MD048 and MG054. MG054 remained distinct from the other datasets in both searches, with higher Heavy-N and acetylated proportions and a lower proportion of peptides with a free N-terminus. Together, these results show that the broader coverage was accompanied by changes in the composition of the identified peptides.

4.5.3. Overlap separated additional identifications from losses

The overall counts did not show whether Carafe2 retained the same Heavy-N peptides or identified different ones. We therefore performed an intersection analysis to divide the N-terminal peptides into shared, baseline-only and Carafe2-only groups (Figure 31). Each Heavy-N N-terminal peptide was counted once across all held-out files within each dataset and search condition. Shared peptides were identified in both searches, but not necessarily in the same file.

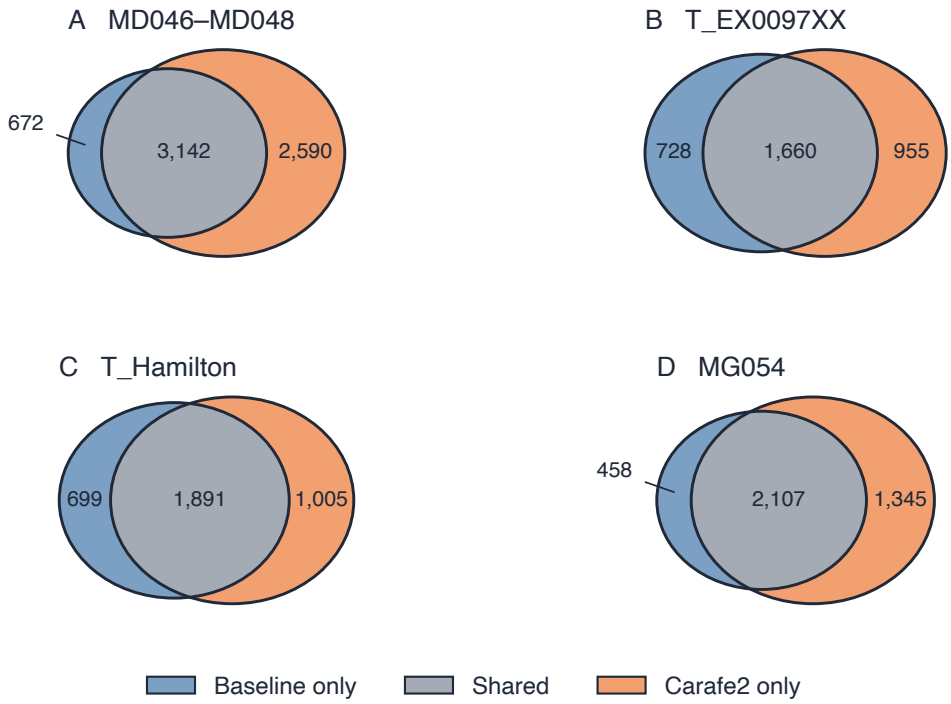


Figure 31.: Shared and exclusive Heavy-N identifications at the N-terminal peptide level. Each peptide was counted once across all held-out files per dataset and search condition, following Table 11. Blue and orange regions show peptides identified only in the baseline or Carafe2 search, respectively; grey regions show peptides identified in both searches. Peptide selection follows Figure 29. Numbers give peptide counts. Areas are proportional to counts within each panel, but diagram sizes cannot be compared between datasets.

Carafe2-only peptides outnumbered baseline-only peptides in all four datasets. The largest differences occurred in MD046–MD048, with 2,590 Carafe2-only and 672 baseline-only peptides, and MG054, with 1,345 and 458, respectively. The exclusive groups were closer in size in T_EX0097XX and T_Hamilton. The higher Heavy-N peptide counts therefore included both additional identifications and losses from the baseline set.

4.5.4. Shared and exclusive Heavy-N identifications differed in sequence properties and charge

The intersection analysis in Section 4.5.3 raised the question of whether baseline-only, shared and Carafe2-only peptides differed in their properties. We therefore compared sequence length,

hydropathy and charge distributions across these groups (Figure 32). We focused on MD046–MD048 and MG054, which showed the largest gains in Heavy-N peptide counts. Sequence length and hydropathy were compared at the N-terminal peptide level, using the groups shown in Figure 31. Hydropathy was expressed as GRAVY, calculated from the Kyte-Doolittle residue scores (Kyte & Doolittle, 1982). Charge distributions were compared at the precursor-group level because this level distinguishes charge states.

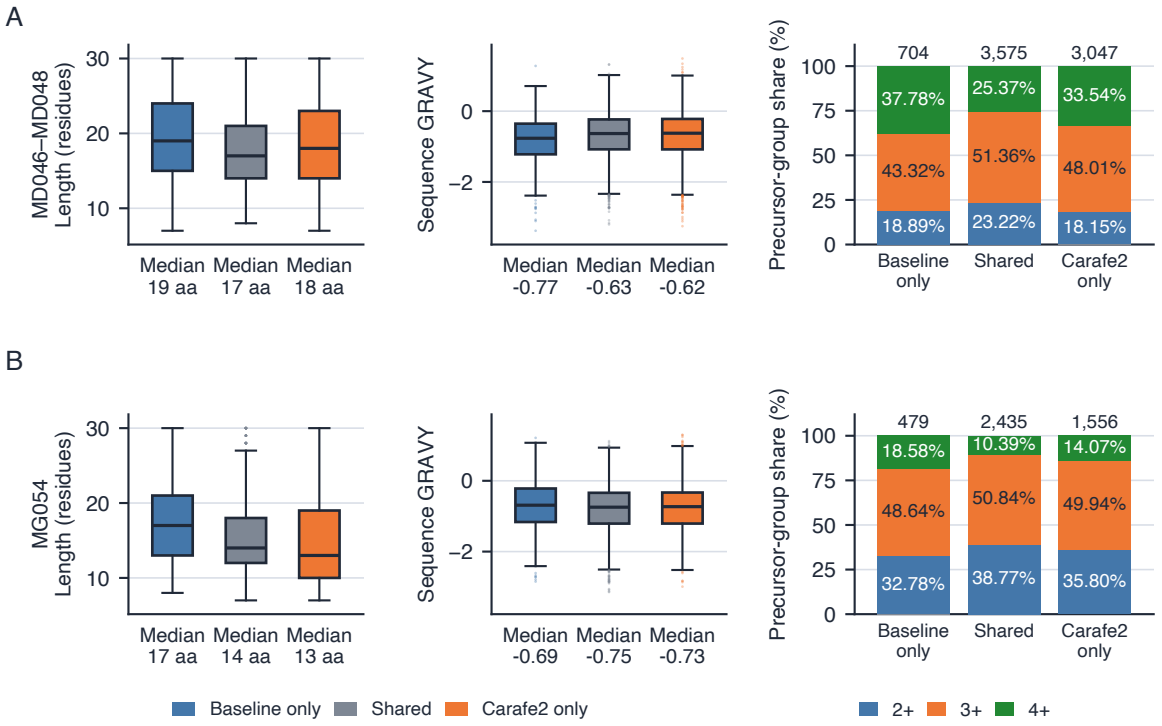


Figure 32.: Sequence properties and charge distributions of Heavy-N identifications. Rows show MD046–MD048 (A) and MG054 (B). Left and middle columns compare length and GRAVY at the N-terminal peptide level. The right column compares charge distributions at the precursor-group level. Shared and exclusive groups were defined separately at each level across all held-out files, following Table 11 and the selection in Figure 29. Sequence properties were calculated from unmodified sequences. One peptide containing X was excluded from GRAVY. Boxes show medians and interquartile ranges; whiskers follow the 1.5-IQR rule, with outliers shown as points. Medians are given below the boxes. Numbers above bars give precursor-group counts. The 1+ fractions were below 0.30% and are omitted from the legend.

Both datasets showed similar patterns in sequence length and charge distribution (Figure 32). Baseline-only peptides were longer than shared peptides, with median lengths of 19 versus 17 residues in MD046–MD048 and 17 versus 14 residues in MG054. Carafe2-only peptides were slightly longer than shared peptides in MD046–MD048, but slightly shorter in MG054. The hydropathy distributions of shared and Carafe2-only peptides largely overlapped. At the precursor-group level, 3+ charges were the most common in all groups in both datasets. The proportion of 4+ charges was highest in baseline-only groups and lowest in shared groups, with Carafe2-only groups in between. For example, in MD046–MD048, 4+ charges accounted for 37.78% of baseline-only groups and 25.37% of shared groups. Overall, shared and exclusive identifications differed more clearly in length and charge distribution than in hydropathy.

4.5.5. Reported RT deviations were larger with Carafe2

The broader Heavy-N coverage raised the question of whether the Carafe2 searches also showed closer agreement between predicted and observed RTs. To compare RT agreement for the same precursors, we used precursors of semi-N peptides carrying the Heavy-N label accepted by both workflows in the same held-out file in MD046–MD048 and MG054 (Figure 33). Pairs matched the same sequence, charge and complete modification pattern within the same held-out file, giving 12,528 paired observations in MD046–MD048 and 7,487 in MG054.

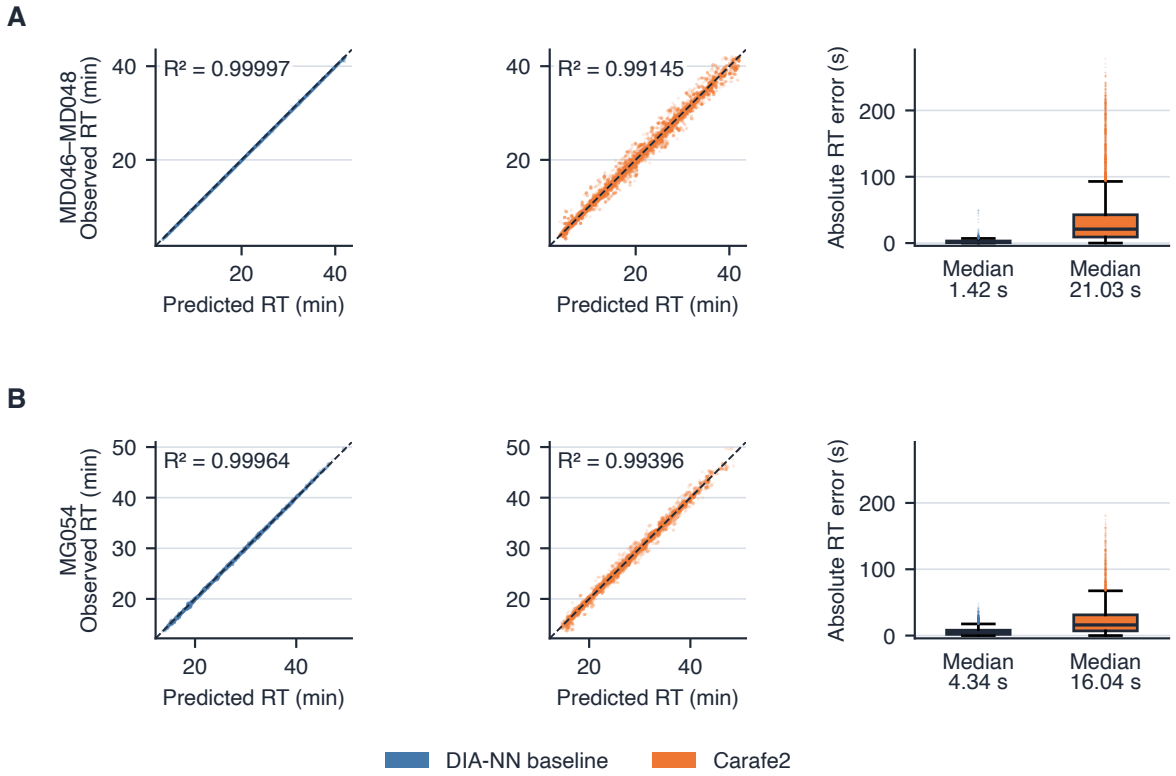


Figure 33.: RT agreement for shared Heavy-N identifications. Rows show MD046–MD048 (A) and MG054 (B). Only precursors of semi-N peptides carrying the Heavy-N label are included. Left and middle columns show predicted versus observed RTs for the unchanged-model baseline and Carafe2 searches, respectively. Dashed lines indicate equal predicted and observed RTs. The right column shows absolute RT errors in seconds. Each observation pairs the same complete precursor form accepted by both workflows within the same held-out file, retaining charge and all modifications. R^2 uses observed RT as the reference. Boxes show medians and interquartile ranges; whiskers follow the 1.5-IQR rule, with outliers shown as points. Medians are given below the boxes.

Predicted and observed RTs followed the same overall trend in both workflows, but differed more with Carafe2 (Figure 33). The median absolute RT deviation increased from 1.42 to 21.03 s in MD046–MD048 and from 4.34 to 16.04 s in MG054. Carafe2 identified more semi-N peptides carrying the Heavy-N label, but predicted and observed RTs differed more for precursors identified by both workflows.

4.6. Controlled external RT fine-tuning

The previous comparisons raised the question of whether more control over RT training could improve predictions for dimethylated peptides and increase Heavy-N coverage. To address this, we fine-tuned the RT models used by DIA-NN and Carafe2 outside their integrated training workflows.

We first compared RT target scales for training on data from several sources. Following this, we examined whether RT predictions improved when training focused on dimethylated peptides or used data from a single source instead of all four sources. Finally, we replaced the RT models in the DIA-NN and Carafe2 workflows with selected externally fine-tuned models. We then used these workflows to generate predicted spectral libraries and search the same held-out files with DIA-NN. We compared Heavy-N coverage with the unchanged-model DIA-NN baseline and the integrated tuning results in Sections 4.4 and 4.5. The external training code is available through the project resources described in Section 3.6.2.

4.6.1. RT alignment reduced prediction errors

For the main external fine-tuning experiments, we combined RT observations from the Carafe2 training data of all four benchmark datasets (Section 3.4.1). These data came from the DIA-NN training-file searches described in the Carafe2 workflow (Section 4.2.2). The aim was to obtain more training examples for fine-tuning and improve RT predictions. The sources covered different organisms and LC experiments. Because measured RTs depended on the chromatographic system and gradient, combining these observations required a common RT target scale.

To choose a common RT target scale, we compared three RT alignment approaches with two controls without alignment. We used data from three sources: MD046–MD048, T_EX0097XX and MG054. Joint-anchor RT and robust linear alignment used a human RT reference, whereas Chronologer HI used model-predicted HI values. Source-native and global 60-min targets served as controls without RT alignment. They only rescaled measured RTs using source-specific or common reference times, respectively (Sections 1.4.4 and 3.4.2).

Figure 34 shows the mean errors across training datasets and sources, expressed in observed minutes. For each RT target, we fine-tuned both model families with the fourteen training-data combinations listed in Table 25. We evaluated prediction errors on holdout observations from the three sources. Results for all eleven RT target definitions are given in Table 29.

Aligned targets gave lower errors than source-native targets or global 60-min scaling in both model families. Joint-anchor RT gave mean MAEs of 1.189 min for DIA-NN and 1.245 min for Carafe2; Chronologer HI and robust linear alignment gave similar errors. In contrast, source-native targets gave 2.635 and 2.728 min, respectively. These results showed that rescaling measured RTs alone was less effective than RT alignment. We kept joint-anchor RT and Chronologer HI for the subsequent external fine-tuning experiments using data from all four benchmark datasets.

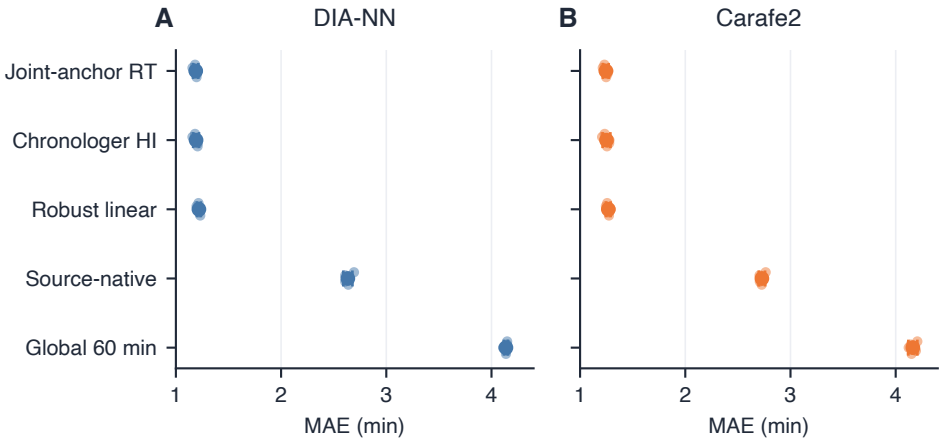


Figure 34.: RT prediction errors with different training targets. Panels show DIA-NN (A) and Carafe2 (B) results for three RT alignment approaches and two controls without alignment. Small points show results for five paired split and training seeds. For each seed, MAEs in observed minutes were averaged equally across the fourteen training-data combinations in Table 25 and all three sources. Large points show the mean across seeds with 95% confidence intervals.

For DIA-NN, we then asked whether aligning the training data to the pretrained model's own RT scale could further reduce prediction errors. To construct this target, we first used the unchanged DIA-NN RT model to predict RT values for unmodified training peptides. Each peptide had both a measured RT and a prediction from the unchanged model. Using these pairs, we fitted a mapping independently for each source using kernel density estimation (KDE) to convert measured RTs to the DIA-NN model's output scale. These mappings were also applied to modified peptidoforms. We termed the resulting scale DIA-NN space.

We then fine-tuned the DIA-NN RT model using DIA-NN space, joint-anchor RT and Chronologer HI on the combined data from all four benchmark datasets. All three targets used the same training, validation and test partitions (Section 3.4.1). RT mappings were fitted on training observations only and then applied to validation and test observations. For each target, we trained 162 models across different loss-weighting settings, sampling settings and learning rates, using three seeds (Table 26). We compared the validation MAEs of models trained with the same settings and seed but different RT targets. MAEs were calculated in observed minutes and averaged equally over the four sources. Heavy-macro was the mean of the source-balanced MAEs for the N-term-only, Lys-only and Heavy at both groups (Section 3.5.2).

Figure 35 summarises the validation-error differences across 162 model pairs. DIA-NN space and Chronologer HI gave nearly identical validation errors overall and for Heavy-macro. Compared with joint-anchor RT, DIA-NN space gave lower errors for N-term-only peptidoforms but higher errors for Lys-only peptidoforms in all 162 model pairs. DIA-NN space therefore did not consistently improve prediction accuracy across the groups.

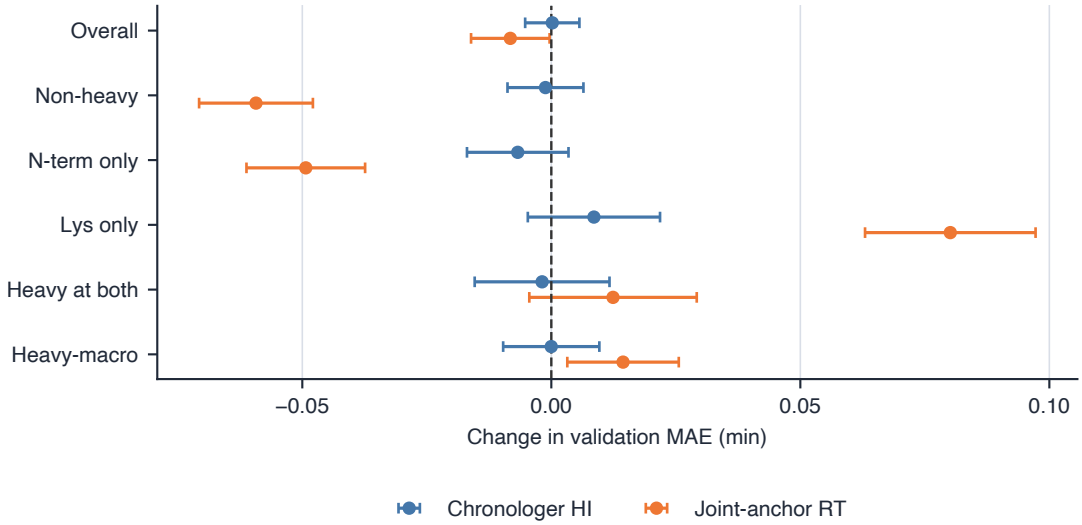


Figure 35.: Validation-error changes with the DIA-NN-space target. Points show the mean difference in validation MAE (DIA-NN space minus the reference target) across 162 model pairs; bars show the standard deviation. Negative values indicate lower error with DIA-NN space. MAEs are in observed minutes, with equal weight for each of the four sources. Overall includes all validation peptidoforms. The other groups distinguish whether heavy dimethylation is absent (Non-heavy), occurs only at the N-terminus (N-term only), only at lysines (Lys only), or at both (Heavy at both). Heavy-macro is the mean MAE of the three groups listed last.

4.6.2. Heavy-focused training gave small improvements in RT prediction

Since dimethylated peptides were the main focus of this study, we asked whether giving them more emphasis during training could improve their RT predictions. We tested two approaches: sampling selected heavy-dimethylated groups more frequently and giving their prediction errors greater weight in the training loss. We evaluated the models on the same test data from all four benchmark datasets, comparing each approach with a control without the additional sampling or loss weighting. Model identities and used training settings are given in Table 27.

Heavy sampling with the DIA-NN RT model

In the first approach, we showed the DIA-NN RT model some peptidoforms more often during training. These carried heavy dimethylation at the N-terminus only or at both the N-terminus and lysines. We compared the sampling model with an unweighted control using the same DIA-NN-space target and learning rate. Both models were fine-tuned by updating all model parameters, including the heavy-dimethyl embeddings. Both models were evaluated on the four-source model-test partition described in Section 3.4.1. Figure 36 shows test MAEs in observed minutes, with equal weight for each source (Section 3.5.2).

The unweighted control gave lower test MAEs overall and for the Lys-only and N-term-only groups, although the N-term-only difference was very small. The heavy-sampling model gave a lower test MAE only for peptidoforms carrying heavy dimethylation at both the N-terminus and lysines, with a reduction of 0.0324 min.

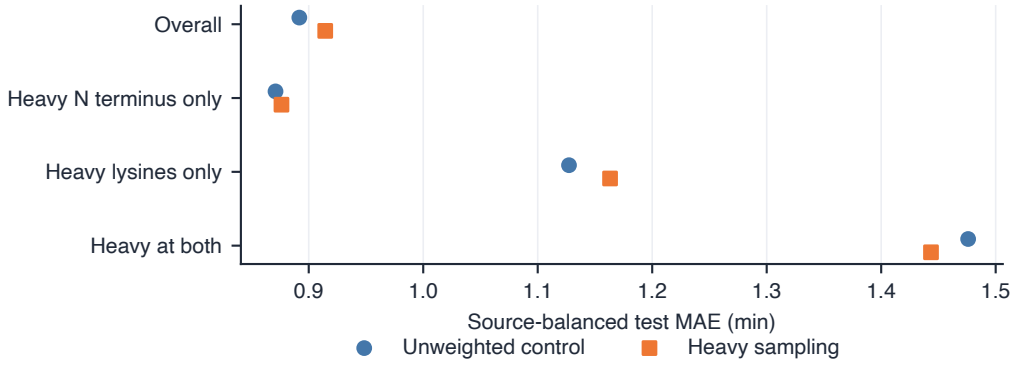


Figure 36.: RT prediction errors with and without heavy sampling. Points compare the unweighted control and heavy-sampling model on the same test data. Test MAEs are in observed minutes, with equal weight for each of the four sources. Rows show all peptidoforms (Overall) or heavy dimethylation at the N-terminus only, at lysines only, or at both.

The differences between modification groups were larger than those between the two models. In both models, test MAEs were lowest for N-term-only peptidoforms, at about 0.87 min, and highest for peptidoforms dimethylated at both positions, at 1.44–1.48 min. Lys-only peptidoforms had intermediate errors. Despite the small improvement with heavy sampling, peptidoforms dimethylated at both positions still had the largest prediction errors.

Loss weighting with the Carafe2 RT model

In the second approach, we tested whether giving selected prediction errors greater weight during training could improve RT predictions with the Carafe2 model. We multiplied the absolute errors for peptidoforms carrying heavy dimethylation at both the N-terminus and lysines by six when calculating the MAE training loss. Errors for the other groups retained their original weight. This increased the penalty for incorrect RT predictions in the selected group, so these errors had a stronger influence on model updates during fine-tuning. We compared this loss weighting with an unweighted control for both joint-anchor RT and Chronologer HI. Each comparison was repeated with three training seeds, using the same learning rate and sampling settings within each target. All model parameters were updated during fine-tuning.

The models were evaluated on the same four-source model-test partition as in the sampling comparison. Figure 37 shows the difference in test MAE between each weighted model and its control, in observed minutes and with equal weight for each source (Section 3.5.2). Negative values indicate lower error with loss weighting.

Loss weighting slightly reduced test MAE for peptidoforms carrying heavy dimethylation at both the N-terminus and lysines. The reduction occurred for all three training seeds with both targets, averaging 0.0467 min for joint-anchor RT and 0.0582 min for Chronologer HI. However, mean overall test MAE changed little and was slightly higher with loss weighting for both targets. Thus, loss weighting gave a small improvement for the selected group without improving overall prediction accuracy.

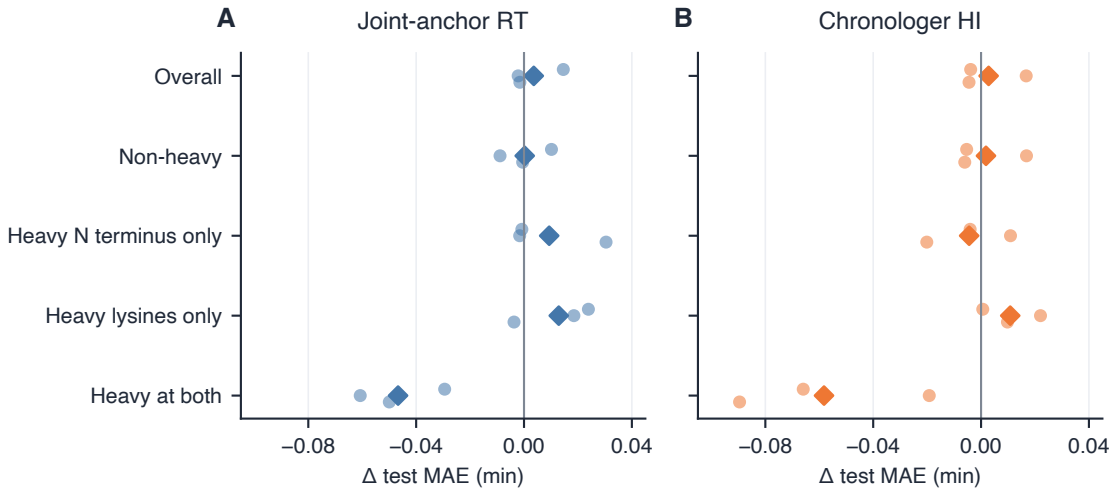


Figure 37.: RT prediction error changes with heavy loss weighting. Panels show joint-anchor RT (A) and Chronologer HI (B). Small points show test MAE differences (weighted minus unweighted) for three training seeds; diamonds show their mean. Test MAEs are in observed minutes, with equal weight for each of the four sources. Negative values indicate lower error with loss weighting.

4.6.3. Source-specific models gave lower errors on their own source

We next asked whether training on a single source could improve RT predictions for that source compared with training on all four sources. For training on all four sources and on each individual source, we tested several training configurations and selected a model based on validation performance. We compared these five DIA-NN models on the same test observations for each source (Figure 38). All models used DIA-NN-space targets, but their learning rates and sampling settings differed.



Figure 38.: Source-wise test errors of DIA-NN RT models trained on all four sources or a single source. Each row shows one selected model, labelled by its training source; columns indicate the data used for testing. Cells give MAE in observed minutes on 5,728, 3,444, 3,649 and 964 source observations, respectively. All models use DIA-NN-space targets, but their learning rates and sampling settings differ.

Each selected source-specific model had a lower MAE on its own source than the model trained on all four sources. Training on all four sources therefore did not give the lowest errors on any individual source. For MD046–MD048, source-specific training reduced the MAE from 1.12 to 0.85 min, but errors increased on all three other sources. For the models trained on the other

three sources, prediction errors were highest on MD046–MD048, with MAEs ranging from 1.55 to 1.96 min.

The two TripleMix sources showed a different pattern. The model trained on T_EX0097XX also predicted RTs for T_Hamilton more accurately than the model trained on all four sources, reducing the MAE from 0.82 to 0.67 min. Likewise, the model trained on T_Hamilton gave a lower MAE on T_EX0097XX, from 0.74 to 0.60 min. Thus, the advantages of source-specific models extended to some other sources, but not consistently across all four. Because learning rates and sampling settings also differed, these results cannot be attributed to the training sources alone.

4.6.4. Full-model fine-tuning gave lower validation errors

We compared fine-tuning the entire RT model with fine-tuning only selected parts while keeping the rest unchanged. The aim was to test whether both approaches could give equally accurate RT predictions. All variants were trained on the combined training data from all four sources, using unweighted MAE loss and unchanged sampling. They used a learning rate of 0.0003, 40 training epochs and three training seeds. DIA-NN used DIA-NN-space targets; Carafe2 used joint-anchor RT. Other training settings were matched within each model family. Table 14 reports validation MAEs for the models selected from each training run.

Table 14.: Validation errors with full-model and restricted fine-tuning. Values are validation MAEs in observed RT minutes, with equal weight for each of the four sources, reported as mean ± SD over three training seeds. Overall includes all validation peptidoforms; Heavy at both includes those carrying heavy dimethylation at both the N-terminus and lysines.

Model	Trainable components	Overall MAE (min)	Heavy at both MAE (min)
DIA-NN	Entire model	0.916 ± 0.0159	1.327 ± 0.0039
DIA-NN	Two heavy-dimethyl embedding rows	1.177 ± 0.0001	1.646 ± 0.0004
DIA-NN	Output layer + two heavy-dimethyl embedding rows	1.043 ± 0.0008	1.537 ± 0.0042
Carafe2	Entire model	0.941 ± 0.0022	1.369 ± 0.0102
Carafe2	Decoder + modification network	1.140 ± 0.0054	1.753 ± 0.0066

Full-model fine-tuning gave lower validation MAEs overall and for the Heavy at both group in both model families. For DIA-NN, updating only the heavy-dimethyl embeddings increased Heavy at both MAE by 0.32 min compared with full-model fine-tuning. For Carafe2, restricting training to the decoder and modification network increased Heavy at both MAE by 0.38 min. Across all variants, Heavy at both MAEs remained higher than overall MAEs. Full-model fine-tuning therefore gave more accurate predictions, but errors remained higher for peptidoforms carrying heavy dimethylation at both positions than for the validation population overall.

4.6.5. MAE training gave the lowest overall validation errors

We next tested whether training with Huber or MSE could improve RT predictions compared with MAE training. DIA-NN was trained with all three losses using DIA-NN-space targets; Carafe2 was trained with MAE or Huber using joint-anchor RT. In all variants, the entire model was fine-tuned on the combined training data from all four sources, without additional loss weighting or changes to sampling. Training ran for 40 epochs and was repeated with three seeds, using learning rates of 0.0002 for DIA-NN and 0.0003 for Carafe2. Other training settings were matched within each model family. All selected models were evaluated using the same validation MAE in observed RT minutes, regardless of the loss used for training. Table 15 reports these errors overall and for the Heavy at both group.

Table 15.: Validation errors with different training losses. Values are unweighted validation MAEs in observed RT minutes, with equal weight for each of the four sources, reported as mean $\pm$ SD over three training seeds. Overall includes all validation peptidoforms; Heavy at both includes those carrying heavy dimethylation at both the N-terminus and lysines.

Model	Training loss	Overall MAE (min)	Heavy at both MAE (min)
DIA-NN	MAE	0.908 ± 0.0021	1.338 ± 0.0053
DIA-NN	Huber ($\delta = 0.03$)	0.925 ± 0.0012	1.354 ± 0.0057
DIA-NN	MSE	0.944 ± 0.0006	1.356 ± 0.0055
Carafe2	MAE	0.941 ± 0.0022	1.369 ± 0.0102
Carafe2	Huber ($\delta \approx 0.02255$)	0.950 ± 0.0024	1.370 ± 0.0074

Models trained with MAE gave the lowest mean overall validation errors in both model families. Compared with these models, MSE training increased overall validation error by 0.04 min for DIA-NN, and Huber training increased it by 0.01 min for Carafe2. For the Heavy at both group, MAE training also gave the lowest errors for DIA-NN, whereas Carafe2 errors were nearly identical with MAE and Huber training. From these comparisons, we conclude that Huber and MSE training offered no advantage over MAE training in terms of validation error.

4.6.6. External fine-tuned RT models were evaluated in DIA searches

We finally evaluated selected externally fine-tuned RT models through the complete DIA-NN and Carafe2 workflows. The resulting predicted spectral libraries were used in the final DIA-NN searches of the held-out files. Performance was assessed by the numbers of identified Heavy-N precursor groups, N-terminal peptides and N-terminal sites. For Carafe2, we also compared predicted and observed RTs for precursors identified by both the integrated and external workflows.

Coverage with selected external RT models

We used the DIA-NN and Carafe2 models selected for overall RT prediction performance (Table 30). The external DIA-NN model was combined with fragmentation models adapted through integrated tuning. External Carafe2 used the unchanged pretrained fragmentation model to predict fragment intensities. The comparisons therefore include differences in fragmentation prediction as well as RT models. Figure 39 compares Heavy-N precursor groups, N-terminal peptides and N-terminal sites with the unchanged-model DIA-NN baseline and the integrated workflows.

In the DIA-NN comparison, the unchanged-model baseline gave the highest Heavy-N counts at all three levels in MD046–MD048, T_EX0097XX and T_Hamilton (Figure 39). MG054 was the exception: the external workflow gave slightly higher counts than the baseline at all three levels, with site counts rising from 674 to 682. External and integrated DIA-NN tuning gave similar counts, with small differences in either direction. For example, external tuning identified 49 fewer sites in MD046–MD048 but 14 more in T_EX0097XX than integrated tuning.

External Carafe2 identified fewer Heavy-N precursor groups, N-terminal peptides and N-terminal sites than integrated Carafe2 in both datasets. In MD046–MD048, it identified 3,432 sites, compared with 4,458 for integrated Carafe2. This remained above the DIA-NN baseline of 2,942 sites. In MG054, precursor-group and peptide counts were slightly higher than with the baseline, whereas site counts were lower. Thus, the selected external RT workflows did not consistently increase Heavy-N coverage over the baseline or the integrated workflows.

Heavy sampling compared with the external control

We next used the same DIA-NN heavy-sampling model and unweighted control as in Section 4.6.2, where we compared their RT test MAEs. Here, we tested whether the heavy-sampling model increased Heavy-N coverage in the final DIA searches. Both RT models used the same learning rate and DIA-NN-space target. The searches also used the same fragmentation models. Figure 40 shows coverage changes relative to the external control.

The heavy-sampling model identified more N-terminal peptides and sites in all four datasets. Site gains ranged from 10 to 40, with MD046–MD048 increasing from 2,690 to 2,730 sites. Precursor-group counts increased in three datasets, but decreased by three in T_EX0097XX. These gains over the external control were small, and site counts remained below the unchanged-model baseline in the first three datasets.

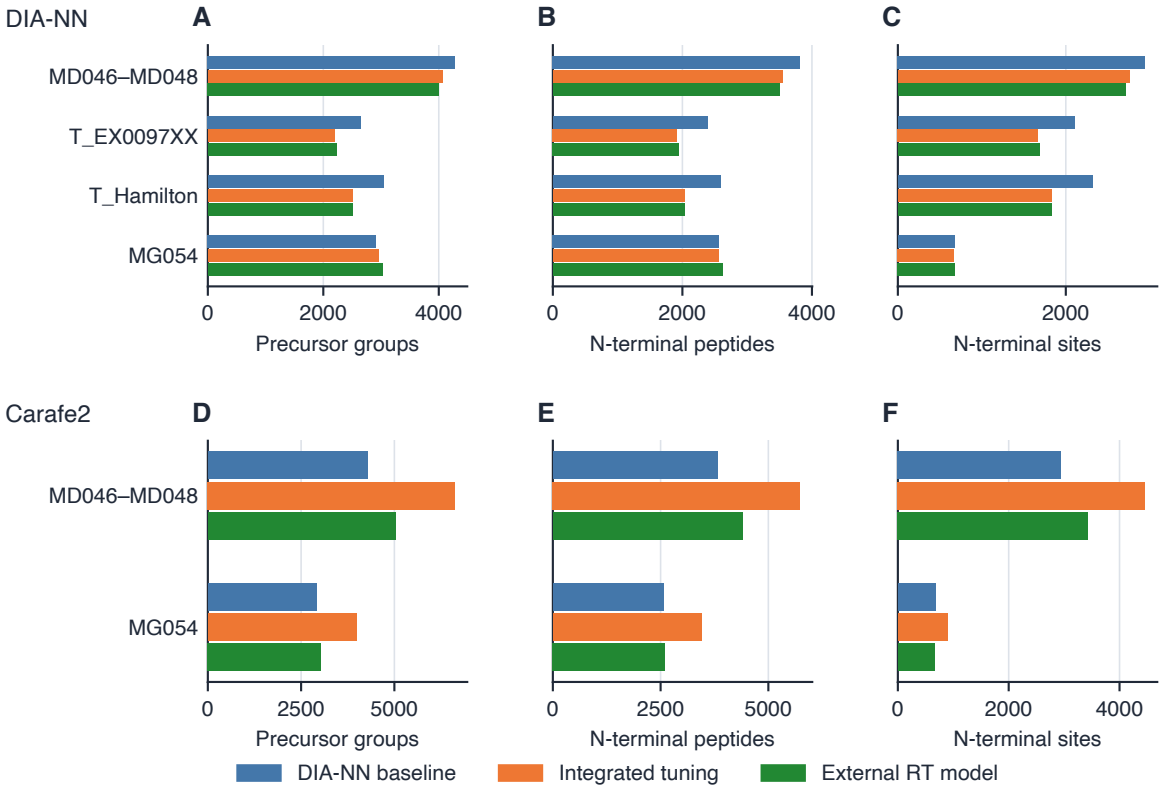


Figure 39.: Heavy-N coverage with integrated and external RT workflows. Panels A–C show DIA-NN comparisons across four datasets; panels D–F show Carafe2 comparisons in MD046–MD048 and MG054. Columns show precursor groups, N-terminal peptides and N-terminal sites. Bars compare the unchanged-model DIA-NN baseline, integrated tuning and the workflow using a selected external RT model. Objects were counted once across the same held-out files per dataset, following Table 11 and the selection in Figure 15. Exact counts are given in Table 28.

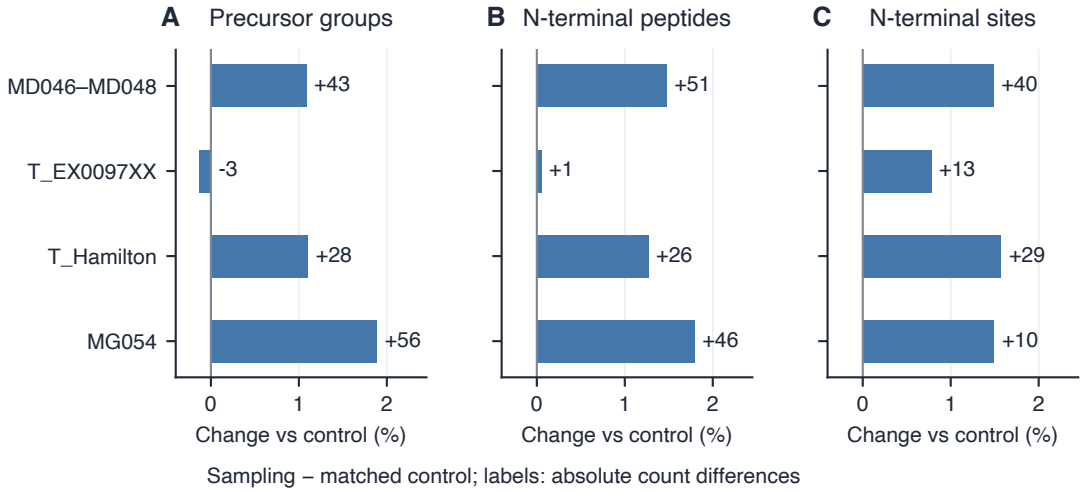


Figure 40.: Heavy-N coverage changes with the selected DIA-NN sampling model. Panels show precursor groups (A), N-terminal peptides (B) and N-terminal sites (C). Bars give percentage changes from the unweighted external control; labels give absolute count differences. Objects were counted once across the same held-out files per dataset and search condition, using the selection in Figure 15.

RT agreement with the external Carafe2 model

To examine RT agreement after external fine-tuning, we compared predicted and observed RTs from the final DIA-NN reports for the integrated and external Carafe2 workflows. As in Section 4.5.5, we used precursors of semi-N peptides carrying the Heavy-N label. Both scatter plots show the same precursors identified by both workflows in the same held-out files. This gave 12,101 paired observations in MD046–MD048 and 6,916 in MG054. Figure 41 shows predicted versus observed RTs and absolute RT errors for the integrated and external Carafe2 workflows.

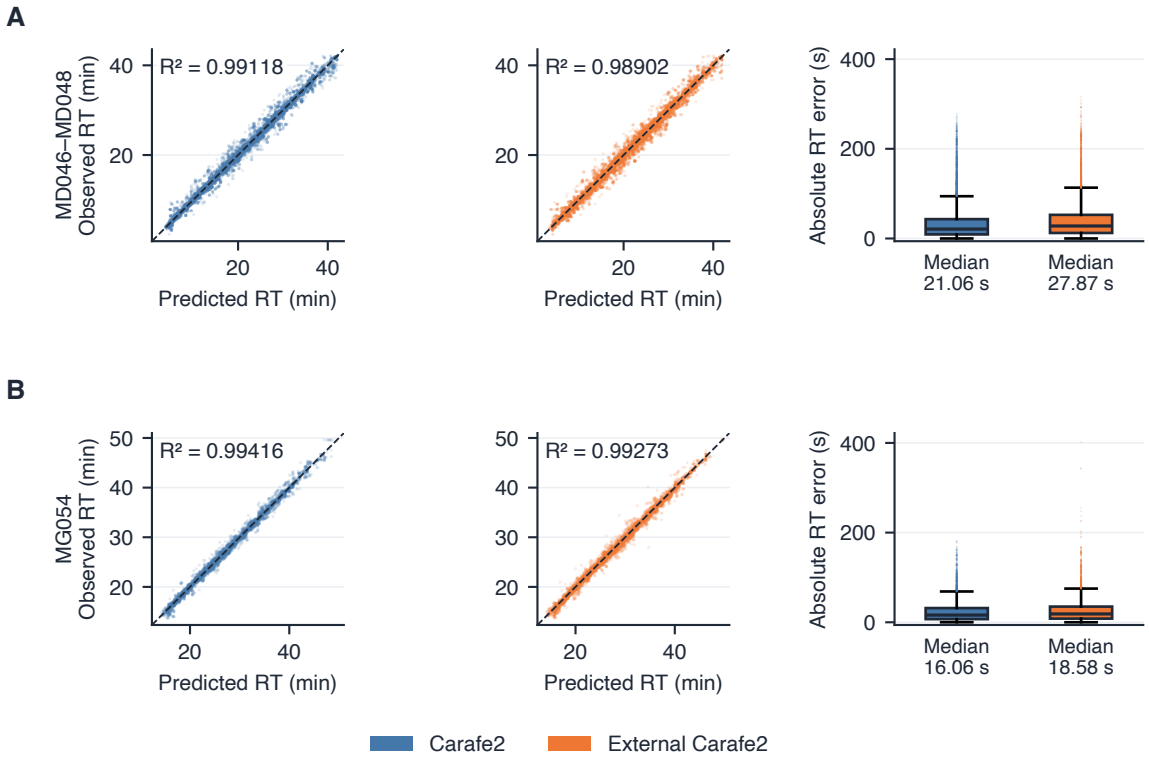


Figure 41.: RT agreement with integrated and external Carafe2 workflows. Rows show MD046–MD048 (A) and MG054 (B). The left and middle columns respectively show predicted versus observed RTs for the integrated and external workflows. Both scatter plots show the same precursors identified in the same held-out files, with identical sequence, charge and modifications. Dashed lines indicate equal predicted and observed RTs. The right column shows absolute RT errors in seconds for these precursors. R^2 uses observed RT as the reference. Boxes show medians and interquartile ranges; whiskers follow the 1.5-IQR rule, with outliers shown as points. Medians are given below the boxes.

Absolute RT errors were larger with external Carafe2 in both datasets (Figure 41). The median absolute RT error increased from 21.06 to 27.87 s in MD046–MD048 and from 16.06 to 18.58 s in MG054. Therefore, the external Carafe2 workflow did not improve RT agreement for precursors identified by both workflows.

5. Discussion

This thesis investigated whether adapting peptide-property models could improve the identification of Heavy-N peptides in HUNTER DIA data. The results differed between workflows, and lower RT prediction errors did not consistently lead to higher Heavy-N coverage. The following discussion places these findings in the context of labelling chemistry, model adaptation and the complete search workflows, and addresses the limitations of the study.

5.1. HUNTER chemistry and search-space design

We observed a median RT shift of +0.41 min in MD048 for N-terminally heavy-dimethylated forms relative to their free-N-terminal counterparts (Section 4.1.1). This later elution is consistent with the increased hydrophobicity of dimethylated peptides described by Cho et al. (2016). The two added methyl groups increase the nonpolar surface of the peptide. The increased hydrophobicity strengthens interactions with the reversed-phase stationary phase and favours later elution. However, the HUNTER protocol used for our datasets applied heavy dimethyl labels. This makes it possible to distinguish dimethylation introduced during sample preparation from endogenous methylation (Demir et al., 2022). The deuterium in these labels can also affect retention. Boutilier et al. (2012) observed slightly earlier elution for most deuterated dimethylated peptides than for their light-labelled counterparts. They linked this shift to weaker interactions with the reversed-phase stationary phase. Peptides carrying heavy dimethyl labels are not necessarily represented in the training data of pretrained RT models. For example, the original AlphaPeptDeep RT model was trained on unmodified peptides and peptides carrying carbamidomethylation, methionine oxidation or protein-N-terminal acetylation (Zeng et al., 2022). Neither light nor heavy dimethylation was included.

HUNTER depletes internal peptides generated during experimental digestion, which increases the relative abundance of N-terminal peptides in the remaining mixture (Weng et al., 2019). This reduces the interfering background of internal peptides and facilitates the identification of N-terminal peptides by LC-MS/MS. However, proteolytic processing can generate these N-termini at different positions within protein sequences. Identifying the corresponding N-terminal peptides therefore still requires a semi-specific search, which substantially increases the candidate space. In our workflow, the resulting candidate space made spectral-library generation computationally demanding. A broad *A. thaliana* library-generation attempt used 415.2 GB of RAM before the system ran out of available memory (Figure 14). Access to bwUniCluster 3.0 provided substantially greater CPU and memory resources for the large DIA-NN benchmark runs (Table 12).

The choice of modifications also strongly affected the size of the candidate space (Figure 13). We needed search settings that worked across the organisms represented in our benchmark and yielded enough Heavy-N identifications to compare model-adaptation workflows. We therefore balanced identification coverage against candidate-space size, memory requirements and runtime, while taking the expected HUNTER labelling chemistry into account. Treating lysine dimethylation as fixed reduced the number of modification combinations. This choice is supported by lysine labelling efficiencies typically above 95% in successful HUNTER experiments (Stockert et al., 2024). Variable N-terminal dimethylation and acetylation retained the main N-terminal states of interest alongside free N-termini. A broader search could also consider N-terminal cyclisation of glutamine, glutamate and carbamidomethylated cysteine, as included in other HUNTER protocols (Stockert et al., 2024). These additional N-terminal states could be relevant in a broader biological characterisation of the N-terminome. Including these additional forms would further expand the search space. We therefore retained the selected settings for our model-adaptation comparisons and did not test whether a broader search would improve identification coverage.

Search-space design also involves a statistical trade-off. Noble (2015) argued that including irrelevant candidates can reduce identification power at a given FDR threshold. However, excluding peptides present in the sample can lead to incorrect matches to remaining candidates (Sticker et al., 2017). Our simplified modification settings have the same potential limitation if excluded peptide forms are present and can match retained candidates. Search-space restrictions should therefore reflect the expected sample chemistry and keep memory use and runtime manageable. We did not test whether these restrictions affected the accuracy of our accepted DIA identifications.

5.2. Integrated tuning and N-terminal coverage

Integrated DIA-NN tuning reduced Heavy-N site coverage in all four datasets, whereas Carafe2 increased it (Figures 15 and 29). In MD046–MD048, Carafe2 identified about 52% more uniquely assigned N-terminal sites than the unchanged-model DIA-NN baseline. The size of the changes varied across sources. The effect of organism identity cannot be assessed separately because sample composition and LC-MS/MS conditions also differed (Tables 2 and 3).

The positive Carafe2 result is consistent with Wen et al. (2025b), who used DIA-derived observations to adapt AlphaPeptDeep RT and fragmentation models. In their DIA-NN searches, adapting both models yielded more precursor identifications than adapting either model alone. In our comparison, the DIA-NN baseline and Carafe2 used DIA-NN 2.5.0 to search the same held-out files with the shared search settings. The spectral libraries were generated using different models and training procedures. To evaluate false-hit behaviour in the DIA-NN baseline, shuffled entrapment proteins were added to the FASTAs for MD046–MD048 and both TripleMix datasets. This also introduced entrapment candidates into the predicted libraries. Carafe2 generated its libraries from FASTAs containing only target proteins. However, Carafe2 also gained sites in MG054, where neither workflow included entrapment candidates. This shows that the entrapment difference was

not required for a gain. The Carafe2 workflow provided higher Heavy-N site coverage under the tested conditions, although the contribution of model adaptation alone remains unresolved.

An important finding of this thesis is that changes in workflow or DIA-NN version altered which Heavy-N identifications were accepted, as well as their total number. In both the DIA-NN tuning and Carafe2 comparisons, each search had exclusive identifications in all four datasets at the precursor-group, N-terminal peptide and N-terminal site levels. Even in MG054, where the Heavy-N peptide count remained nearly unchanged after DIA-NN tuning, each search identified around 700 peptides absent from the other (Figure 17). Carafe2 identified more Heavy-N peptides overall in all four datasets, but also lost some identifications from the DIA-NN baseline (Figure 31). In addition, the MG054 version comparisons showed both additional and lost Heavy-N peptides (Figure 43). Workflow evaluation should therefore consider which identifications are retained, added and lost alongside total coverage. This complementarity suggests that combining results from several search configurations could increase the total number of distinct Heavy-N identifications. In DDA proteomics, Kwon et al. (2011) investigated such integration and showed that simply merging separately FDR-filtered search results could exceed the target FDR. Combining our DIA search results would therefore require an appropriate FDR-control procedure for the combined identifications.

Comparing the identified precursors revealed two further differences between the workflows. Across the Heavy-N precursor groups compared here, DIA-NN tuning increased the proportion of 2+ charges and reduced that of 4+, whereas Carafe2 largely retained the DIA-NN baseline distribution (Figures 18 and 32). In the paired RT comparison, Carafe2 showed larger reported RT deviations for shared accepted precursors of semi-N Heavy-N peptides, despite higher overall Heavy-N site coverage (Figure 33). Our paired analysis directly compares reported RT agreement for the same complete precursor forms accepted by both searches in the same file. However, it does not cover precursors that were not identified by one or both searches and cannot explain why they were not accepted by DIA-NN. In a controlled experiment, Bruderer et al. (2016) showed that adding noise to library iRT values reduced peptide identification in Spectronaut. In our workflow comparison, however, Carafe2 achieved higher Heavy-N site coverage despite larger reported RT deviations among shared precursors. One possible explanation is that RT agreement is only one of several signals used by DIA-NN to score candidates, alongside fragment agreement (Demichev et al., 2020). Other signals could have supported additional identifications despite larger RT deviations, although their contribution to the Carafe2 gain was not assessed separately.

The biological relevance of the coverage gain depends on whether it includes additional N-terminal positions. Different peptide forms and charge states can support the same site, while peptides matching several protein positions cannot always be assigned unambiguously. Our unique-site counts therefore provide a conservative measure of the positions that could be assigned under the chosen FASTA. The Carafe2 gain extended to these uniquely assigned sites, providing a broader set of protein N-termini for subsequent biological interpretation.

5.3. Practical limitations and transparency of DIA-NN

Our integrated DIA-NN tuning experiments did not improve Heavy-N site coverage in any of the four datasets (Figure 15). The additional tuning and search controls also did not restore the matched DIA-NN baseline site coverage (Sections 4.4.6 and 4.4.8). These results raised the question of why tuning had not helped and how the workflow could be improved. However, the available documentation offered limited support for investigating this outcome. It provides commands for generating training libraries and applying tuned models, and recommends verifying that tuning improves on the unchanged model (Demichev, 2026b). However, the instructions did not describe the workflow in enough detail for us to follow each step directly or check our pipeline against a complete example. In addition, we found no published comparison showing that DIA-NN's integrated tuning increased identifications of peptides carrying a specific modification relative to the unchanged models. Developing and investigating our pipeline therefore required us to piece together information from GitHub issues, direct developer replies and our own exploratory tests. Despite these checks, we cannot rule out having overlooked relevant settings or not fully understood their effects on tuning.

To understand why the unchanged-model DIA-NN baseline achieved high Heavy-N coverage, we investigated how the default predictor handled our heavy dimethyl label when its modification tokens were absent from the default dictionary. Because we found no explanation of this handling in the repository documentation, we asked the developer directly. He explained that unknown modification tokens are typically replaced by those of the corresponding unmodified residues and recommended using UniMod:255 while retaining the heavy-label mass (Demichev, 2026d). Following this recommendation, we tested UniMod:255 with the same heavy-label mass, but obtained substantially fewer Heavy-N identifications than with the UniMod:510 baseline (Figure 27). This left us unable to explain why the UniMod:510 baseline performed better, and the exact internal representation of its heavy dimethyl label remains unclear. Software-version comparisons showed another limit to our understanding of the workflow: accepted identifications changed even though the audited baseline libraries contained identical precursor forms and stored RT and fragment predictions (Table 24). We could establish that the identification sets differed, but could not identify the internal changes responsible.

The training outputs also offered limited guidance for evaluating and selecting models. Suitable training settings require considering hyperparameters such as LR and the number of training epochs. However, how DIA-NN determined training duration and selected the model checkpoint remained unclear to us (Demichev, 2026b). The reported loss also required clarification; the developer described it as probably training loss per batch (Demichev, 2026c). At most tested LRs, losses decreased and reached similar final values (Figure 28). Without validation-loss curves, these trends could not show whether performance on unseen data improved or which LR was most suitable. Only one LR was evaluated in final searches. Why tuning reduced coverage remained unclear from both the logs and the identification counts. This led us to develop an external RT framework, where we had control over training settings and could evaluate model performance through separate validation.

5.4. External RT training and model performance

Our external RT training used peptide assignments and measured RTs from the initial DIA-NN training searches. These observations served as training data but provided no independent ground truth, as the peptide assignments had not been independently verified. To improve comparability, we selected only datasets acquired on the same Orbitrap Exploris 480, although their LC systems and acquisition settings still differed (Table 3). Pooling these sources broadened the training data but required comparable RT targets. Alignment addressed these chromatographic differences but did not verify the peptide assignments. Alignment reduced prediction errors relative to rescaling alone (Figure 34), yet used only one mapping per source. In MD046–MD048, this pooled three experiments acquired at different times without separately aligning their runs. RTs can drift over time, even when nominal LC settings remain unchanged (Remes et al., 2020). Such drift was not established in our data, but remaining differences within sources could have limited pooled training. Source-specific models were more accurate on their own source than the model trained on all four sources (Figure 38). Differences in measurement conditions could have limited the benefit of pooled training, but the peptides used for training and the training settings could also have affected accuracy. Fine-tuning DeepLC on experiment-specific data improved RT predictions, consistent with the better accuracy of our source-specific models (Bouwmeester et al., 2026). Together with our results, this suggests that the match between training data and measurement conditions matters alongside the amount of training data.

We used sampling or loss weighting to give selected heavy-dimethylated groups more influence during training. Both approaches slightly improved predictions for peptidoforms carrying heavy dimethylation at both the N-terminus and lysines, while slightly increasing overall errors (Figures 36 and 37). With DIA-NN sampling, errors also increased for peptidoforms carrying heavy dimethylation at only one of these positions. The selected sampling model nevertheless identified 10–40 additional Heavy-N sites per source compared with its matched external control (Figure 40). This raises the question of how much loss in accuracy for other groups is acceptable when prioritising the target group. One option would be to retain a general RT model and use a specialised model for peptidoforms dimethylated at both positions. Whether this two-model approach would preserve accuracy across groups and improve coverage remains untested. Limited GPU capacity on Wynaut prevented final searches with the weighted Carafe2 models, leaving their identification benefit unknown.

The gain over the external control did not imply a consistent advantage over the DIA-NN baseline or integrated workflows (Figure 39). External DIA-NN searches unintentionally used integrated tuned fragmentation models alongside the external RT models. External Carafe2 used pretrained fragmentation models, whereas integrated Carafe2 adapted RT and fragmentation. These comparisons therefore assess the complete workflows and do not isolate the contribution of RT adaptation to Heavy-N coverage.

5.5. Outlook

Given the central role of data quality in machine learning, better data for training and evaluation offer a promising starting point for further model development. One option would be to add suitable iRT standards to the prepared HUNTER samples before LC-MS/MS. Their measured RTs and assigned iRT values could be used to map sample RTs onto a common reference scale (Escher et al., 2012). This could provide more comparable training targets across measurements, with calibration to each model's RT scale where needed. Reliably identified sample peptides could then serve as additional anchors to align further datasets to this reference. Run-level alignment tools could also help address remaining differences before pooling training data. Synthetic Heavy-N peptides with independently verified sequences and modification positions could provide ground truth for selected target peptides. Separate RT and fragment-spectrum measurements, followed by controlled addition to complex samples, could test prediction accuracy and reveal which added reference peptides were not identified. This would require synthesising the reference peptides and independently verifying their sequences and modification positions. Measurements of these reference peptides could assess identification accuracy, but would not establish whether other peptides in the sample were correctly identified.

Further comparisons should establish which adaptations improve Heavy-N identification. DIA-NN RT-plus-fragment tuning identified more sites than RT-only tuning, although both remained below the baseline, motivating a fragmentation-only test with the unchanged RT model. External RT models should be compared with the same fragmentation model to isolate their contribution to coverage. The weighted Carafe2 RT models slightly improved predictions for peptidoforms dimethylated at both the N-terminus and lysines, but were not tested in final searches. Such searches could show whether this improvement leads to additional Heavy-N sites. These model comparisons should use the same candidate sets and search settings, with unchanged models as controls.

To explore broader N-terminal coverage, searches could include N-terminal cyclisation and peptidoforms containing unmodified lysines, which were excluded here. This extension should be assessed by the sites gained and retained, error diagnostics, memory use and runtime. Further organism datasets, such as pure Arabidopsis samples or moss, could be used to test whether the search and adaptation strategies perform differently. LC conditions should be kept comparable to distinguish measurement effects from organism-related differences. Model adaptation could also target N-terminally acetylated peptides to test whether it improves their identification.

Advances in sequence modelling and machine learning could also help improve peptide-property prediction. ESM-2 was trained on a large collection of protein sequences to learn patterns and relationships between amino acids. These learned representations supported protein structure prediction (Lin et al., 2023). These learned representations could support RT and fragment-intensity prediction through an added prediction head, possibly with task-specific fine-tuning. For HUNTER peptides, this would require suitable encoding of heavy dimethylation. Adapting these models to HUNTER could help identify N-terminal peptides that current workflows leave undetected.

6. Conclusion

This thesis investigated how peptide-property models could be adapted to HUNTER DIA data and whether these adaptations improved Heavy-N identification. An automated DIA-NN benchmark pipeline, a HUNTER-compatible Carafe2 workflow and an external RT fine-tuning framework were developed for this purpose. Together, they enabled reproducible comparisons across four datasets using separate training and held-out search files. Evaluation at the precursor, peptide and N-terminal site levels related the identification results to N-terminal coverage.

The integrated workflows differed in their identification performance. DIA-NN tuning reduced Heavy-N site coverage in all four datasets, whereas the integrated Carafe2 workflow increased it relative to the unchanged-model DIA-NN baseline. The Carafe2 gain extended from Heavy-N peptides and precursor groups to uniquely assigned N-terminal sites, providing broader coverage of protein N-termini for biological interpretation. However, differences in models, spectral libraries and database configurations mean that this gain reflects the complete workflow rather than the isolated effect of model adaptation. Both workflows also changed which Heavy-N peptides were identified, with additional identifications accompanied by losses from the baseline set. Even similar total counts after DIA-NN tuning concealed substantially different peptide sets.

External RT fine-tuning provided greater control over training and model evaluation. Heavy-focused sampling and loss weighting slightly improved RT predictions for selected dimethylation groups, and sampling yielded small site gains over the unweighted external control. Nevertheless, the selected external workflows did not consistently improve Heavy-N coverage over the baseline or the corresponding integrated workflow. These findings show that the benefit of adaptation depends on the workflow and cannot be assessed from RT prediction error alone. The developed workflows provide a basis for evaluating model accuracy alongside Heavy-N peptide identification and N-terminal site coverage in HUNTER DIA analyses.

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A. Appendix

A.1. Thesis resources

The code is available through the HUNTER DIA model-adaptation overview repository, which links to the individual software repositories (Section 3.6.2): `https://github.com/mmmddd98/hunter-dia-model-adaptation`. Additional clones of these repositories will be deposited on the [redacted] together with the plot package and FASTA databases.

Table 16 lists the code, dataset and result locations. The designated thesis archive is [redacted] Measurement files and experiment results will remain at their existing locations. Share paths start at [redacted] Access requires permission to use the corresponding storage.

Table 16.: Locations of code, data and results.

Resource	Location
Code and plot package	
Repository clones	[redacted]
Plot package	[redacted]
Datasets	
Measurement files (RAW)	[redacted]
FASTA databases	[redacted]
Results	
Cluster search results	[redacted]
Carafe2 libraries	[redacted]
DIA-NN RT training	[redacted]
Carafe2 RT training	[redacted]

The plot package contains the plotting code, required derived data, figure PDFs and short instructions for each figure. The dataset YAML files linked through the overview repository define the input files and training/test assignments. The listed measurement-file directory contains the RAW files; the assignments are also listed below. The FASTA archive will contain the native and entrapment databases. The FASTA used for each search will also be retained with its pipeline results.

The result folders cover the search and training outputs, start commands and logs.

A.2. Raw-file allocations

Tables 17 and 18 list all measurement files assigned to training or held-out testing in the four final benchmark datasets. File identifiers refer to the original Thermo RAW files; the corresponding mzML files were used for the searches. Assignments follow the explicit `train_files` and `test_files` lists in the dataset YAML files linked through Section 3.6.2 (analysis-repository commit `9f08db4b934c`).

Table 17.: Complete raw-file allocations for the combined MD046–MD048 benchmark.

Experiment	Training raw-file IDs	Test raw-file IDs
MD046	EX011972, EX011973, EX011974, EX011975, EX011976, EX011977, EX011979, EX011980, EX011981, EX011982, EX011983, EX011984	-
MD047	EX012096, EX012097, EX012098, EX012099, EX012100, EX012101, EX012103, EX012104, EX012105, EX012106, EX012107, EX012108	-
MD048	EX012131, EX012132, EX012133, EX012138, EX012139, EX012140	EX012134, EX012135, EX012136, EX012141, EX012142, EX012143

Note. - indicates no files assigned to that partition. All six test files originated from MD048.

Table 18.: Complete raw-file allocations for the T_EX0097XX, T_Hamilton and MG054 benchmarks.

Dataset	Training raw-file IDs	Test raw-file IDs
T_EX0097XX	EX009759, EX009760, EX009766, EX009767	EX009761, EX009768
T_Hamilton	EX011062, EX011064, EX011065, EX011067, EX011068, EX011069, EX011072, EX011074, EX011075, EX011077, EX011078, EX011079, EX011082, EX011084, EX011085, EX011087, EX011088, EX011089	EX011063, EX011070, EX011073, EX011080, EX011083, EX011090
MG054	EX009121, EX009122, EX009128, EX009129, EX009133, EX009134	EX009123, EX009124, EX009126, EX009127, EX009131, EX009132

A.3. Supplementary files

The complete raw-file allocation list is provided as [redacted] The source YAML files and benchmark FASTAs are indexed through the overview repository described in Section 3.6.2.

A.3.1. Supplementary integrated DIA-NN comparisons

Table 19.: Site-mapping audit for Heavy-N N-terminal peptides. Counts include only the three retained R* specificity classes and are distinct across the held-out files per dataset and arm. Unique denotes one protein and starting-position assignment; ambiguous denotes multiple candidate sites. Several uniquely assigned peptides may support one N-terminal site. Mapping ambiguity is separate from cleavage-class ambiguity.

Dataset	Search	Unique assignments	Ambiguous assignments
MD046–MD048	Baseline	2,997	817
	Tuned	2,803	745
T_EX0097XX	Baseline	2,201	187
	Tuned	1,762	155
T_Hamilton	Baseline	2,386	204
	Tuned	1,894	136
MG054	Baseline	678	1,887
	Tuned	672	1,889

Table 20.: R* cleavage classes of Heavy-N N-terminal peptides. Each peptide is counted once across held-out files. The three classes sum to the denominator in Figure 16A. Peptides with conflicting or unclassifiable cleavage assignments are excluded.

Dataset	Search	R*-specific	Semi-N	Semi-C
MD046–MD048	Baseline	353	3,408	53
	Tuned	347	3,143	58
T_EX0097XX	Baseline	482	1,773	133
	Tuned	410	1,383	124
T_Hamilton	Baseline	370	2,129	91
	Tuned	316	1,634	80
MG054	Baseline	356	2,168	41
	Tuned	298	2,242	21

Table 21.: Modification groups at the N-terminal peptide level. Each sequence-N-terminal-status combination is counted once across held-out files per arm. Heavy-N and acetylated denote leading UniMod:510 and UniMod:1; free denotes an unmodified peptide N-terminus. Heavy-N peptides without a retained R* class are excluded, as in Figure 16B.

Dataset	Search	Heavy-N	N-terminally acetylated	Free N-terminus
MD046–MD048	Baseline	3,814	1,524	33,959
	Tuned	3,548	1,429	30,605
T_EX0097XX	Baseline	2,388	1,596	28,076
	Tuned	1,917	1,512	23,961
T_Hamilton	Baseline	2,590	2,102	30,402
	Tuned	2,030	1,847	27,256
MG054	Baseline	2,565	2,761	4,247
	Tuned	2,561	2,688	3,821

Table 22.: Heavy-N coverage at three levels across MD048 ablation contexts. All counts are distinct across the same six held-out files. A, B and C refer to the contexts in Figure 22; training-file counts apply only to tuned arms. The selection follows Figure 15.

Context	Search	Precursor groups	N-terminal peptides	N-terminal sites
A	Baseline	4,610	4,119	3,172
A	RT only	3,778	3,284	2,555
A	RT + fragment	4,243	3,694	2,868
B	Baseline	4,610	4,119	3,172
B	RT only	3,826	3,324	2,584
B	RT + fragment	4,179	3,622	2,822
C	Baseline	4,279	3,814	2,942
C	RT only	3,533	3,071	2,378
C	RT + fragment	4,067	3,548	2,759

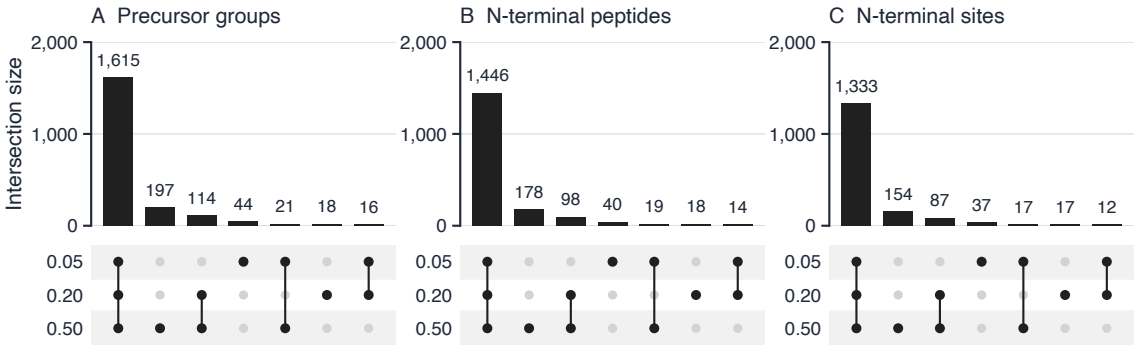


Figure 42.: Shared and exclusive Heavy-N identifications across library-export q-value thresholds. Panels show precursor groups (A), N-terminal peptides (B) and N-terminal sites (C). Bars give counts for the exact threshold combinations marked by filled dots. Each object was counted once across the two held-out MD048 files, using the selection in Figure 25.

Table 23.: Heavy-N coverage across DIA-NN versions in MG054. Counts correspond to Figure 26. No completed tuned search was available for version 2.3.0.

DIA-NN version	Search	Precursor groups	N-terminal peptides	N-terminal sites
2.3.0	Baseline	3,133	2,737	706
2.5.0	Baseline	2,914	2,565	674
2.5.0	RT + fragment tuning	2,952	2,561	669
2.6.0	Baseline	2,941	2,590	671
2.6.0	RT + fragment tuning	2,883	2,498	660
2.7.0	Baseline	3,410	2,981	783
2.7.0	RT + fragment tuning	3,041	2,624	691

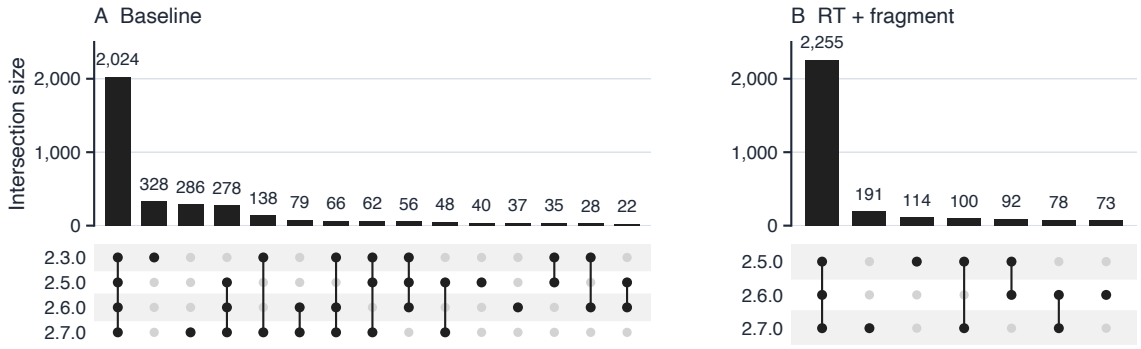


Figure 43.: Shared and exclusive Heavy-N N-terminal peptides across DIA-NN versions in MG054. Panels show the unchanged-model baseline (A) and RT-plus-fragment tuning (B). Bars give counts for the exact version combinations marked by filled dots. No completed tuned search was available for version 2.3.0. Each peptide was counted once across the six held-out files, using the selection in Figure 26.

A.3.2. Audit of baseline predicted spectral libraries

A custom binary parser compared the MG054 baseline predicted spectral libraries by modified peptide sequence and charge, independently of record order. All six version pairs gave the same results (Table 24).

Table 24.: Content comparison of MG054 baseline predicted spectral libraries across DIA-NN versions. All six pairs of versions 2.3.0, 2.5.0, 2.6.0 and 2.7.0 were compared.

Compared content	Result
Complete precursor forms	47,130,003 shared target forms; no version-specific forms or decoys
RT predictions	Bit-identical stored values for every shared form
Fragment predictions	562,122,060 fragments per library; identical fragment identities and multiplicities, with bit-identical m/z and relative intensities

The complete files differed; original record order and the trailing index block were not assessed. Code, detailed results and source-file hashes are provided under [redacted]

For the UniMod:255 controls, the archived library-prediction commands document the mass declaration; a standalone final-search command was not retained beside the reports.

A.4. Supporting comparisons for external RT fine-tuning

The tables below document training settings, selected models, search counts and the complete RT-target comparison. Model-test MAEs retain full peptidoforms and are distinct from search coverage. The resource package retains complete prediction rows, canonical object keys and source provenance. The unavailable integrated Carafe2 model-test prediction export is not replaced by report-level RT comparisons.

Table 25.: Training-data combinations for the RT target comparison. Percentages specify the fraction of distinct peptide sequences sampled within each source after holdout exclusion; all retained peptidoforms of selected sequences were included. Both model families used these conditions for all eleven RT target definitions.

Condition	MD046–MD048	T_EX0097XX	MG054	Condition	MD046–MD048	T_EX0097XX	MG054
1	10	0	0	8	100	100	0
2	25	0	0	9	100	0	25
3	50	0	0	10	100	0	50
4	75	0	0	11	100	0	100
5	100	0	0	12	100	25	25
6	100	25	0	13	100	50	50
7	100	50	0	14	100	100	100

Note. Fractions apply separately to each source. Each condition used five paired split and training seeds (2024–2028); models were evaluated on holdout data from all three sources.

Table 26.: Training settings for the matched DIA-NN RT target comparison. The upper block shows how strongly each group contributed to the training loss; the lower block shows its relative sampling weight. All combinations were used for DIA-NN space, joint-anchor RT and Chronologer HI.

Loss weighting	Non-heavy (%)	N-term only (%)	Lys only (%)	Heavy at both (%)
Unweighted	Every peptidoform has equal loss weight			
99% heavy, equal	1	33	33	33
95% heavy, equal	5	31.67	31.67	31.67
90% heavy, equal	10	30	30	30
95% heavy, N-term focus	5	47.5	23.75	23.75
95% heavy, Heavy at both focus	5	23.75	23.75	47.5
Sampling factors				
Profile	Non-heavy	N-term only	Lys only	Heavy at both
Unchanged	1	1	1	1
Increased heavy	1	10.17	1.7	15.04
Stronger heavy	1	14.23	2.38	21.06

Note. Loss percentages refer to whole groups. Sampling weights above 1 increase how often a group's peptidoforms are drawn; values are rounded. Learning rates: 0.00005, 0.00010 and 0.00015; seeds: 2024–2026. These combinations gave 162 models per target. Checkpoints were selected by the lowest validation MAE averaged equally across the three heavy-dimethylated groups.

Table 27.: Selected external RT training recipes and model-test errors. DIA-NN rows describe single exported checkpoints on DIA-NN-space targets; Carafe2 rows give means across three optimisation seeds on the fixed four-source partition. All rows use complete-model adaptation. MAEs are in observed minutes, averaged equally over sources; Heavy-macro additionally averages the three heavy-labelled strata equally. Sampling factors and loss weights are different interventions.

Family	Model/recipe	LR	Overall	Heavy-macro
DIA-NN	Unweighted control	0.00015	0.8918	1.1581
DIA-NN	Heavy sampling	0.00015	0.9144	1.1610
Carafe2	Joint-anchor, control	0.00045	0.9441	1.2081
Carafe2	Joint-anchor, weighted	0.00045	0.9477	1.1999
Carafe2	Chronologer, control	0.00090	0.9634	1.2582
Carafe2	Chronologer, weighted	0.00090	0.9662	1.2409

DIA-NN sampling factors are 3.05/1/4.51 for the N-term-only/Lys-only/Heavy at both groups; the control uses 1/1/1. The control and sampling checkpoints were selected at epochs 24 and 34, respectively. Carafe2 loss factors are 1/1/6, with unchanged sampling. The final Carafe2 search used a separate general external model, not either weighted recipe.

Table 28.: Heavy-N counts for the external RT search comparisons. Blocks A and B give the counts in Figure 39; block C gives the counts underlying Figure 40. Objects were counted once across all held-out files per dataset and search condition, using the selection in Figure 15.

Dataset	Model/workflow	Precursor groups	N-terminal peptides	N-terminal sites
A. DIA-NN main comparison				
MD046–MD048	DIA-NN baseline	4,279	3,814	2,942
	Integrated tuning	4,067	3,548	2,759
	External RT model	4,002	3,497	2,710
T_EX0097XX	DIA-NN baseline	2,651	2,388	2,106
	Integrated tuning	2,195	1,917	1,669
	External RT model	2,225	1,947	1,683
T_Hamilton	DIA-NN baseline	3,043	2,590	2,314
	Integrated tuning	2,504	2,030	1,827
	External RT model	2,510	2,032	1,826
MG054	DIA-NN baseline	2,914	2,565	674
	Integrated tuning	2,952	2,561	669
	External RT model	3,025	2,615	682
B. Carafe2 main comparison				
MD046–MD048	DIA-NN baseline	4,279	3,814	2,942
	Integrated tuning	6,622	5,732	4,458
	External RT model	5,039	4,406	3,432
MG054	DIA-NN baseline	2,914	2,565	674
	Integrated tuning	3,991	3,452	899
	External RT model	3,030	2,601	662
C. DIA-NN heavy-sampling comparison				
MD046–MD048	Unweighted external control	3,970	3,457	2,690
	Heavy sampling	4,013	3,508	2,730
T_EX0097XX	Unweighted external control	2,206	1,924	1,666
	Heavy sampling	2,203	1,925	1,679
T_Hamilton	Unweighted external control	2,535	2,050	1,843
	Heavy sampling	2,563	2,076	1,872
MG054	Unweighted external control	2,968	2,565	669
	Heavy sampling	3,024	2,611	679

Table 29.: Complete RT-target comparison. Mean holdout MAEs in observed minutes for all eleven RT target definitions. Errors were averaged equally across the training-data combinations in Table 25 and the three sources. Intervals are 95% confidence intervals across five paired split and training seeds. These holdouts also informed target selection.

Model	RT target	Mean MAE (min)	95% CI (min)
Carafe2	Joint-anchor RT	1.245	1.218–1.272
Carafe2	Chronologer HI	1.251	1.214–1.288
Carafe2	Robust linear	1.268	1.247–1.290
Carafe2	LOWESS	1.323	1.256–1.390
Carafe2	KDE ridge	1.362	1.316–1.408
Carafe2	Calib-RT	1.364	1.298–1.429
Carafe2	Isotonic	1.390	1.325–1.455
Carafe2	Monotonic spline	1.463	1.409–1.517
Carafe2	Source-native	2.728	2.699–2.758
Carafe2	Relative gradient RT	2.728	2.699–2.758
Carafe2	Global 60 min	4.163	4.120–4.206
DIA-NN	Joint-anchor RT	1.189	1.164–1.213
DIA-NN	Chronologer HI	1.192	1.164–1.219
DIA-NN	Robust linear	1.219	1.201–1.237
DIA-NN	LOWESS	1.274	1.204–1.345
DIA-NN	KDE ridge	1.304	1.261–1.347
DIA-NN	Calib-RT	1.322	1.245–1.399
DIA-NN	Isotonic	1.356	1.288–1.424
DIA-NN	Monotonic spline	1.435	1.384–1.486
DIA-NN	Source-native	2.635	2.592–2.678
DIA-NN	Relative gradient RT	2.635	2.592–2.678
DIA-NN	Global 60 min	4.137	4.118–4.157

Table 30.: Training settings of the external RT models used in the main search comparison. Columns describe the two external RT models used in Figure 39.

Setting	DIA-NN	Carafe2
Model identifier	`diannrt_c6d44b8f51a6`	`carafe2rt_709a44ce0f5c`
Training data	All four sources	All four sources
RT target	DIA-NN space	Chronologer HI
Trainable components	Entire RT model	Entire RT model
Training loss	Unweighted MAE	Unweighted MAE
Heavy sampling	None	None
Learning rate	0.00045	0.00045
Training epochs	40	40
Batch size	512	512
Exported training seed	2024	2024
Checkpoint selection	Lowest validation MAE on the training target scale	Lowest validation L1 loss over batches

Note. Both used Adam, zero weight decay and 10 warm-up epochs followed by cosine decay. The exported seed was chosen by the validation MAE closest to the mean across three seeds; each run supplied its validation-selected checkpoint.